

The Intention Experiment (full text) by Lynne McTaggart. In HTML for free access. Part 3 of 4.

 metallicman.com/laoban4site/the-intention-experiment-full-text-by-lynn-mctaggart-in-html-for-free-access-part-1-of-3-2-2

congjing yu

June 8, 2020



This is part 3 of 4.

This is a complete reprint of the book titled “The Intention Experiment” by Lynne McTaggart. It is a non-fiction book, and it is groundbreaking. In this book, the author has compiled all those studies about the reality of ESP, and PSI, and compiled the results. The results are pretty damning. Something is going on, and Newtonian physics cannot explain it. It can only be explained with quantum physics.

What is going on is that quantum physics is working and weaving it’s magic throughout our lives, and rather than discount things as “superstition” and out-dated religion, this book connects actual scientific studies with the quantum physics principles involved. It explains so many thing that have been discounted as pure superstition.

Thus it’s placement in my blog.

This is for those people who want nice and clean answers to what is going on, yet cannot shake off the Newtonian physics that they learned in High School. This book

teaches you that there is a deeper reality behind everything and as such, it helps explain some elements of paranormal and religion that are often discounted as primitive nonsense.

Welcome to the world of quantum physics and how all those things about prayer, intention, and spirituality actually does have a scientific foundation that they are based upon.

CHAPTER EIGHT

The Right Place

IN 1997, WILLIAM TILLER had been helping a Californian company develop product to eliminate electromagnetic pollution. The product contained a quartz crystal, which was why they had consulted him. Tiller, a physicist and professor emeritus of materials science and engineering at Stanford University, had carved out an influential niche for himself in the science of crystallization; he had written three textbooks on the subject and more than 250 scientific papers.[1](#)

The product consisted of a simple black box, about the size of a remote control. Inside its casing were three oscillators of 1–10 megahertz, barely a microwatt's worth of power when the device was turned on. The box also contained an electrically erasable, programmable, read-only memory (EEPROM) component unconventionally connected in the circuit. It seemed to be able to screen incoming electromagnetic energy, possibly through the quartz oscillators also contained inside the box: quartz was thought to modulate quantum information by rotating the direction of waves.

As Tiller examined the equipment, an outrageous idea struck him. Fascinated by evidence that remote influence worked, Tiller had been carrying out a number of his own experiments and had formulated an entire theory about 'subtle energy' in living systems. Perhaps the little box he held in his hand might help him put intention to the supreme test. If thoughts were just another form of energy, what if he attempted to 'charge' this simple low-tech machine with a human intention and then use it to try to affect a chemical process? His experiment would rest on the unthinkable assumption that thoughts could be imprisoned in a bit of electronic memory and later 'released' to affect the physical world.[2](#)

This fanciful idea would lead to a bizarre experimental result, offering convincing evidence that there is such a thing as the right place, as well as the right time, for carrying out intentions.

Tiller borrowed some lab space at the Terman Engineering building at Stanford from one of his tolerant colleagues in civil engineering, and some other space in the

biology department, made some adjustments to the commercial device, and began designing his experiments.

He wanted to go for broke, to see if this 'caged' intention could affect actual live test subjects. He realized he could not yet try his experiments on human beings, because they presented too many random, uncontrollable variables. But he could experiment on what scientists consider the next best thing to a human being: the fruit fly.

In the laboratory among the experimental animal population, the fruit fly is prom queen. Scientists have considered *Drosophila melanogaster* a model organism for more than a century, largely because its life cycle is so short. Within six days a fruit fly will completely remodel itself from larval grub to six-legged, winged insect and die just two weeks later.

Tiller had in mind an experiment that would speed up their entire developmental process even further. His Stanford colleague Michael Kohane an expert in fruit flies, had been studying the effects of supplements of nicotinamide adenine dinucleotide (NAD) on his fruit fly specimens. An important cofactor for enzymes, NAD helps in energy metabolism within cells by transporting hydrogen which is essential in setting the fly's built-in timer for larval development. Energy availability also directly affects an organism's fitness.[3](#)

NAD marshals electrons into the pathway needed to maximize energy production and metabolism; low levels of NAD adversely affect the production of adenosine triphosphate (ATP).

Every cell uses oxygen and glucose to convert ADP (adenine diphosphate) and phosphoric acid into ATP, a molecule that slow-drips energy to power most cellular processes. ADP and ATP are the equivalent of chemical energy storage tanks. Each molecule hoards a tiny supply of energy deep within its phosphorus-oxygen bond. Increasing the supply of NAD will increase the ratio of ATP to ADP, causing the cellular processes to work harder and faster, fast-forwarding larval development.

As the fruit fly develops, the higher the ATP/ADP ratio, the more energy available to the cells, and the fitter the fly. The net effect of NAD is to increase a fruit fly's overall level of health, from cradle to grave.

Electromagnetic fields can have a profound effect on cellular energy metabolism, particularly the synthesis of ATP. [4](#) Human thoughts could be construed as a similar form of energy, Tiller reasoned. But could the energy of a thought interact with the transport chain of electrons to stoke up the metabolic fire?

To carry out the protocol he had in mind, Tiller needed a second lab. He set up one near the benefactor who was going to fund the studies in a small facility in Minnesota, just north of Excelsior. There he installed Michael Kohane and Wal Dibble, one of his former graduate students.

One morning in early January 1997, Tiller gathered his four participants, including himself, his wife Jean, and two friends, all highly experienced meditators, around a table. He unwrapped the first black box, placed it in the middle of the table and turned it on.

At the signal, Tiller told them all to enter a deep meditative state. After mentally 'cleansing' the environment and the equipment itself, he stood before them, a tall, lanky figure with bright, irreverent eyes and a wispy white beard, and read aloud the intention he had scripted earlier:

Our intention is to synergistically influence (a) the availability of oxygen, protons, and ADP (b) the activity of the available concentration of NAD plus (c) the activity of the available enzymes, dehydrogenase and ATP-synthase, in the mitochondria so that the production of ATP in the fruit fly larvae is significantly increased (as much as possible without harming the life function of the larvae) and thus the larval development time significantly reduced relative to that with the control device.

Although the intention boiled down to significantly increasing the ratio of the ATP to ADP, Tiller had purposefully made the intention highly specific, so there would be no possible misunderstanding. He suspected that the more specific the thought, the more likely it was to have an effect, and so was careful, with each experiment, to pinpoint its aims. He had added 'without harming the life function of the larvae' because he suspected that if they tried to push things too far, they might well kill the tiny creatures.

The meditators held the intention for 15 minutes, before abruptly releasing it, at Tiller's signal, then they focused for a final 5 minutes on a closing intention, to mentally 'seal the intention' into the device.

Tiller had prepared an identical control box that had not been 'imprinted' with intention by wrapping it in aluminium foil and placing it in an electrically grounded Faraday cage, in order to screen out electromagnetic frequencies of all magnitudes.

He wrapped the imprinted black box, or the 'Intention-Imprinted Electronic Device', as he now called it, in aluminium foil and placed it in another Faraday cage until ready for shipping. On separate days he shipped each box via FedEx to the Minnesota laboratory, some 1500 miles away. He had been careful to blind the experiment so

that neither Dibble nor Kohane would know which device contained the intention and which the control when the two devices arrived.

The Excelsior scientists prepared several groups consisting of eight vials of fruit fly larvae and placed three of the groups of vials inside Faraday cages. They then placed both black boxes inside two of the cages with the vials and turned them on.

Over the next eight months, they carried out experiments on 10,000 larvae and 7000 adult flies, in each instance tracking the ATP/ADP ratio. After compiling their data and mapping it on a graph, Tiller and Kohane discovered not only that the ratio of ATP to ADP had increased, but also that those larvae exposed to the imprinted devices developed 15 per cent faster than normal.[5](#)

Furthermore, once the larvae had reached their adult stages they were healthier than normal, as were their descendants.[6](#) The intention not only had a positive effect on the flies themselves; it also appeared to affect the genealogical line.

By that time, Tiller had tried out other black boxes on a great number of other subjects, choosing his experimental targets with care. He needed tests like that of the fruit fly co-enzyme ratio that would show a genuine, measurable change. He decided on two new targets: the pH of water and the increase in the activity of a liver enzyme called alkaline phosphatase (ALP). He chose the pH test because water pH – the measure of acidity or alkalinity in a solution – remains fairly static and tiny changes of one-hundredth or even one-thousandth of a unit on the pH scale can be measured; a change of a full unit or more on the pH scale would represent an enormous shift that was unlikely to be the result of an incorrect measurement. ALP is another ideal test target because its activity proceeds at an unvarying rate.

In both instances, his meditators imprinted intentions into the black boxes to change the pH of water both up and down by a full pH unit and to increase by a 'significant factor' the activity of ALP. Tiller then sent off both imprinted and control boxes to Dibble, who made use of a similar study design as the fly experiment. Both experiments were extraordinarily successful.[7](#)

In the water experiments, their intentions managed to change the pH up and down by one unit, and the ALP activity was significantly increased.[8](#)

Tiller was in the midst of some of his black-box experiments when he noticed something strange. After three months, the results of his studies began to improve; the more he repeated the experiment, the stronger and quicker the effects.

Tiller decided to try to isolate the aspect of the environment that had changed. He took readings of the air temperature, in and outside the Faraday cages, and

discovered that the temperature appeared to be going up and down according to a regular rhythm or oscillation, dipping and climbing at regular intervals. He had first taken the temperature readings with an ordinary mercury thermometer. In case these results had something to do with the instrumentation, he switched to a computerized, low-resolution thermistor-based digital thermometer.

Then he tried a high-resolution thermometer. All three recorded the same readings. When he plotted it, he saw that the temperature change was oscillating at a precise rhythm every 45 minutes or so, varying by some 4°C.[9](#)

Tiller then measured the pH of water in the lab and measured its capacity to conduct electricity. He observed the same phenomenon as he had with the temperature: periodic oscillations of at least one-quarter of a unit on the pH scale, and regular dips and peaks in the water's ability to conduct electricity. Tiller was especially intrigued by the changes in pH. The acid/alkaline balance in any substance is highly sensitive to change; if the pH of a person's blood shifts up or down by just a half a pH unit, it means that they are dying or already dead.

A pattern was developing: as the temperature of the air rose, the pH fell, and vice versa, in near perfect harmonic rhythm. The water's electrical conductivity showed a similar harmonic cycle.[10](#) Somehow his lab was beginning to manifest different material properties, almost as if it were a specially charged environment.

The effects also continually increased. No matter which experiment he carried out, the longer the imprinted devices were in the room, the larger the rhythmic fluctuations of the temperature and pH.[11](#)

These fluctuations remained unaffected by the opening of doors or windows, the operation of air conditioners or heaters, and even the presence or movement of humans or objects around their immediate vicinity. When he compared graphs of air and water temperature readings, they again mapped in perfect harmony.

Every corner of the room that was measured registered the same result. Each aspect of the physical space appeared to be in some sort of rhythmic, energetic harmony.

By this time, Tiller and his colleagues had set up four labs, each separated from the others by between 35 and 280 metres. Once enough experiments had been carried out, every other site also began to evidence these rhythmic fluctuations.

Tiller had never observed these types of 'organized' oscillations in his conventional science labs at Stanford. Indeed, they had never been observed anywhere else before. Just to be sure that this phenomenon was not being caused by the boxes themselves, he and his colleagues carried out three control experiments, in which

devices that had not been imprinted with intention were placed in the spaces and turned on. In those cases, all the readings of air and water behaved normally.

Tiller still puzzled over the meaning of the effects, and whether they might be due to some physical disturbance. He wondered whether having two large fans in the room would affect the oscillations in the air and water. Ordinarily, forced air convection from a fan should cause oscillations in temperature to disappear. He placed a desk fan and a floor fan in strategic places near a line of temperature probes. Even when the fans were turned up high enough to scatter pieces of paper, the original temperature oscillations remained.

What exactly was going on? This could be a magnetic effect, Tiller thought. Perhaps he should check out the magnetic field of the water. He placed an ordinary bar magnet under a jar of water for three days, with the north pole of the magnet pointing upwards, and measured the water's pH.

Then he turned the magnet over so that the south pole faced upwards under the jar for the same period. In normal circumstances, when ordinary water is exposed to this kind of weak magnet, which has a field strength of less than 500 gauss, the pH will be the same, no matter which side of the magnet is exposed to the water.

The world as we know it is magnetically symmetrical. Quantum physicists speak in terms of gauge theory and symmetry to explain the relationships between forces and particles, which include electric and magnetic charge. We are believed to exist in a state of electromagnetic $U(1)$ -gauge symmetry – a rather complicated scenario in which magnetic force is proportional to the gradient of the square of the magnetic field. This boils down to a simple truism: no matter where in a given field you measure the electromagnetic property, you get the same reading. The electromagnetic laws of nature are the same wherever you look.

If you raise the electromagnetic pull in one area, you will find you have changed the electromagnetic pull by the same degree everywhere else. In *The Cosmic Code*,^{[12](#)} Heinz Pagels likens the universe to an infinite piece of paper, painted grey. If you change the colour to a different shade of grey or 'change the gauge', you still don't change the gauge symmetry, because all the rest of the paper will be changed to the exact same shade of grey, so that it is even impossible to distinguish where exactly you are on the paper. A state of symmetrical magnetism is referred to as a magnetic 'dipole'.

But the pH of the water in Tiller's lab was significantly different with one polarity as compared with another, with huge differences of 1–1.5 pH units. Exposing the water to the south pole would send the pH soaring upward, while turning the magnet over to the north pole would cause the pH to decrease.

At two of his experimental sites, the pH of the water, when exposed to the south-pole polarity, continued to change with the passage of time, peaking after about six days. When the water was exposed to the north pole of the magnet, however, the rhythmic changes in pH that he had been recording dwindled away.[13](#)

Orthodox science maintains that monopoles only exist in electricity (as a positive or negative charge), but not in magnetism, which creates only dipoles from spinning or orbiting electrical charge.[14](#) Governments around the world have spent billions of dollars looking for magnetic monopoles everywhere on earth, without success.[15](#) Somehow, Tiller had managed to access a magnetic monopole in his crude lab. This phenomenon appeared to be a system-wide effect. In any lab of his exposed to the intention-imprinted black boxes, instruments recorded magnetic monopole type of behaviour.

It dawned on Tiller that he was witnessing the most astonishing result of all: human intention captured in his little black boxes were somehow 'conditioning' the spaces where the experiments were carried out.

Tiller wondered whether this phenomenon would still be present if he altered anything about the space. When he removed one element, such as a computer, the oscillations disappeared for about 10 hours before returning. The arrival of any new materials in his lab also caused the effects to disappear for several weeks, although, once again, they eventually returned.

It was as though the space had become an exquisitely tuned configuration, and no disturbance or change would destroy this higher state. Even when Tiller shielded the intention-imprinted devices in aluminium foil and Faraday cages, all the vibrations in water and air temperature continued. One of the sites, a converted barn, recorded oscillating air temperatures on and off for six months; in another site, an office lab, for a full year.[16](#)

After the imprinted boxes had been turned on for a while, the effect became relatively 'permanent'; the target, whether water pH, ALP or fruit flies, would continue to be affected even if the device was not in the lab. Tiller decided to see what would happen when he removed all the elements of the experiment.

He dismantled the Faraday cage and the water vessels and removed them from his lab, then recorded the air temperature of the place where the cages had been.

Even though the experimental vessel was no longer there, his thermometers continued to record periodic oscillations in temperature of 2–3°C. Although this influence decayed very slowly over time, Tiller's laboratories appeared to have

undergone some long-term thermodynamic transformation. The energy from intention appeared to 'charge' the environment and create a domino effect of order.[17](#)

The only other phenomena Tiller could think of that had similar effects on the environment were those of highly complex chemical reactions. But all he was working with was ordinary air and purified water.

According to the laws of conventional thermodynamics, air and water are supposed to exist in a state extraordinarily close to equilibrium, which is to say that they remain more or less static. These types of results had never been recorded in any lab in the world.

He suspected that he had been witnessing a quantum effect. The constant replaying of ordered thoughts seemed to be changing the physical reality of the room, and making the quantum virtual particles of empty space more 'ordered'. And then, like a domino effect, the 'order' of the space appeared to assist the outcome of the experiment. Carrying out the intentions in one particular space appeared to enhance their effects over time.

Somehow, in these charged spaces he and his colleagues had managed to create a $SU\{2\}$ -gauge space, where electric and magnetic monopoles coexisted – similar to the reality supposedly present in the supersymmetry states of exotic physics. In these conditioned spaces, the very law about the proportion of magnetic force had altered.

A basic property of physics had completely changed. The only way to get such a polarity effect was to produce some element of $SU\{2\}$ -gauge symmetry.[18](#)

This change in the gauge symmetry of the space meant that profound changes had occurred in the ambient Zero Point Field. In a $U\{1\}$ -gauge symmetry, the random fluctuations of the Field have no effect on the physical universe. However, in $SU\{2\}$ -gauge symmetry states, the Field has become more ordered and produces changes in the tiniest elements of matter – which add up to a profound alteration in the very fabric of physical reality.

Tiller felt as though he had somehow entered into a twilight zone of higher energy and that he was witness to a system with an extraordinary ability to self-organize. Indeed, the oscillations he had measured had all the hallmarks of a Bose–Einstein condensate – a higher state of coherence. Up until then, scientists had created a Bose–Einstein condensate only in highly controlled environments and at temperatures approaching absolute zero. But he had managed to create the same effects at room temperature, and from a thought process captured in a rudimentary piece of equipment.

Other scientists have witnessed a similar 'charging' of intention space. In one series of meticulous studies, for instance, researcher Graham Watkins and his wife Anita recruited human participants, many known for their psychic ability, and asked them to attempt to mentally influence anaesthetized mice to revive more quickly than usual. The experimental mice were drawn from a batch that had demonstrated similar waking times when placed under anaesthesia; the chosen group were divided in two, with half acting as controls.

In the first batch of studies, the experimental group woke up about 4 seconds earlier than the controls, a result considered only slightly significant. However, in subsequent studies the wake-up times of the experimental mice improved, and continued to do so with every study.

The Watkinses repeated their experiment seven times. They discovered that the healing had a 'linger effect'; if a mouse were simply placed on the spot on a table where another mouse had received a psychic's intentions, the second one would also revive more quickly than usual. The space appeared to have developed a healing 'charge', affecting anything that happened to occupy that space.[19](#)

Biologist Bernard Grad at McGill University in Montreal, Canada, observed similar phenomenon during experiments with Hungarian healer Oscar Estabany: once the Hungarian healer touched something – even simple fabric – it appeared to hold a phantom charge. The material could be used successfully for healing in place of Estabany's healing hands.[20](#)

The idea of 'conditioned space' was also explored by former PEAR scientist Dr Roger Nelson at sacred sites. Nelson was intrigued by these sacred spaces and whether their special purpose, or even some inherent quality about the site, had 'charged' the space with an energetic resonance that might register on a REG machine. He had run a number of experiments suggesting that a 'field consciousness' in a highly charged atmosphere, such as an intense gathering, affected the machines and made them more 'ordered'.

He carried around a portable REG, to register any changes in the randomness in the ambient field at various sites: Wounded Knee, the site of the massacre of an entire Sioux tribe; Devil's Tower in Wyoming; the Queen's Chamber in the Great Pyramid of Giza.

Nelson registered highly significant evidence of increased order on REGs at some sites, as if the location itself contained a lingering vortex of coherent energy, from all the people who had prayed or died there.[21](#)

Dean Radin used REGs to investigate whether healing can condition the place where it is carried out. He placed three REGs near a culture of human brain cells then asked a group of healers to send intentions for the culture to grow more quickly, and to engage in traditional space-conditioning meditations.

Any deviation from the random activity of the REGs would indicate the probable presence of greater coherence. Radin also prepared a control batch of cells, which were not to be sent intention.

After three days, there was no overall difference in the growth between the treated cells and the controls. Nevertheless, as the experiment progressed, the treated cells began to grow faster.

On the third day, all three of the REGs began moving away from random activity and became more ordered. The intention of the healers also appeared to have effects on background ionizing radiation.[22](#)

Like Nelson's readings at sacred sites, Radin's experiment offers tantalizing hints about the nature of the 'linger' effect of intention. The REGs' registering of movement away from randomness to greater order implies that the Zero Point energy of empty space has shifted into a state of greater coherence.

The 'charge' of intention may have a domino effect on its environment, causing greater quantum order in empty space, which would enhance the effectiveness of its aim.[23](#) Russian scientists have observed a similar phenomenon in water, which retains a memory of applied electromagnetic fields for hours, even days.[24](#)

The effect is like that of a laser; when waves of the ambient Field become more ordered, an intention may ripple through it like one powerful, highly targeted bolt of light.

With magnetic monopoles, Tiller was out on a ledge shared by few of his colleagues, even in consciousness research. His studies needed to be replicated by other, independent laboratories. But if his body of work does stand up over time, it will demonstrate the extent to which the energy of human thought can alter its environment.

The ordering process of intention appears to carry on, perpetuating, possibly even intensifying its charge.

The strange, almost unbelievable events occurring during Tiller's experiments made me wonder whether setting aside a particular room for carrying out intention might be an important consideration. Perhaps we each need our own 'temple' to which we return, if only in our mind's eye, every time we send a directed thought.

Chapter 8: The Right Place

1. William Tiller's major books on crystallization include: *An Introduction to Computer Simulation in Applied Science*, New York: Plenum, 1992; *The Science of Crystallization: Microscopic Interfacial Phenomena*, Cambridge: Cambridge University Press, 1991; *The Science of Crystallization: Macroscopic Phenomena and Defect Generation*, Cambridge: Cambridge University Press, 1992.
2. All personal details about William Tiller have resulted from multiple interviews, April 2005–January 2006.
3. O. Warburg, *New Methods of Cell Physiology Applied to Cancer and Mechanism of X-ray Action*, New York: John Wiley and Sons, 1962, as quoted in W. Tiller et al., *Conscious Acts of Creation: The Emergence of a New Physics*, Walnut Creek, Calif.: Pavior Publishing, 2001: 144–6. All description of experiment derived from interview with Dr Tiller, Boulder, Colorado, April 29, 2005, plus information from *Conscious Acts* and W. Tiller et al., *Some Science Adventures with Real Magic*, Walnut Creek, Calif.: Pavior Publishing, 2005.
4. M. J. Kohane, 'Energy, development and fitness in *Drosophila melanogaster*', *Proceedings of the Royal Society (B)*, 1994; 257: 185–91, in Tiller et al., *Conscious Acts*, op. cit.: 147.
5. William A. Tiller and Walter E. Dibble, Jr., 'New experimental data revealing an unexpected dimension to materials science and engineering', *Material Research Innovation*, 2001; 5: 21–34.
6. Tiller and Dibble, 'New experimental data', op. cit.
7. Ibid.
8. Ibid.
9. Tiller et al., *Conscious Acts*, op. cit.: 180.
10. Tiller et al., *Conscious Acts*, op. cit.: 175.
11. Tiller et al., *Conscious Acts*, op. cit.: 216.
12. H. Pagels, *The Cosmic Code*, New York: Simon and Schuster, 1982.
13. Tiller et al., *Conscious Acts*, op. cit.: 216.
14. Tiller et al., *Science Adventures*, op. cit.: 34.
15. Interview with W. Tiller, April 2005.
16. Tiller et al., *Conscious Acts*, op. cit.: 182.
17. Correspondence between Tiller and Michael Kohane, 2005.
18. Tiller and Dibble, 'New experimental data', op. cit.
19. G. K. Watkins and A. M. Watkins, 'Possible PK influence on the resuscitation of anesthetized mice', *Journal of Parapsychology*, 1971; 35: 257–72; G. K. Watkins et al., 'Further studies on the resuscitation of anesthetized mice', in W. G. Roll, R. L. Morris and J. Morris (eds.) *Research in Parapsychology*, Metuchen, NJ: Scarecrow Press, 1973: 157–9.

20. R. Wells and J. Klein, 'A replication of a "psychic healing" paradigm', *Journal of Parapsychology*, 1972; 36: 144–9.
21. See McTaggart, *The Field*, op. cit.: 205–7.
22. D. Radin, 'Beyond belief: Exploring interaction among body and environment', *Subtle Energies and Energy Medicine*, 1992; 2 (3): 1–40; D. Radin, 'Environmental modulation and statistical equilibrium in mind-matter interaction', *Subtle Energies and Energy Medicine*, 1993; 4 (1): 1–30.
23. D. Radin et al., 'Effects of healing intention on cultured cells and truly random events', *The Journal of Alternative and Complementary Medicine*, 2004; 10: 103–12. Notes 301
24. L. P. Semikhina and V. P. Kiselev, 'Effect of weak magnetic fields on the properties of water and ice', *Zabedenii, Fizika*, 1988; 5: 13–17; S. Sasaki et al., 'Changes of water conductivity induced by non-inductive coil', *Society for Mind-Body Science*, 1992; 1: 23; Tiller et al., *Conscious Acts*, op. cit.: 62.

PART THREE

The Power of Your Thoughts

Baseball is 90 per cent mental. The other half is physical.
-Yogi Berra

CHAPTER NINE

Mental Blueprints

SEVEN WEEKS BEFORE MUHAMMAD ALI met World Heavywei Champion George Foreman for their 'rumble in the jungle' at Kinshasa in 1974, he practised his punches as if he couldn't care less, taking a few desultory swipes at his sparring partner as if distractedly popping a speed bag. Mostly he would lie against the ropes and allow his opponent to pound away at him from every angle.

In the latter years of his fighting career, Ali spent much of his training time learning how to take punches. He studied how to shift his head by just a hair a microsecond before the connection was made, or where in his body he could mentally deflect the punch, so that it would no longer hurt. He was not training his body to win. He was training his mind not to lose, at the point when deep fatigue sets in around the twelfth round and most boxers cave in.¹ The most important work was being done, not in the ring, but in his armchair. He was fighting the fight in his head.

Ali was a master of intention. He developed a set of mental skills that eventually altered his performance in the ring. Before a fight, Ali used every self- motivational technique out there: affirmation; visualization; mental rehearsal; self- confirmation;

and perhaps the most powerful epigram of personal worth ever uttered: 'I am the greatest'. Ali also made public statements of his intentions. His constant barrage of rhyming couplets and quatrains, seemingly so innocuous, were highly specific intentions in disguise:

Archie Moore

Is sure
To hug the floor
By the end of four
Now Clay swings with a right
What a beautiful swing
And the punch knocks the Bear Clear out of the ring.

Before a fight, Ali repeated these little rhymes like a mantra – to the press, to his opponent, and even in the ring – until he himself accepted them as fact.

When they met in Kinshasa, Foreman was seven years younger than Ali and among the most savage fighters in the ring. Just two months earlier, he had left Ken Norton for dead with five blows to the head after only two rounds.

Nevertheless, in the weeks before the fight, when reporters pressed Ali about the two-to-one odds against him, Ali had rewritten the history of the Norton–Foreman fight, which he repeated, virtually verbatim, to every journalist who interviewed him. 'He's got a hard-push punch but he can't hit,' he would say, punching the air in front of the reporter's nose. 'Foreman just pushes people down. He just got slow punches, take a year to get there. You think that's going to bother me? This is going to

be the greatest upset in the history of boxing.'

Ali's intention came to pass in the jungle. He also made masterful use of intention to beat Joe Frazier in the Philippines later that year, in perhaps the most brutal and stunning display of boxing of all time.

This time, he created a voodoo doll. Ali turned his ferocious opponent into a tiny rubber gorilla, which he carried around with him in his top pocket, taking a swipe at it with his right from time to time for the television cameras: 'It's gonna be a thrilla and a chilla and a killa when I get the gorilla in Manila.' By the time Frazier entered the ring, he had been reduced in his own mind to something less than human.

Besides these verbalized intentions, Ali carried out mental intentions by rehearsing every moment of the fight in his head: the fatigue in his legs, the sweat pouring off his body, the pain to his kidneys and bruises on his face, the flash of the photographers, the exultant screams of the crowd, even the moment when the referee lifts his arm in victory against Frazier. He sent an intention to his body to win and his body responded by following orders.

To take intention out of the laboratory, I began to sift through the data from people or groups who were using intention successfully in real life. I wanted to study their techniques, the particular thought processes they underwent when sending intention, and would try to extrapolate from their experiences some tools that all the rest of us could use when sending intention. I was also curious about the extent of their mental reach – just how far people had been able to push their intentions.

The most instructive examples came from sports, not only from the greatest athlete of all time, but also from other elite sportsmen and women. Athletes of all varieties now routinely practise what is variously termed ‘mental rehearsals’, ‘implicit practice’[3](#) or even ‘covert rehearsal’. Focused intention is now deemed essential to alter and improve performance. Swimmers, skaters, weightlifters and football players employ intention to enhance their level of performance and consistency. It is even being used in leisure sports, such as golf and rock climbing.

Any modern coach of a competitive sport routinely offers training in some form of mental rehearsal, and often it is touted as the decisive element separating the elite sportsperson from the second-division player.

National-level soccer players, for instance, are more likely to use imagery than those who remain at the provincial or local levels.[4](#)

Virtually all Canadian Olympic athletes use mental imagery.

Psychologist Allan Paivio, professor emeritus of the University of Western Ontario, first proposed that the brain uses ‘dual coding’ to process verbal and non-verbal information simultaneously.[5](#)

Mental practice has been shown to work just as well as physical practice for patterns and timing.[6](#)

Paivio’s model has been largely adapted to help athletes with motivation or in learning or improving a certain skill set.[7](#)

The techniques involved in mental rehearsal have been exhaustively studied and written about in scientific literature and popular publications,[8](#) and their credibility was given a further boost in 1990, when the National Academy of Sciences examined all the scientific studies to date on these methods and declared them effective.[9](#)

Athletic mental rehearsal has been incorrectly considered synonymous with

visualization. ‘Visualization’ implies that you observe yourself in the situation, as if watching a mental video featuring yourself or seeing yourself through another pair of eyes. Although this may be useful in other areas of life, visualizing oneself from an

external perspective in a sports event can hamper athletic performance. Mental rehearsal also differs from positive thinking; happy thoughts on their own do not work in competitive sports.[10](#)

The most successful internal rehearsal involves imagining the sports event from the athlete's perspective *as though he or she is actually competing*. It amounts to a mental trial run – Ali imagining his right fist at the moment of impact on Frazier's left eye.

The athlete envisages the future in minute detail *as it is unfolding*. Champion athletes forecast and rehearse every aspect of the situation, and the steps they should take to overcome any possible setbacks.

Tracy Caulkin used intention to land a third gold medal in the 1984 Olympics. Caulkin had already broken 5 world records and 63 American records, and at the age of 23 was considered the best American swimmer who had ever lived. All she needed to complete her trophy wall was a few Olympic golds.

At the time, electronic touchpads had just replaced stopwatches. Whereas the watch could only distinguish differences of hundredths of a second, the new electronic technology could distinguish the winning lead within a thousandth of a second – 400 times faster than the blink of an eye. In the Olympics relay swimmers are given two-hundredths of a second of grace to leave their block before their previous team mate hits the touchpad. This kind of fine timing is critical; even a single coat of paint on one side of the pool can make a swimmer's lane one-thousandth of a second longer to swim and give another swimmer the leading edge.

During the four-woman 400-metre relay race, Tracy took the lead by diving in one-hundredth of a second before her returning team mate hit the touchpad.

Although all her competitors had a similar level of fitness, Caulkin had one enormous advantage. She already knew every moment of her swim, from the dive and the cool rush of water past her head to the very moment when she would lunge out in front.

Tracy had practiced that hair's width lead, the precise moment when she would leave the block a hundredth of a second earlier than her opponents, every night inside her head. The conclusion of the Olympic relay had entirely depended upon the specificity of her intention.

The most successful athletes break down their performances into tiny component parts and work on improving specific aspects. For general mastery of their sport, they imagine a flawless performance.[11](#) They concentrate on the most difficult moments

and work out good coping strategies – how to stay in control in the face of adversity, such as a pulled muscle or an umpire's adverse call.

Different intention is employed, depending on whether they are first learning a skill or simply wishing to reinforce and improve their technique. Like Muhammad Ali, elite athletes learn how to block out images representing doubt. If an image of difficulty pops into their heads, they become extremely adept at changing the internal movie, quickly editing the scene to imagine success.[12](#)

Winning depends on how specifically you can mentally rehearse. Seasoned athletes use vivid, highly detailed internal images and run-throughs of the entire performance.[13](#)

The most important aspect of the intention is to rehearse the victory, which appears to help secure it. Successful competitors rehearse their own feelings, particularly their elation and emotional response to winning: the reactions of their parents, the medals, the post-match celebration and the residual rewards like sponsorships.[14](#)

They imagine that the crowd is cheering for their performance alone.

Experienced athletes engage all their senses in their mental rehearsal. They not only have a visual, internal image of the future event, they also hear it, feel it, smell it and taste it – the ambience, the competitors, the sweat of their bodies, the applause.

Of all the sensations, the most vital for athletes appears to be mentally rehearsing the 'feel', or kinaesthetic sensations in their bodies.[15](#)

The more experienced the athletes, the better they are at imagining the feel of their bodies when engaged in their sport.[16](#)

Champion rowers are most successful when they can forecast the 'feel' of every part of the race, from the drag on the oar to the strain on their muscles.[17](#)

Some athletes find that it helps to study the actual setting where the sporting event is to take place first and then to imagine themselves there. Those who can combine the knowledge of the sports venue with mental rehearsal tend to be more successful than those who simply use mental rehearsal on its own.[18](#)

Rocky Bleier, former running back for the Pittsburgh Steelers, used intention to help the Steelers win the Super Bowl. His technique was to saturate his mind with the details of specific plays. He carried out mental rehearsals in the morning, before the team meal and last thing before drifting off to sleep every day of the two weeks before a game.

He also found it reassuring to run through the entire catalogue of moves one final time just before play. While sitting on the bench, he again rehearsed some 30 runs and 30 passes. No matter what the field threw up to him that day, he was determined to be ready.[19](#)

Techniques differ among the various sports. Those mental rehearsals that work best for sports requiring aerobic ability and fast, coordinated movement tended to fail with strength training. Weight lifters are most successful after carrying out a mental intention that galvanizes them to lift an impossibly heavy object.[20](#)

Conventional wisdom has it that the best state for performance is a state of relaxation, but as I found with masters of intention, a relaxed state is not necessarily optimum. In a study of karate, using relaxation techniques before carrying out the intentions did not improve performance.[21](#)

It was only useful if the participant was nervous and needed to be calmed down in order to perform better.[22](#)

Relaxation and hypnosis used with intention have worked to improve aim – say, for basketball shots or accuracy in chipping in golf.[23](#)

But as with Davidson's Buddhists, the most successful athletes manage to work themselves into peak intensity – a state of calm hyperawareness.

But how can simply thinking about a future performance actually affect the day of the event? Some clues come from intriguing brain research with electromyography (EMG). EMG offers a real-time snapshot of the brain's instructions to the body when and where it tells it to move – by recording every electrical impulse sent from motor neurons to specific muscles to cause a contraction. Ordinarily, EMG offers doctors a useful tool to diagnose neuromuscular disease and to test whether muscles respond appropriately to stimulation.

But EMG has also been employed to solve an interesting scientific conundrum whether the brain differentiates between a thought and an action. Does the thought of an action create the same pattern in neurotransmission as the action itself? This very question was tested by wiring a group of skiers to EMG equipment while they were carrying out mental rehearsals.

As the skiers mentally rehearsed the downhill runs, the electrical impulses heading to their muscles were just the same as those they used to make turns and jumps actually skiing the run.[24](#) The brain sent the same instructions to the body, whether the skiers were simply *thinking* of a particular movement or actually carrying it out. *Thought produced the same mental instructions as action.*

Research with EEGs has shown that the electrical activity produced by the brain is identical, whether we are *thinking* about doing something or actually doing it. In weightlifters, for instance, EEG patterns in the brain that would be activated to produce the actual motor skills are activated while the skill is simply being simulated mentally.[25](#) *Just the thought is enough to produce the neural instructions to carry out the physical act.*

Based on this research, scientists have posited some interesting theories of how mental rehearsal works. One school of thought proposes that mental rehearsal creates the neural patterns necessary for the real thing. As though the brain were simply another muscle, these rehearsals train the brain to facilitate the moves more easily during the actual performance.[26](#)

When an athlete performs, the nerves that signal to the muscles along a particular pathway are stimulated and the chemicals that have been produced remain there for a short period. Any future stimulation along the same pathways is made easier by the residual effects of the earlier connections. We get better at physical tasks because our signalling from intention to action has already been forged. It is not unlike a train track laid down through wild, inhospitable country. Future performances improve because your brain already knows the route and follows the track already laid down. Because the brain does not distinguish between doing something specific and just *thinking* about doing it, mental rehearsal lays down the tracks just as well as physical practice does. The nerves and muscles create a pathway just as sound as one produced through repeated practice.

Nevertheless, there are a few important differences between mental and physical practice. With physical practice, when you practise too much, you become fatigued, which causes electrical interference and blockage along the tracks. With mental intention, no road blocks ever appear, no matter how much you practise in your head.

The other difference concerns the size of the effect; the neuromuscular pattern laid down with mental practice may be slightly smaller than that of physical practice. Although both types of practice create the same muscle patterns, the imagined performances have smaller magnitude.[27](#)

To derive any benefit, mental rehearsal must replicate the real thing – at normal speed. Although it might seem logical that a rehearsal would work best in slow motion, with particular attention to specific moves, that is not borne out by research.

When skiers monitored by EMGs imagined their performance in slow motion, they produced a different muscle response pattern from that produced when carrying out the skill at an ordinary pace. In fact, the brain–muscle activity of rehearsing at slow

motion is identical to the brain–muscle pattern when the skiing itself is carried out in slow motion.

This accords with what scientists understand about the neural patterns involved in slow motion, compared with those of normal speed. The same task carried out in slow motion produces completely different neuromuscular patterning than when it is done at normal speed.[28](#)

There is no such thing as cross-training in mental rehearsal; intention facilitates just the type of athletic event that is being mentally rehearsed and is not transferable to other sports, even those involving similar muscle groups.

This was apparent in a fascinating study involving sprinters. The researchers had divided a group of runners into four groups and asked them to do one of four types of preparation: to imagine themselves in a 40-metre sprint; to engage in power training on a stationary bicycle; to combine imagery and power training; or, as the controls, to do no training in any form.

After six weeks of training the athletes were asked to perform two tests – to cycle their hardest while their effort was recorded on a cycle ergometer, which tests for cycling power, and to run a 40-metre sprint. Both activities require much the same motor ability and leg muscles.

In the cycling test, those groups who had used power training alone showed improvement. However, when it came to the sprint, only the groups who had mentally practised sprinting had significantly improved. Specific imagery enhanced *only the specific task that had been imagined*. It did not simply build muscles generally. The motor neuron training was highly specific, and only affected the actual performance visualized in the mind.[29](#)

Beside improving performance, mental intention can produce actual physiological changes, and not only in athletes' bodies. Guang Yue, an exercise psychologist at Cleveland Clinic Foundation in Ohio, carried out research comparing participants who went to the gym with those who carried out a virtual workout in their heads.

Those who regularly visited the gym were able to increase their muscle strength by 30 per cent. But even those who remained in their armchairs and ran through a mental rehearsal of the weight training in their minds increased muscle power by almost half as much.

Volunteers between 20 and 35 years old imagined flexing one of their biceps as hard as they could during daily training sessions carried out five times a week. After ensuring that the participants were not doing any actual exercise, including tensing

their muscles, the researchers discovered an astonishing 13.5 per cent increase in muscle size and strength after just a few weeks, an advantage that remained for three months after the mental training stopped.[30](#)

In 1997, Dr David Smith at Chester College came up with similar results participants who worked out could achieve 30 per cent increases in strength, while those who just imagined themselves doing the training achieved a 16 per cent increase.[31](#)

Pure directed thought can give you the burn almost as well as any workout.

Thinking of changing an aspect of the body in other ways can also work – which might prove comforting to anyone who is not happy with his or her body shape.

One study demonstrated that, under hypnosis, women increased the dimensions of their breasts simply by visualizing themselves on the beach with the sun's rays warming their chests.[32](#)

The kinds of vivid visualization techniques used by athletes are also highly effective in treating illness. Patients have boosted treatment of an array of acute and chronic conditions, from coronary artery disease[33](#) and high blood pressure to low- back pain and musculoskeletal diseases,[34](#) including fibromyalgia,[35](#) by using mental pictures or metaphoric representations of their bodies fighting the illness. Visualization has also improved postsurgical outcomes,[36](#) helped with pain management[37](#) and minimized the side-effects of chemotherapy.[38](#)

Indeed, the outcome of a patient's illness has been predicted by examining the types of visualizations used to combat them. Psychologist Jeanne Achterberg, who healed herself of a rare cancer of the eye through imagery, went on to study a group of cancer patients who were using visualization to fight their own disease. She predicted with 93 per cent accuracy which patients would completely recover and which would get worse or die, simply by examining their visualizations and rating them. Those who were successful had a greater ability to visualize vividly, with powerful imagery and symbols, and could hold a clear visual intention imagining themselves overpowering the cancer and the medical treatment being effective. The successful patients also practiced their visualizations regularly.[39](#)

If the brain cannot distinguish between a thought and an action, would the body follow mental instructions of any sort? If I send my body a mental intention to calm down or speed up, will it necessarily listen to me?

Literature about biofeedback and mind–body medicine indicates that it will. In 1961, Neal Miller, a behavioural neuroscientist at Yale University, first proposed that people can be taught to mentally influence their autonomic nervous system and control

mechanisms such as blood pressure and bowel movements, much as a child learns to ride a bicycle.

He conducted a series of remarkable conditioning-and-reward experiments on rats. Miller discovered that, if he stimulated the pleasure centre in the brain, his rats could be trained to decrease their heart rate at will, control the rate at which urine filled their kidneys, even create different dilations in the blood vessels of each ear.[40](#) If relatively simple animals like rats could achieve this remarkable level of internal control, Miller figured, couldn't human beings, with their greater intelligence, regulate more bodily processes?

After these early revelations, many scientists found that information about the autonomic nervous system could be fed back to a person as 'biofeedback' to pinpoint where a person should send intention to his body. In the 1960s, John Basmajian, a professor of medicine at McMaster University in Ontario and a specialist in rehabilitative science, began training people with spinal-cord injuries to use EMG feedback to regain control over single cells in their spinal cords.[41](#)

At roughly the same time, psychologist Elmer Green at the Menninger Institute pioneered a method of biofeedback to treat migraine, after discovering that a migraine patient of his could make her headaches go away whenever she practiced a structured form of relaxation. Green went on to use biofeedback to help patients cure their own migraines, and it is now an accepted form of therapy.[42](#)

Biofeedback is particularly useful to treat Raynaud's disease, a vascular condition in which blood vessels are constricted when exposed to cold, causing extremities to grow cold, pale, and even blue.[43](#)

During a biofeedback treatment, a patient is hooked up to a computer. Transducers applied to different parts of his or her body send information to a visual display, which registers activities of the autonomic nervous system, such as brain waves, blood pressure and heart rate, or muscle contractions.

The audio or visual information fed back to the patient depends on the condition; in the case of Raynaud's, as soon as the arteries to the hands constrict, the machines record a drop in skin temperature, a light bulb flashes or a beeper sounds. The feedback prompts the patient to send an intention to his body to adjust the process in question – in the case of Raynaud's, the patient sends an intention to warm up his hands.

Since those early days, biofeedback has become well established as a therapy for virtually every chronic condition, from attention deficit hyperactivity disorder (ADHD) to menopausal hot flushes. Stroke patients and victims of spinal-cord injuries now

use biofeedback to rehabilitate or regain the use of paralysed muscles. It has proved invaluable in eliminating the pain felt in a phantom limb.[44](#) Astronauts have even used biofeedback to cure motion sickness while journeying to outer space.[45](#)

The more conventional view of biofeedback maintains that it has something to do with relaxation – learning to calm down the fight-or-flight responses of our autonomic nervous systems. However, the sheer breadth of control would argue that the mechanism has more to do with the power of intention. Virtually every bodily process measurable on a machine – even a single nerve cell controlling a muscle fibre – appears to be within an individual's control. Volunteers in studies have achieved total mental mastery over the temperature in their bodies,[46](#) or even the direction of blood flow to the brain.[47](#)

Like biofeedback, Autogenic Training, the technique developed by a German psychiatrist named Johannes Schultz to relax the body and slow the breathing and heart rate, also demonstrates that a wide variety of the body's functions are under our conscious control. Those who practise the technique are able to lower blood pressure, raise temperature in extremities, and slow heartbeat and breathing. Autogenic Training has also been used to treat many chronic conditions besides stress, such as asthma, gastritis and ulcers, high blood pressure and thyroid problems.[48](#) There is even evidence that Autogenic Training can work effectively in groups.[49](#)

For a cat, nirvana is the food bowl just around the corner.

Dr Jaak Panksepp professor emeritus of psychology at Bowling Green University, theorizes that this anticipatory joy has to do with the 'seeking' mode of the brain – one of the five primitive emotions that humans share with members of the animal kingdom.[50](#)

The seeking system helps animals investigate and work out the meaning of their environment. The seeking circuits are fully engaged when an animal is involved in high anticipation, intense interest or insatiable curiosity. As Panksepp was astonished to discover, the most emotionally arresting part for any animal is the hunt, not the catch.[51](#)

When animals are curious, the hypothalamus lights up and the 'feel-good' neurotransmitter dopamine is produced. Scientists used to believe that the chemical itself caused the pleasure, until it was discovered that the chemical's true purpose is to arouse a certain neural pathway. What actually feels good is the activation of the seeking portion of the brain.

Forty years ago, Barry Sterman, professor emeritus of the departments of Neurobiology and Biobehavioral Psychiatry at UCLA, accidentally discovered that this anticipatory emotion sent cats into a meditative state; their brains slowed to an EEG rhythm of 8–13 hertz, corresponding to human alpha brain frequencies, moments before they got their reward.[52](#)

Eventually, he was able to get the cats to re-create this state at will, not simply when they were awaiting food. It was tantamount to the animals being able to control their own brain waves.

But could a human being do the same?

To test this, Sterman needed to test someone whose brain waves were so out of the ordinary that any change would be apparent immediately. He located a woman troubled by periodic epileptic seizures, which are caused by the brain firing theta brain waves at inappropriate moments. Sterman constructed a biofeedback EEG machine that would flash a red light in the presence of a theta wave and a green light during an alpha state.

After a while, his patient was able to change her state at will and reduce the amount and intensity of her epileptic fits. Sterman spent the next 10 years of his life studying epileptics and training them to reduce their own fits.[53](#)

In the 1980s, two American psychologists, Eugene Peniston and Paul Kulkosky made use of Sterman's findings to reform alcoholics. With their brain-wave biofeedback, alcoholic patients concentrated on damping down high beta brain waves, which tend to be predominant during moments of craving and dependency, and increasing the alpha and theta wave frequencies, which help one to relax and establish greater brain-wave coherence.

Some 80 per cent of the alcoholics were able to control their cravings and stay off alcohol. The training also seemed to affect their blood chemistry, increasing their levels of beta-endorphin, another 'feel-good' brain chemical. Biofeedback, combined with work on their self-image, eventually eliminated much of their dysfunctional behaviour and transformed them into better people.[54](#)

Joe Kamiya, a psychologist at the University of Chicago, demonstrated the amazing specificity of brain-wave biofeedback through some remarkable brain research. He attached EEG electrodes to the rear sides of the scalps of several volunteers, over the portion of the brain where alpha brain waves are most prominent. At the sound of a tone, his participants had to guess whether their brain waves were predominantly alpha. After comparing their answers with the information recorded on the EEG machines, Kamiya let them know whether they were right or wrong. By the

second day, one of his participants was able to guess correctly two-thirds of the time, and two days after that, virtually all the time. A second participant discovered a means of putting himself into a particular brain-wave state on cue.[55](#)

EEG biofeedback has now developed into a sophisticated means of controlling the range and type of frequencies emitted by the brain. It works particularly well with trauma patients suffering from depression,[56](#) helps students concentrate, and enhances creativity and focus. It may well be that intention can be used to control the brain, brain wave by brain wave.

Hypnosis is also a type of intention – an instruction to the brain during an altered state. Hypnotists continually demonstrate that the brain or body is susceptible to the power of directed thought.

One dramatic example of the power of mental suggestion concerned a small group of people with a mysterious congenital illness called ichthyosiform erythroderma, known disparagingly as fish-skin disease because unsightly fish-like scales cover most of the body. In one study, five patients were hypnotized and told to focus on a part of their body and visualize the skin becoming normal. Within just a few weeks, 80 per cent of each patient's body had completely healed. The skin remained smooth and clear.[57](#)

Through hypnotic intention, spinal-surgery patients about to undergo their operations have reduced blood loss by nearly half, simply by directing their blood supply away from the site of the surgery.[58](#)

Pregnant women have been able to turn their babies from breech positions, burn victims have sped up their healing; and people suffering haemorrhages in the gastrointestinal tract have willed their bleeding to stop.[59](#)

Clearly, during an altered state, roughly corresponding to the hyperalert state of intense meditation, conscious thought can convince the body to endure pain, cure many serious diseases and change virtually any condition.

Surgeon Dr Angel Escudero of Valencia, Spain, has carried out more than 900 cases of complex surgery without anaesthesia. BBC cameras were invited into his operating room and captured on film a woman who was having such an operation without anaesthetic. All she had to do was keep her mouth full of saliva and keep repeating to herself, 'My leg is anaesthetized.' An affirmation like hers is another form of intention. A dry mouth is one of the mind's first warning signals of danger. When the mouth is kept lubricated, the brain relaxes, assumes all is well, and turns off its pain receptors, assured that anaesthetics had been given.[60](#)

A fascinating study by David Spiegel, a professor of psychiatry and behavioural sciences at Stanford University, offers a glimpse of what happens to the brain when an intention is given under hypnosis.

His participants were shown a coloured grid painting, similar to a Mondrian, and were asked to imagine the colour draining from the picture, leaving only black and white. Through the use of positron emission tomography (PET) scans, which record physical activity in the brain, Spiegel showed that blood flow and activity were noticeably diminishing in the part of the brain dealing with the perception of colour, while the areas that process black, white and grey images were being stimulated.

When the experiment was reversed, and the participants in the study were asked to imagine grey images turning into colour, the opposite changes in brain-perception patterns resulted.[61](#)

This illustrated another instance in which the brain was the maidservant of thoughts. The brain's visual cortex, the area responsible for processing images, could not distinguish between a real image and an imagined one. *The mental instructions were more important than the actual visual image.*

The placebo effect has shown that beliefs are powerful, even when the belief is false. The placebo is a form of intention – an instance of intention trickery. When a doctor gives a patient a placebo, or sugar pill, he or she is counting on the patient's belief that the drug will work.

It is well documented that belief in a placebo will create the same physiological effects as that of an active agent – so much so that it causes the pharmaceutical industry enormous difficulty when designing drug trials. So many patients receive the same relief and even the same side effects with a placebo as with the drug itself that a placebo is not a true control.

Our bodies do not distinguish between a chemical process and the *thought* of a chemical process. Indeed, a recent analysis of 46,000 heart patients, half of whom were taking a placebo, made the astonishing discovery that patients taking a placebo fared as well as those on the heart drug. The only factor determining survival seemed to be belief that the therapy will work and a willingness to follow it religiously.

Those who stuck to doctor's orders to take their drug three times a day fared equally well whether they were taking a drug or just a sugar pill. Patients who tended not to survive were those who had been lax with their regime, regardless of whether they had been given a placebo or an actual drug.[62](#)

The power of the placebo was best illustrated by a group of patients treated for Parkinson's disease, a motor system disorder in which the body's system for releasing the brain chemical dopamine is faulty. The standard treatment for Parkinson's is a synthetic form of dopamine.

In a study at the University of British Columbia, a team of doctors demonstrated with PET scanning that, when patients given placebos were told they had received dopamine, their brains substantially increased the release of their own stores of the chemical.[63](#)

In another dramatic instance, at Methodist Hospital in Houston, Dr Bruce Moseley, a specialist in orthopaedics, recruited 150 patients with severe osteoarthritis of the knee and divided them into three groups. Two-thirds were either given arthroscopic lavage (which washes out degenerative tissue and debris with the aid of a little viewing tube) or another form of debridement (which sucks it out with a tiny vacuum cleaner).

The third group were given a sham operation: the patients were surgically prepared, placed under anaesthesia and wheeled into the operating room. Incisions were made in their knees, but no procedure carried out.

Over the next two years, during which time none of the patients knew who had received the real operations and who had received the placebo treatment, all three groups reported moderate improvements in pain and function. In fact, the placebo group reported better results than some who had received the actual operation.[64](#) The mental expectation of healing was enough to marshal the body's healing mechanisms. The intention, brought about by the expectation of a successful operation, produced the physical change.

Extreme instances of intention and expectation can also manifest physically. The phenomenon of stigmata, in which religious fervour produces blood, bruising or wounds on people's hands, feet and sides that mirror the wounds of Christ during his crucifixion, are a form of intention.

The Association for the Scientific Study of Anomalous Phenomena has recorded at least 350 such instances of stigmata resulting from identification with Christ. Saybrook University psychologist Stanley Krippner and his colleagues witnessed this first hand with Brazilian sensitive Amyr Amiden.

As soon as their talk turned to Jesus Christ, red spots and drops of blood appeared on the backs of each of Amiden's hands and on his palms and forehead.[65](#) A similar situation occurred during the three weeks before Easter Sunday with a young African-

American Baptist girl, who had been profoundly moved by a television movie about the crucifixion and was preoccupied with Christ's suffering.

She manifested bleeding on the palm of her left hand two to six times a day.[66](#)
Krippner knew of three Anglicans who regularly evidenced stigmata.[67](#)

Cases of spontaneous cures are an instance of an extreme intention that reverses almost certain death. A person with what is considered a terminal illness defies the textbook description of his disease progression and the prognoses of his doctors and beats it virtually overnight, without the aid of the tools of modern medicine.

The Institute of Noetic Sciences has gathered together all scientifically recorded cases of so-called miracle cures.[68](#) Although the received wisdom is that these cases are rare, a scan of the medical literature is instructive. One in eight skin cancers spontaneously heal, as do nearly one in five of genitourinary cancers.

Virtually all types of illnesses, including diabetes, Addison's disease and atherosclerosis, where vital organs or body parts are supposedly irretrievably damaged, have spontaneously healed.[69](#) A small body of research concerns terminal cancer patients, who with little or no medical intervention, end up beating the odds.

Although these cases are labelled instances of 'spontaneous remission', as though the illness has suddenly decided to go into hiding but might suddenly spring out at any moment, in many instances they represent another example of the body's ability to self-correct through the power of intention. Case after case of spontaneous remission describes people up against a major road block in their lives: unremitting stress, unresolved trauma, prolonged hostility, marked isolation, profound dissatisfaction or quiet despair.[70](#) They often describe people who have lost their role as the central protagonist in their own life drama.[71](#)

Many cases of spontaneous remission seem to occur after someone makes a massive psychological shift, and recreates a life that is engaging and purposeful. In these instances, the patient gets rid of the source of the psychological heartache[72](#) and takes full responsibility for his illness and treatment.[73](#)

Some people, this would suggest, get ill because they lose all hope of life ever being good – because they are thinking the wrong thoughts. These cases of spontaneous remission suggested to me that casual thoughts that run through our minds every day together become our life's intention.

We can use intention to gain control over virtually any bodily process and perhaps even life-threatening illnesses. But can our thoughts about others be as potent as our thoughts about ourselves?

Psychologist William Braud is one of the few scientists who has examined this question. He gathered a group of volunteers and asked them to carry out biofeedback on themselves.

After pairing off the group, he attached one member of each pair to the biofeedback equipment, but asked the other partner to respond to the readings and carry out the sending of mental instructions. According to Braud's evidence, the results were equivalent to those that occurred when the patients on the equipment used biofeedback on themselves. Somebody else's good intentions for you may be as powerful as your own.[74](#)

Braud's other studies also suggested that we can most influence others to become more 'ordered' when we ourselves are ordered. For instance, in his studies, calm people were the most successful at sending mental influence to calm down highly nervous people, and focused people the best at helping distracted people focus.[75](#)

Braud's work also suggests that the greatest effects occur when the person most needs help.[76](#)

Scientific evidence also reveals that we can affect virtually any other living thing as well. The enormous body of research on healing gathered by Dr Daniel Benor shows that thoughts can have powerful effects on a variety of plants, seeds, single-celled organisms such as bacteria and yeast, and insects and other small animals.[77](#)

Most recently, a series of double-blind experiments carried out over two years by Dr Serena Roney-Dougal in Somerset showed that lettuce seeds that were sent intention yielded 10 per cent more crops with significantly less fungal disease than those grown conventionally.[78](#)

The evidence convinced me that we can improve our health, enhance our performance in every area of our lives, and possibly even affect the future by consciously using intention. The intention should be a highly specific aim or goal, which you should visualize in your mind's eye as having already occurred while you are in a state of concentrated focus and hyperawareness. When you imagine this future event, hold a mental picture of it as if it were occurring to you at that moment. Engage all five senses to visualize it in detail. The centrepiece of this mental picture should be the moment you achieve the goal.

A doctor might improve the survival rate of his patients by never giving a negative diagnosis.[79](#) A surgeon could improve his patients' recovery by mentally rehearsing the surgery before heading into the operating theatre. Indeed, we might no longer need drugs, but simply good intentions.

Since intention has been shown to affect the chemistry in our bodies, we should be able to speed up, slow down or improve any physiological processes. We might develop many more breakthrough medicines by mentally targeting their effectiveness and minimizing their side effects.

We could raise the quality of our daily endeavours just by carrying out a detailed mental rehearsal. At home, we might be able to send intentions to our children to perform better at school or be more loving to their friends. Human intention might be powerful enough to affect every element of our lives.

All of these possibilities suggest that we have an awesome level of responsibility when generating our thoughts. Each of us is a potential Frankenstein, with an extraordinary power to affect the living world around us. How many of us, after all, are sending out mostly positive thoughts?

Notes – Chapter 9: Mental Blueprints

1. All description of Ali's fighting techniques from N. Mailer, *The Fight*, London and New York: Penguin, 2000.
2. Ibid.
3. A. Richardson, 'Mental practice: A review and discussion, Part I', *Research Quarterly*, 1967; 38: 95–107; A. Richardson, 'Mental practice: A review and discussion. Part II', *Research Quarterly*, 1967; 38: 264–73
4. A. Paivio, *Mental Representations: A Dual Coding Approach*, New York and London: Oxford University Press, 1986.
5. B. S. Rushall and L. G. Lippman, 'The role of imagery in physical performance', *International Journal for Sport Psychology*, 1997; 29: 57–72.
6. A. Paivio, 'Cognitive and motivational functions of imagery in human performance', *Canadian Journal of Applied Sport Sciences*, 1985; 10 (4): 22S–28S.
7. K. E. Hinshaw, 'The effects of mental practice on motor skill performance: Critical evaluation and meta-analysis', *Imagination, Cognition and Personality*, 1991–2; 11: 3–35.
8. J. A. Swets and R. A. Bjork, 'Enhancing human performance: An evaluation of "New Age" techniques considered by the U. S. Army', *Psychological Science*, 1990; 1: 85–96; D. L. Feltz et al., 'A revised meta-analysis of the mental practice literature on motor skill learning', in D. Druckman and J. A. Swets (eds.), *Enhancing Human Performance: Issues, Theories, and Techniques*, Washington, DC: National Academy Press, 1988: 274.
9. R. J. Rotella et al., 'Cognitions and coping strategies of elite skiers: a exploratory study of young developing athletes', *Journal of Sport Psychology*, 1980; 2: 350–4.

10. R. S. Burhans et al., 'Mental imagery training: effects on running speed performance', *International Journal of Sport Psychology*, 1988; 19: 26–37.
11. B. S. Rushall, 'Covert modeling as a procedure for altering an elite athlete's psychological state', *Sport Psychologist*, 1988; 2: 131–40; B. S. Rushall 'The restoration of performance capacity by cognitive restructuring and covert positive reinforcement in an elite athlete', in J. R. Cautela and A. J. Kearney (eds.), *Covert Conditioning Casebook*. Boston, Mass.: Thomson Brooks/Cole, 1993.
12. M. Denis, 'Visual imagery and the use of mental practice in the development of motor skills', *Canadian Journal of Applied Sport Sciences*, 1985; 10: 4S– 16S.
13. Paivio, 'Cognitive and motivational functions of imagery', op. cit.
14. J. R. Cautela and A. J. Kearney (eds.), *Covert Conditioning Casebook*. Boston, Mass.: Thomson Brooks/Cole, 1993: 30–1.
15. B. Mumford and C. Hall, 'The effects of internal and external imagery o performing figures in figure skating', *Canadian Journal of Applied Spor Sciences*, 1985; 10: 171–7.
16. K. Barr and C. Hall, 'The use of imagery by rowers', *International Journal of Sport Psychology*, 1992; 23: 243–61.
17. S. C. Minas, 'Mental practice of a complex perceptual-motor skill', *Journal of Human Movement Studies*, 1978; 4: 102–7.
18. R. Bleier, *Fighting Back*, New York: Stein and Day, 1975.
19. R. L. Wilkes and J. J. Summers, 'Cognitions, mediating variables an strength performance', *Journal of Sport Psychology*, 1984; 6: 351–9.
20. R. S. Weinberg et al., 'Effects of visuo-motor behavior rehearsal, relaxation, and imagery on karate performance', *Journal of Sport Psychology*, 1981; 3: 228–38.
21. Cautela and Kearney, *Covert Conditioning*, op. cit.
22. J. Pates et al., 'The effects of hypnosis on flow states and three-poin shooting in basketball players', *The Sport Psychologist*, 2002; 16: 34–47; J. Pates and I. Maynard, 'Effects of hypnosis on flow states and gol performance', *Perceptual and Motor Skills*, 2000; 9: 1057–75.
23. R. M. Suinn, 'Imagery rehearsal applications to performance enhancement' *The Behavior Therapist*, 1985; 8: 155–9.
24. L. Baroga, 'Influence on the sporting result of the concentration of attention process and time taken in the case of weight lifters', in *Proceedings of the 3rd World Congress of the International Society of Sports Psychology*, Volume 3. Madrid, Spain: Instituto Nacional de Educacion Fisica Y Deportes, 1973.
25. A. Fujita, 'An experimental study on the theoretical basis of mental training', in *Proceedings of the 3rd World Congress of the International Society of Sports Psychology*, Volume Abstracts. Madrid, Spain: Instituto Nacional de Educacion Fisica Y Deportes, 1973: 37–8.
26. Ibid.

27. Rushall and Lippman, 'The role of imagery in physical performance', op cit.
28. G. H. Van Gyn et al., 'Imagery as a method of enhancing transfer from training to performance', *Journal of Sport and Exercise Science*, 1990; 12: 366–75.
29. G. H. Yue and K. J. Cole, 'Strength increases from the motor program Comparison of training with maximal voluntary and imagined muscle contractions', *Journal of Neurophysiology*, 1992; 67: 114–23; V. K. Ranganathan et al., 'Increasing muscle strength by training the central nervous system without physical exercise', *Society for Neuroscience Abstracts*, 2001; 31: 17; V. K. Ranganathan et al., 'Level of mental effort determines training-induced strength increases', *Society of Neuroscience Abstracts*, 2002; 32: 768; P. Cohen, 'Mental gymnastics', *New Scientist*, November 24, 2001; 172 (2318): 17.
30. D. Smith et al., 'The effect of mental practice on muscle strength and EMG activity', *Proceedings of British Psychological Society* annual conference, 1998; 6 (2): 116.
31. T. X. Barber, 'Changing "unchangeable" bodily processes by (hypnotic) suggestions: A new look at hypnosis, cognitions, imagining and the mind- body problem', in A. A. Sheikh (ed.), *Imagination and Healing*, Farmingdale, NY: Baywood Publishing Co., 1984. Also published in *Advances*, Spring 1984.
32. F. M. Luskin et al., 'A review of mind-body therapies in the treatment of cardiovascular disease, Part 1: Implications for the elderly', *Alternative Therapies in Health and Medicine*, 1998; 4 (3): 46–61.
33. F. M. Luskin et al., 'A review of mind/body therapies in the treatment of musculoskeletal disorders with implications for the elderly', *Alternative Therapies in Health and Medicine*. 2000; 6 (2): 46–56.
34. V. A. Hadhazy et al., 'Mind-body therapies for the treatment of fibromyalgia. A systematic review', *Journal of Rheumatology*, 2000; 27 (12): 2911–18.
35. J. A. Astin et al., 'Mind-body medicine: State of the science: Implications for practice', *Journal of the American Board of Family Practitioners*, 2003; 16 (2): 131–47.
36. J. A. Astin, 'Mind-body therapies for the management of pain', *Clinical Journal of Pain*, 2004; 20 (1): 27–32.
37. L. S. Eller, 'Guided imagery interventions for symptom management' *Annual Review of Nursing Research*, 1999; 17, 57–84.
38. J. Achterberg and G. F. Lawlis, *Bridges of the Bodymind: Behavioral Approaches for Health Care*, Champaign, Ill.: Institute for Personality and Ability Testing, 1980.
39. N. E. Miller and L. DiCara, 'Instrumental learning of heart rate changes in curarized rats: Shaping and specificity to discriminative stimulus', *Journal of Comparative and Physiological Psychology*, 1967; 63: 12–19; N. E. Miller, 'Learning of visceral and glandular responses', *Science*, 1969; 163: 434–45.

40. J. V. Basmajian, *Muscles Alive: Their Functions Revealed by Electromyography*. Baltimore, Md.: Williams and Wilkins, 1967.
41. E. Green, 'Feedback technique for deep relaxation', *Psychophysiology*, 1969; 6 (3): 371–7; E. Green et al., 'Self-regulation of internal states', in J Rose (ed.), *Progress of Cybernetics: Proceedings of the First International Congress of Cybernetics, London, September 1969*. London: Gordon and Breach Science Publishers, 1970: 1299–318; E. Green et al., 'Voluntary control of internal states: Psychological and physiological', *Journal of Transpersonal Psychology*, 1970; 2: 1–26; D. Satinsky, 'Biofeedback treatment for headache: A two-year follow-up study', *American Journal of Clinical Biofeedback*, 1981; 4 (1): 62–5; B. V. Silver et al., 'Temperature biofeedback and relaxation training in the treatment of migraine headaches: One-year follow-up', *Biofeedback and Self Regulation*, 1979; 4 (4): 359–66.
42. B. M. Kappes, 'Sequence effects of relaxation training, EMG, an temperature biofeedback on anxiety, symptom report, and self-concept', *Journal of Clinical Psychology*, 1983; 39 (2): 203–8; G. Rose et al., 'The behavioral treatment of Raynaud's disease: A review', *Biofeedback and Self Regulation*, 1987; 12 (4): 257–72.
43. W. T. Tsushima, 'Treatment of phantom limb pain with EMG and temperature biofeedback: A case study', *American Journal of Clinical Biofeedback*, 1982; 5 (2): 150–3.
44. T. G. Dobie, 'A comparison of two methods of training resistance to visually-induced motion sickness.' Paper presented at VII International Man in Space Symposium: Physiologic adaptation of man in space, Houston Texas, 1986. *Aviation, Space, and Environmental Medicine*, 1987; 58 (9) Sect. 2: 34–41.
45. A. Ikemi et al., 'Thermographical analysis of the warmth of the hands during the practice of self-regulation method', *Psychotherapy and Psychosomatics*, 1988; 50 (1): 22–8.
46. J. L. Claghorn, 'Directional effects of skin temperature self-regulation on regional cerebral blood flow in normal subjects and migraine patients', *American Journal of Psychiatry*, 1981; 138 (9): 1182–7.
47. M. Davis et al., *The Relaxation and Stress Reduction Workbook*, 5th edn, Oakland, Calif.: New Harbinger, 2000: 83–90.
48. J. K. Lashley et al., 'An empirical account of temperature biofeedback applied in groups', *Psychological Reports*, 1987; 60 (2): 379–88; S. Fahrion et al., 'Biobehavioral treatment of essential hypertension: A group outcome study', *Biofeedback and Self Regulation*, 1986; 11 (4): 257–77.
49. J. Panksepp, 'The anatomy of emotions', in R. Plutchik (ed.), *Emotion: Theory, Research and Experience Vol. III. Biological Foundations of Emotions*, New York: Academic Press, 1986: 91–124.

50. J. Panksepp, 'The neurobiology of emotions: Of animal brains and human feelings', in T. Manstead and H. Wagner (eds.), *Handbook of Psychophysiology*, Chichester: John Wiley & Sons, 1989: 5–26.
51. C. D. Clemente et al., 'Postreinforcement EEG synchronization during alimentary behavior', *Electroencephalography and Clinical Neurophysiology*, 1964; 16: 335–65; M. H. Chase et al., 'Afferent vagal stimulation: Neurographic correlates of induced EEG synchronization and desynchronization', *Brain Research*, 1967; 5: 236–49.
52. M. B. Stermann, 'Neurophysiological and clinical studies of sensorimotor EEG biofeedback training: Some effects on epilepsy', *Seminars in Psychiatry*, 1973; 5 (4): 507–25; M. B. Stermann, 'Neurophysiological and clinical studies of sensorimotor EEG biofeedback training: Some effects on epilepsy', in L. Birk (ed.), *Biofeedback: Behavioral Medicine*. New York: Grune and Stratton, 1973: 147–65; M. B. Stermann, 'Epilepsy and its treatment with EEG feedback therapy', *Annals of Behavioral Medicine*, 1986; 8: 21–5; M. B. Stermann, 'The challenge of EEG biofeedback in the treatment of epilepsy: A view from the trenches', *Biofeedback*, 1997; 25 (1): 6–7; M. B. Stermann, 'Basic concepts and clinical findings in the treatment of seizure disorders with EEG operant conditioning', *Clinical Electroencephalography*, 2000; 31 (1): 45–55.
53. E. Peniston and P. J. Kulkosky, 'Alpha-theta brainwave training and beta-endorphin levels in alcoholics', *Alcoholism: Clinical and Experimental Research*, 1989; 13: 271–9; E. Peniston and P. J. Kulkosky, 'Alcoholic personality and alpha-theta brainwave training', *Medical Psychotherapy*, 1990; 3: 37–55.
54. J. Kamiya, 'Operant control of the EEG alpha rhythm', in C. Tart (ed.) *Altered States of Consciousness*, New York: John Wiley & Sons, 1969, J. Kamiya, 'Conscious control of brain waves', *Psychology Today*, April 1968: 7.
55. N. E. Schoenberger et al., 'Flexyx neurotherapy system in the treatment of traumatic brain injury: An initial evaluation', *Journal of Head Trauma Rehabilitation*, 2001; 16 (3): 260–74.
56. C. B. Kidd, 'Congenital ichthyosiform erythroderma treated by hypnosis' *British Journal of Dermatology*, 1966; 78: 101–5, as cited in Barber, 'Changing "unchangeable" bodily processes', op. cit.
57. H. Bennett, 'Behavioral anesthesia', *Advances*, 1985; 2 (4): 11–21, as reported in H. Dienstfrey, 'Mind and mindlessness in mind-body research', in M. Schlitz et al., *Consciousness and Healing: Integral Approaches to Mind Body Healing*, St Louis, Mo.: Elsevier Churchill Livingstone, 2005: 56.
58. H. Dienstfrey, 'Mind and mindlessness', op. cit.: 51–60.
59. Dr Angel Escudero was featured on the BBC's *Your Life in Their Hands* series, May 1991. In the film, Escudero made incisions, sawed, drilled and hammered in order to break and reset the deformed leg of his fully conscious patient using his 'Noesitherapy' technique of pain control.

60. S. M. Kosslyn et al., 'Hypnotic visual illusion alters color processing in the brain', *American Journal of Psychiatry*, 2000; 157: 1279–84; Mark Henderson, 'Hypnosis really does turn black into white', *The Times*, 18 February 2002.
61. S. H. Simpson et al., 'A meta-analysis of the association between adherence to drug therapy and mortality', *British Medical Journal*, 2006; 333: 15–19.
62. Raúl de la Fuente-Fernández et al., 'Expectation and dopamine release Mechanism of the placebo effect in Parkinson's disease', *Science*, 2001; 293 (5532): 1164–6.
63. J. B. Moseley et al., 'A controlled trial of arthroscopic surgery for osteoarthritis of the knee', *New England Journal of Medicine*, 2002; 347: 81–8.
64. S. Krippner, 'Stigmatic phenomenon: An alleged case in Brazil', *Journal of Scientific Exploration*, 2002; 16 (2): 207–24.
65. L. F. Early and J. E. Kirsch, 'A case of stigmata', *Archives of General Psychiatry*, 1974; 30: 197–200.
66. T. Harrison, *Stigmata: A Medieval Mystery in a Modern Age*, New York: St Martin's Press, 1994, as referenced in S. Krippner, 'Stigmatic phenomenon', op. cit.
67. B. O'Regan and Caryle Hirshberg, *Spontaneous Remission: An Annotated Bibliography*, Petaluma, Calif.: Institute of Noetic Sciences, 1993.
68. Ibid.
69. L. L. LeShan and M. L. Gassmann, 'Some observations on psychotherapy with patients with neoplastic disease', *American Journal of Psychotherapy*, 1958; 12: 723.
70. D. C. Ban Baalen et al., 'Psychosocial correlates of "spontaneous regression of cancer"', *Humane Medicine*, April 1987.
71. R. T. D. Oliver, 'Surveillance as a possible option for management of metastatic renal cell carcinoma', *Seminars in Urology*, 1989; 7: 149–52.
72. P. C. Raud, 'Psychospiritual dimensions of extraordinary survival', *Journal of Humanistic Psychology*, 1989; 29: 59–83.
73. McTaggart, *The Field*, op. cit.: 132.
74. W. Braud and M. Schlitz, 'Psychokinetic influence on electrodermal activity', *Journal of Parapsychology*, 1983; 47 (2): 95–119.
75. Interview with William Braud, October, 1999.
76. Benor, *Healing Research*, op. cit.
77. S. M. Roney-Dougal and J. Solvvin, 'Field study of an enhancement effect on lettuce seeds – Replication study', *Journal of Parapsychology*, 2003; 67 (2): 279–98.

78. Dr Larry Dossey calls negative diagnoses ‘medical hexing’, and there is anecdotal evidence that patients often live up to their doctor’s gloomy prognosis, even when there is no physical evidence that they should do so. For a potent example see the story of a leukaemia patient who was thriving until he happened to find out what he had. He was dead within a week once his illness had the label of a potentially terminal illness: L. McTaggart, *What Doctors Don’t Tell You*, London: HarperCollins, 2005: 343.

CHAPTER TEN

The Voodoo Effect

DICK BLASBAND WAS DRAWN TO THE IDEA that there might be a way amplify and direct life energy, like holding up a magnifying glass to focus the rays of the sun.

Blasband, a psychologist, was intrigued by the theories of Wilhelm Reich, the Austrian psychiatrist and one-time protégé of Sigmund Freud, who thought it possible to trap ‘orgone’ – the name he gave to what he believed to be omnipresent cosmic energy – in an orgone energy ‘accumulator’. An accumulator, a box-like enclosure of any size, could be made of alternating layers of any metal and non-metallic materials, such as cotton cloth or felt.

Reich believed that atmospheric energy would be first attracted, then instantly repelled by the metal and eventually absorbed by the non-metallic substance. Because the enclosure was layered, orgone energy would continuously flow between the atmosphere and the box, like a current of air, and so constantly ‘accumulate’. Reich had early encouraging results with animals and plants placed in the boxes, which lay the basis for his later claims that accumulated energy had an immense capacity to heal.

It occurred to Blasband that Reich’s ideas about energy fields were not dissimilar to those of his colleague Fritz-Albert Popp and his work on biophotons. Perhaps the best means of testing an accumulator was to measure its effect on the emission of these tiny specks of light from a living thing.

In August 1993, Blasband travelled to Popp’s International Institute, then in Kaiserslautern, Germany. He and Popp created a variety of orgone accumulators then chose a number of plants in Popp’s laboratory – cress seeds, cress seedlings and *Acetabularia crenulata*, a primitive form of marine algae – to be the experimental population. Popp’s photomultipliers would count the light emissions of all the plants inside and outside the orgone boxes and record any differences.

Blasband carried out four experiments – placing the algae in the accumulator first for one hour, and then continuously for two weeks – with no result. Popp’s equipment recorded not the slightest alteration in the light emissions. Blasband wondered if this was because the plants were already so healthy that the boxes could not improve their state of health. Perhaps he would see a larger change in a subject that needed help or improvement. He and Popp decided to try making the *Acetabularia* ‘ill’ by depriving it of most of its vitamin supply for 24 hours before treatment. It appeared to make no difference. The plants’ biophotons didn’t change. No amount of exposure to an accumulator of any variety seemed to make one bit of difference to the health or well-being of any of the plants.

Blasband and Popp then decided to test whether a mental intention could boost the action of the accumulators. In his new series of experiments, Blasband sent an intention for the energy within the accumulator to be beneficial to certain seedlings and harmful to others. These results were disappointing, too. There was only one significant difference in the number or quality of biophoton emissions before and after treatment of any of the plants: the only effective intention appeared to be the one he had sent to stunt their growth.¹ In both experiments, negative intention was more powerful than positive intention. Thoughts to harm had the greatest effect.

Blasband’s little study highlights perhaps the most disturbing consideration of all about intention: that bad thoughts, as well as good ones, can have an effect on the world, and indeed may be the more powerful of the two. After all, in many native cultures, prayer and intention have a shadow component in hexes, voodoo and curses, which are reported to be highly effective forms of negative intention.

Many healers routinely use a negative means to a positive end.

As Dr Larry Dossey, author of *Be Careful What You Pray For* ...² has noted, negative intention is the very foundation of most healing. Healing from an infectious agent or a rogue cell line such as cancer requires intent to harm.³ It works from a desire to kill something: to inhibit bacterial enzymes, alter cell membrane permeability, or interfere with the nutrition given to the cell or the synthesis of DNA.⁴ In order for the patient to get better, the offending agent has to die.

Many pioneers of mind–body medicine in the treatment of cancer, such as Dr Bernie Siegel, Dr Carl Simonton, and Australian psychiatrist Ainslie Meares encouraged their patients to use vivid forms of mental imagery – a metaphoric representation of their illness – to enhance their healing.⁵ The majority of the cancer patients who first made use of visualization techniques imagined a battlefield, on which good (the patient) is pitted against evil (the cancer), with the cancer patient possessing the larger weapon.

Some patients imagined their white blood cells as an army killing the cancer cells or a 'tap' containing the blood that feeds the cancer cells, which they can turn off. Some patients visualized themselves as participating in a violent video game. When Simonton first introduced this technique to his patients in the 1970s, Pac-Man was the most popular video game of the time. He encouraged his patients to imagine a little Pac-Man inside their bodies, gobbling up cancer cells in its path. But whatever the particulars of the imagery, the intention itself needed to be murderous; the patient had to want to annihilate the enemy.

Research on negative mental influence presents a number of obstacles to scientists. One basic problem, as Cleve Backster found in his research, is finding a living thing that no one objects to having killed. Many choose to study the most basic life forms, such as paramecia or fungi, or to experiment with seeds or small plants.[6](#)

Another problem is avoiding an unintended 'spray' effect: what if a healer's aim is slightly off one day and the negative intent gets sent to the host instead?

The Canadian healer Olga Worrell refused to carry out negative intention on infectious diseases for exactly that reason. She worried that her negative intent might move beyond the bacteria and accidentally target the person she was trying to heal.[7](#)

One of the earliest experiments using negative intention was conducted by Jean Barry, president of the Institut Métapsychique International, who studied bacteria and fungi. As insignificant as these lowly organisms appear, Barry, a general practitioner, understood their pivotal role in maintaining health and causing illness. If it could be shown that intention has the power to eliminate these small organisms, humans might be able to exert greater control over their own health.

Barry decided to test the effect of negative intention on a fungus called *Rhizoctonia solani*. *Rhizoctonia*, a thready filament and a distant relative of the common mushroom, is an enemy to 500 types of crops. Farmers call it pod rot or root rot, as it commonly attacks the pods and roots, stunting growth and eventually consuming the plant. No one would object to a means of controlling this garden menace.

Barry set up a batch of experimental Petri dishes and matched them with a set of controls of the identical type of fungus growing in the same conditions. He enlisted ten volunteers and assigned five experimental Petri dishes and five controls to each person. At the appointed time, each volunteer was asked to send intentions to slow the growth of the fungi in the experimental Petri dishes. After the experiment, the lab assistant measured the growth of each sample of *Rhizoctonia* by outlining its boundary on tracing paper. Of the 195 dishes involved in the negative intention, 151, or 77 per cent, were smaller than the average size of the controls.[8](#)

Barry's study was successfully replicated by researchers from the University of Tennessee, although their study also tested the effect of remote influence; this time, the volunteers sending the intention were 15 miles away from the fungus samples.[9](#)

Similar research was conducted by Carroll Nash, the director of the parapsychology department at St Joseph's University in Philadelphia, but on *Escherichia coli*, microbes with a direct impact on human beings. Millions of these bacteria, which help to digest food and keep hostile bacteria at bay, peacefully reside in the gut. *E. coli* also metabolizes lactose, the enzyme present in milk. Yet, as with many microbes, *E. coli* can suddenly turn unfriendly by migrating out of the digestive tract or mutating into a virulent form that causes illness. Many toxic strains are also present in food. *E. coli* represented an interesting choice for Nash. If humans could control its growth, they might avoid serious *E. coli* infections and improve their general digestive health.

Nash decided to test whether mental influence could affect the mutation rates of *coli* bacteria. Usually, an *E. coli* population starts life unable to ferment lactose (and so is 'lactose-negative'), but after it mutates, over numerous generations, the new population can do so (at which point it become 'lactose-positive'). This process ordinarily occurs at a predictable rate. Nash wanted to see whether his volunteers could slow it down or speed it up. To work out the growth rates of these tiny organisms, Nash employed an electrophotometer, which counts the microbes by measuring the slightest differences in the density of the media in which they are suspended.

Each of his 60 student participants received nine test-tubes containing both lactose-negative and lactose-positive strains of *E. coli* culture. The students were asked to mentally encourage the transformation of the unmutated bacteria in the first three test-tubes from lactose-negative to lactose-positive. With the next three test-tubes, they were to attempt to inhibit the process of mutation. The final three, the controls, would not be exposed to influence of any kind. When he tallied the results, Nash discovered more mutation than normal in the test-tubes that had received the positive intentions to mutate, and fewer than normal in those for which the intentions were to inhibit the process, although the greatest effect occurred with negative intention.

Nash's study had an interesting coda: he had not stipulated any particular location where the mental influence had to originate; the volunteers were allowed to send their thoughts from the place of their choosing, whether the lab or elsewhere. When Nash examined the differences in the results according to the place from where the intentions had been sent, an interesting pattern emerged. Those students assigned the task of sending positive intentions had the best success if they sent their thoughts while in the lab; those with negative intentions had the best result if they waited until they had left. The Tennessee researchers who replicated Barry's study also

discovered that negative intention was most effective when it was sent from a remote site. Positive intent appeared to work best in the presence of its object, whereas a negative intent worked best when the object of ill will was not anywhere in the line of sight.[10](#)

These early studies revealed several important aspects of intention. Thoughts take aim with great accuracy; although their effects on living things can drastically differ depending on the nature of the intention – whether it is positive or negative. Where we position ourselves when sending out a thought might also have a bearing on our success. Being near the target while sending a positive intention or away from the target when sending a negative intention might magnify its effect.

* * *

The next best experimental subject to a live human being is its cells. If you can prove an effect on an essential component of a living thing, it is likely that the same effect will occur with the entire organism. Dr John Kmetz, a colleague of William Braud's in San Antonio, Texas, decided to test the effect of negative intention on cancer. Although he could not test his theory on a live human being, he settled for a sample of cervical cancer cells, and enlisted Matthew Manning, a gifted British healer.

Manning sent negative intentions either by touching the beaker of cells or from a distance, inside an electromagnetically shielded room. Kmetz then used special equipment to count how many cancer cells were in the culture medium. Ordinarily, a cancer cell, which has a positive charge, will grasp the side of a plastic beaker, attracted to its negative electrostatic charge. An injury to the cell will cause it to drop off the side and into the culture medium. Kmetz's equipment demonstrated that Manning had fatally injured the culture.[11](#) Manning's extraordinary healing ability had been turned on its head; in this study he had become a killing machine.

Practitioners of *Qigong* openly acknowledge that intention has the power to enhance or destroy – indeed, the Chinese term for sending positive *Qi*, or life energy, through intention translates into English as 'peaceful mind', while sending negative *Qi* is referred to as 'destroying mind'.[12](#)

A host of studies of *Qigong* carried out in China have been collated on the *Qigong* Database®, many of which claim to offer evidence that 'destroying mind' can kill human cancer cells or tumours in mice, decrease the growth rate of *E. coli* and inhibit activity of amylase, a digestive enzyme used to help digest carbohydrates.[13](#) Nonetheless, some Western scientists maintain quiet reservations about the database; few of these studies have been replicated in the West.

One study on plants conducted at the First World Conference of Academic Exchange of Medical *Qigong*, in Beijing in 1988, examined whether sending *Qi* could affect the growth of a confederate spiderwort plant by concentrating on its process of replication. A *Qi* master was asked to damage one of the plant's self-destruct mechanisms, which would cause the plant to live longer than normal.[14](#) The master had to target his negative intention precisely, so that it would injure only one aspect of the plant while the rest would thrive.

To record any subtle effects on the health of the plant samples during the experiment, any increases or decreases in certain cells after replication, the researchers used a micronuclear method developed at Western Illinois State University. During the study, the *Qigong* master displayed a remarkable ability to send precise instructions to specific parts of the plant, some of which were damaging, some beneficial.[15](#)

A similar study was carried out by researchers at the National Yang Ming Medical College and National Research Institute of Chinese Medicine in Taipei Taiwan. In this instance, the *Qigong* master alternately sent positive and negative intention to boar sperm cells and human fibroblast cells, which make up the connective tissue of the body.

After 2 minutes of negative intention, the growth rates and protein synthesis of the cells decreased dramatically by 22–53 per cent. When the *Qigong* master reversed his intention and sent 10 minutes of positive intention, all the activity of the cell increased by 5–28 per cent.[16](#) In another well-controlled study by the Mt Sinai School of Medicine, two *Qigong* masters were able to inhibit the process involved in the contraction of muscles by as much as 23 per cent.[17](#)

These studies raise the obvious question: which is more powerful, a positive or a negative thought? In some studies, the will to harm appears to be the stronger of the two intentions, but that makes sense in a study like Blasband's, where it is probably far easier to damage a healthy system than to make a healthy system even healthier, or indeed to fix something that is broken, or to order a disordered system.[18](#)

Nevertheless, effective intention of any variety is likely to require order and deliberately focused thought. How many negative intentions are sent by someone as ordered as a *Qigong* master?

Although negative intention appears capable of disrupting the most fundamental biological processes when precisely targeted,[19](#) one study suggests that healing does not necessarily require negative intention. Leonard Laskow, an American gynaecologist and healer, was recruited by American biologist Glen Rein to test the most effective healing strategy for inhibiting the growth of cancer cells. In his own practice, Laskow believed in establishing an emotional connectedness with his

subject – even with cancer cells – before sending out healing. Rein prepared five different Petri dishes containing identical numbers of cancer cells and then asked Laskow to send out a different intention while holding each one. Laskow's first intention was that the natural order be reinstated and the cells' growth rate return to normal.

With the next Petri dish he was to adopt a Taoist visualization that entails imagining that only three of the cancer cells remained in the Petri dish. For the third dish he was not to have an intention, but simply to ask God to have His will flow through Laskow's hands. He offered unconditional love to the cancer cells of the fourth dish, which involved meditating on a state of love and compassion, much as Davidson's Buddhists had done. For the final dish of cancer cells, Laskow carried out his only truly destructive intention, by visualizing the cells dematerializing, either going into the light or the 'void'. Rein gave Laskow a choice of imagery largely because he was uncertain which visualization would be most effective in obliterating something. Was it more effective to release an entity by offering it an endpoint (the light), or simply to give it a full range of potential (the void)? As a yardstick of Laskow's effectiveness, Rein would measure the amount of radioactive thymidine absorbed by the cancer cells – an indicator of the growth rate of malignant cells.

Laskow's various intentions had quite different effects. The most powerful were undirected intentions asking the cells to return to the natural order, which inhibited the cancer cells' growth by 39 per cent. Acquiescing to God's will with no specific request was about half as effective, inhibiting the cells by 21 per cent, as was the Taoist visualization. An unconditional acceptance of the way things were had no effect either way, nor did imagining the cells dematerializing. In these two instances, the problem may have been that the thought was simply not focused enough.

In a follow-up study, Rein asked Laskow to limit himself to two possibilities, the Taoist visualization and a request for the cells to return to the natural order. This time, he achieved an identical result with both intentions; the cancer cell growth was inhibited by 20 per cent. The strongest effect of all occurred when he combined the two approaches, mixing an intention to return to the natural order while imagining only three cells left; his rate of cell inhibition doubled, to 40 per cent. Clearly the combination of asking the universe to restore order while imagining a specific outcome exerted a powerful effect. Rein asked Laskow to repeat this combined approach, but to target the medium in which the cancer cells grew, rather than the cells themselves. Laskow achieved the same result as when he had focused directly on the cells themselves.

Finally, Rein instructed Laskow to hold each of his five states of mind in turn while grasping one of five vials of water, which would later be used to make up the tissue-

culture medium of the cancer cells. The water treated with the 'natural-order' intention again had the greatest effect, inhibiting the growth of the cancer cells by 28 per cent. In this case, water apparently 'stored' and transferred the intentions to the culture medium and on to the cancer cells.

Laskow's approach was instructive. The most effective healing intention had been framed as a request, combined with a highly specific visualization of the outcome, but not necessarily a destructive one.[20](#) With healing, the most effective approach may not be to destroy the source of the illness, but, as with other forms of intention, to move aside, let go of the outcome, and allow a greater intelligence to restore order.

* * *

Most research about negative intention concerns a conscious desire to harm something. I wondered about those moments when negative intention is unconscious. Suppose you don't like someone and harbor an unconscious ill will towards him? Do you unwittingly send out a negative intention? Or, what about those moments when you explode in anger? Is it possible that your momentary anger causes unintentional harm?

An overenthusiastic cleaner of mine once accidentally stripped off all the chrome on every fixture in our bathrooms. When I discovered the damage, a few hours after she had left our house, I was so overwhelmed with anger that I had to lie down. I had only just finished a five-month long renovation project on our newly purchased family home and had lovingly overseen the entire project, which had cost a good deal of hard-earned savings. I later learned, to my horror, that at about the time I had given voice to my fury, she had fallen off the bus and broken her leg. At another time, I was irrationally overwhelmed with anger at our bank manager, after discovering that our bank, now run by computers, had not recorded a deposit and had bounced several of our cheques. Later, I was horrified to learn that at roughly the moment I had vented my spleen, she had tripped on a pavement and broken most of her front teeth.

I had always felt guilty and curious about both these incidents. Was their misfortune my doing? Was it possible to curse people through your thoughts? I considered the effect of the everyday negative thoughts that swim through everyone's mind every day. A negative thought about yourself ('I'm untalented and lazy') or your children ('He's such a slob'; 'She's lousy at maths') might ultimately manifest as a physical energy and become a self-fulfilling prophecy. Indeed, moments when you feel an aversion to someone or something that you cannot rationally explain may simply be an instance when you are picking up a negative intention towards you. Even times when you are depressed could have a physical effect on the people and other living things around you.

Bernard Grad, the Canadian biologist, addressed many of these issues in a study that tested the power of a negative frame of mind on the growth of plants. He planted four groups of 18 pots, each containing 20 barley seeds. Each pot was to be watered with 1 per cent saline solution, slightly stronger than the kind used by hospitals when giving intravenous infusions to patients, which can stunt a plant's growth. Three batches of the plants were to receive watering with the salt water, but only after the water had been held by one of three people for half an hour. The control batch would be watered with the solution that had not been exposed to anyone.

The first vial was held by a healer with green fingers and a passion for plants. The other two vials were held by two depressed patients – a man diagnosed as a psychotic depressive and a woman who was neurotically depressed – chosen from the Canadian hospital where Grad worked. The man was so depressed that he didn't even ask what was in the bottle, but simply assumed that Grad, who wore a white coat, was just another in the procession of doctors preparing him for periodic electric shock therapy. While holding the bottle, he repeatedly protested that he didn't need an ECT treatment. The woman, on the other hand, visibly lifted when Grad told her that the bottle was part of an experiment. Half an hour later, when he came to retrieve the bottle from her, he discovered that she had been cradling it as if it were a baby.

This unforeseen turn of events worried Grad, as the woman had been chosen precisely because he believed she would be in a negative state of mind. She had suddenly appeared to regain her *joie de vivre*, simply at the thought of her involvement in the experiment. After carefully creating a multi-blind system so that he could not know or be influenced by who had done what, Grad poured the water over the seeds.

Several weeks later, he was pleased to see that the result more or less followed his prediction. The plants watered by the man with the psychotic depression grew the slowest, followed by the control plants, whose bottle had not been held by anyone. The fastest growing plants had been watered by the green-fingered healer, followed by those of the depressed woman, which was a surprise. It seemed that her plants had grown faster because of her own enthusiasm about the experiment.[21](#)

Carroll Nash tried a similar experiment, asking a group of psychotics to hold individual sealed glass bottles of a solution of dextrose and sodium chloride for half an hour. Nash then removed 6 millilitres of the solutions from each bottle and poured them into fermentation tubes. Similar solutions that had not been 'charged' by the psychotics were poured into control test-tubes.

All 24 test-tubes received a suspension of yeast. After two hours Nash measured the amount of carbon dioxide produced in each of the tubes and took periodic

measurements for the next six weeks. When he compared the tubes containing the 'held' solutions with the controls, he discovered that the solution held by the psychotics had marginally prevented the yeast from growing.[22](#)

Even deeply buried feelings might have an effect on people we purport to care about. In 1966, Dr Scott Walker of the University of New Mexico School of Medicine conducted a study of alcoholics in the midst of rehabilitation. He divided the group randomly and had members of the Albuquerque Faith Initiative pray for them each day for six months. Half of the participants (some from the treatment group and some from the controls) knew they were being prayed for by family members.

At the end of the six months, Scott discovered that those in both groups whose relatives and friends were praying for them were drinking *more* heavily than the others. Prayer from those who supposedly had the patients' best interests at heart was having the opposite effect.

Scott came up with an interesting interpretation. The across-the-board negative effect of prayer by relatives may reflect their complicated, unconscious feelings towards alcoholics. Although consciously they might wish for their loved one to recover, they might actually wish for them to carry on drinking, if the person praying is a fellow drinker and does not wish to lose a drinking buddy. Or, perhaps the boorish, selfish behaviour of an alcoholic has so hurt the relatives that they unconsciously wish for the alcoholic to die.

All these studies are small, but they carry a huge implication: *even your current state of mind carries an intention that has an effect on life around you*. The mind continues affecting its surroundings whether or not we are consciously sending an intention. To think is to affect. When we are consciously attempting to affect someone else with our thoughts, we may want to search our hearts about our true feelings to ensure that we are not sending tainted love.

These studies also raise the possibility that the thoughts that spill out of us at every moment also affect inanimate objects within our reach. Some people have a reputation for having a positive or negative effect on electronic equipment – they are either an 'angel' or a 'gremlin'. One of the fathers of quantum theory, the brilliant theoretical physicist Wolfgang Pauli, was widely known to possess a powerfully negative force field. Whenever he arrived at his laboratory, mechanisms would freeze, collapse or even be set alight.[23](#)

I am a gremlin of the first order. In those rare moments when I am crashing around in a bad mood, all the computers in our office begin crashing in unison. Once, during a day of extreme agitation, after I had broken my computer and printer at home, I headed off for work and tried to work on a variety of computers around my company's

office. One by one, they died in my hands. When one of our laser copier printers also froze the moment I tried to photocopy a page, my team firmly but politely escorted me off the premises.

The late Jacques Benveniste discovered the gremlin effect first hand when he carried out experiments on electromagnetic signalling between cells. From 1991, after his noted 'memory of water' studies, Benveniste understood that the basic signalling between molecules was not chemical but electromagnetic. Within a living cell, molecules communicate, not by chemicals but by electromagnetic signalling at low frequencies, and each molecule has its own signature frequency.[24](#)

Until the end of his life in 2005, Benveniste explored the possibility that these molecular signals could be transferred simply by using an amplifier and electromagnetic coils. He demonstrated that it was possible to effect a molecular reaction without the presence of the molecule in question simply by playing the molecule's unique 'sound'.

One of Benveniste's many experiments with cellular signalling concerned the interruption of the coagulation of plasma, the yellowish medium of the blood. Ordinarily caused by the presence of calcium in the liquid, the clotting capacity of plasma can be precisely controlled by first chemically removing all existing calcium in the plasma, then adding back particular amounts of the mineral. By also adding heparin, an anticoagulant drug, the plasma is prevented from clotting, even in the presence of calcium.

In his study Benveniste would remove calcium from the plasma and add calcium to water, but instead of adding the actual heparin to the calcium water, he simply exposed the water containing calcium to the 'sound' of heparin transmitted through the digitized electromagnetic frequency of heparin that he had discovered. As with all his other experiments, the signature frequency of heparin worked as though the molecules of heparin itself were there: in its presence, the blood was less able to coagulate.

Benveniste had a robot built to carry out this experiment, largely to silence his critics by eliminating the potential bias of human interference. The robot was a box with an arm that moved in three directions, mechanically exposing the water containing calcium to the heparin in several easy steps.

After hundreds of such experiments, Benveniste discovered that it usually worked well except on days that a certain woman – an otherwise experienced scientist – was present. Benveniste suspected that the woman must be emitting some form of waves that were blocking the signals. He developed a means of testing for this, and discovered that the woman emitted powerful, highly coherent electromagnetic fields

that appeared to interfere with the communication signalling of his experiment. Somehow, the woman acted as a frequency scrambler. To test this further, he asked the woman to hold a tube of homeopathic granules in her hand for 5 minutes. When he later tested the tube with his equipment, all molecular signalling had been erased.

Since the problem was likely to be electromagnetic, the obvious next step was to protect the machine from EMFs by building a shield. But once the shield was in place, the machine stopped producing good results. Benveniste pondered this development for some days. Perhaps it had to do with positive effects of the environment, and not simply the absence of negative effects. He opened the shield and asked the man who had been in charge of the lab for many years to stand in front of the robot. Immediately, the robot began again to crank out perfect results. As soon as the man left and the shield was put up, the robot no longer produced decent data. This suggested that, just as some people inhibited equipment, others enhanced it. The shield, originally erected to stop negative influences, had blocked positive ones as well.

Benveniste reasoned that the only substance near the robot capable of picking up positive or negative activity was the tube of water, so he asked the head lab technician to hold the tube in his pocket for two hours. He then put the tube into the machine, removed the man from the room and put up the shield. After that, the robot's experiments worked virtually 100 per cent of the time.[25](#)

These anecdotal stories of the gremlin effect are not so farfetched when you consider the mountains of data generated by the PEAR laboratory, demonstrating that human intention has the ability to make the random output of computers more orderly even when the intention is not conscious or deliberate. Living consciousness might have a major effect on microprocessor technology, which is now exquisitely sensitive. The tiniest disturbances in a quantum process can be highly disruptive. My own gremlin effect appears to be linked to moments of extreme stress or agitation but for some people it may be the very nature of their thought system.

The idea that we can 'charge' an inanimate object with our thoughts is the basis of the dark arts of many native cultures, which infuse effigies and voodoo dolls with negative intention and then use them to target enemies. There is a rich tradition of using effigies, but not much scientific study of them. Dean Radin once designed an experiment to test the effectiveness of voodoo dolls as an instrument of positive intention. He constructed a tiny effigy of a person known to a group of volunteers, who then directed their prayers to the doll. The prayers turned out to have demonstrable effects – an instance of benevolent voodoo.[26](#)

If we can be unwitting recipients of negative influence, should we take steps to block it or ward it off? Many psychics recommend using visualization to create a mental image of protection, such as imagining yourself in a giant bubble. Marilyn Schlitz and William Braud tested this idea in a variation on their staring studies with 300 volunteers divided into pairs in separate rooms. One member of each pair (the sender) was asked to use a mixture of imagery and self-regulation techniques like relaxation or Autogenic Training to relax or energize themselves. They were then asked to send an intention to reproduce a similar state in their partner (the receiver) which would be recorded with a polygraph pen. Comparisons of the EDA readings of both senders and receivers showed that the senders had an effect – when they were relaxed or activated, so were their receivers.

The receivers were then asked to visualize a variety of images that would act as a psychological ‘shield’ to block the senders’ influences; any image – a shield, a huge concrete wall, a steel fence, pulsating white light – was suitable, so long as it felt protective. These strategies proved highly successful in blocking one of the unwanted influences.[27](#)

Then, other scientists from the University of Edinburgh attempted to replicate the EDA studies under more rigorous conditions. The senders alternately attempted to calm or activate the receivers, who were to be open to being influenced for half the session and then to ‘block’ the influence attempt for the other half, by imagining themselves wrapped up in a ‘shielding cocoon’ or adopting a stubborn and uncooperative frame of mind. Nevertheless, during the times of attempted influence, the receivers recorded the same EDA readings, regardless of whether they were ‘allowing’ or blocking it. If anything, there was a slightly larger effect during the blocking sessions. This suggests that ordinary mental strategies of isolating or protecting ourselves may not be enough to successfully resist unwanted influence.[28](#)

Qigong practitioners undergo lengthy training to learn techniques enabling them to ‘disguise’ or make their energy fields temporarily ‘invisible’ in order to ward off unwanted influence. Creating a psychic shield around yourself to prevent a barrage of negative influences – whether from your boss, a well-meaning but interfering professional, that unfriendly neighbour, even the stranger staring at you in the supermarket queue – is likely to require more than an attitude of resistance or a bit of internal imagery.

Larry Dossey once wrote that the most powerful antidote to negative intention was the line in the Lord’s Prayer: ‘deliver us from evil’. I came across another more

ecumenical instance, from the work of Dr John Diamond, who discovered a simple means of grounding yourself against unwelcome influences. Diamond, a psychiatrist

and holistic healer, was inspired by George Goodheart, creator of applied kinesiology, which tests the effect of various substances on the body. Goodheart developed the technique of 'muscle testing', now a feature of applied kinesiology. He would ask a patient to stand facing him, with her left arm out, parallel to the floor: he placed his left arm on the patient's shoulder to steady her, and then asked her to resist with all her strength while he pushed on her arm. In most instances, the arm would spring back and resist the force of Goodheart's push. However when Goodheart exposed that person to noxious substances, such as food additives or allergens, the person's left arm would be unable to resist the pressure of Goodheart's push and easily be overcome.

Diamond applied this muscle testing to toxic thoughts. When a person was exposed to any unpleasant thought, the 'indicator muscle' would test weak. Diamond called it 'behavioral kinesiology' and has tested it on thousands of subjects over many years as a means of instantly taking stock of a person's thoughts and most secret desires.[29](#)

Diamond discovered one thought that could overcome any sort of negative influence, or debilitating idea or situation. He called it a 'homing thought', because it reminded him of his youth in Sydney, Australia, swimming in the surf. Whenever a large wave threatened, he and his friends would dive to the bottom of the water and hold on to the sand with their fingertips. 'We had learned that as soon as we were faced with this situation of stress, we could dive down, grab on to our securing handhold and hang on to our "rock" until the stress passed,' he writes.[30](#)

The homing thought that each of us can hold on to, Diamond realized, was our ultimate aspiration or purpose in life. He has also referred to it as 'cantillation': each person's special gift or talent that not only gives one a sense of joy but also union with the Absolute. The term 'homing thought' also reminded him of the direction finder that lost aeroplane pilots use to find their way home. The homing thought can act as a homing beacon for everyone, particularly during the most difficult moments. 'It holds us steadfast,' he once wrote, 'on our course.'

Diamond's ideas have not been subjected to scientific scrutiny, but the sheer weight of his anecdotal evidence in using behavioural kinesiology on thousands of patients lends them a certain significance. Whenever we are besieged by the darkest of intentions, we might best protect ourselves when holding on to the thought of what we have been born to do.

Note – Chapter 10: The Voodoo Effect

1. A. Blasband and Gottfried Martin, 'Biophoton emission in "orgon energy" treated cress seeds, seedlings and *Acetabularia*', *International Consciousness Research Laboratory, ICRL Report No 93.6*.
2. Dossey, *Be Careful What You Pray For*, op. cit.: 171–2.
3. *Ibid*.
4. Benor, *Healing Research*, op. cit.: 261.
5. C. O. Simonton et al., *Getting Well Again*, New York: Bantam, 1980; B. Siegel, *Love, Medicine and Miracles: Lessons Learned about Self-Healing from a Surgeon's Experience with Exceptional Patients*, London: HarperCollins, 1990; A. Meares, *The Wealth Within: Self-Help Through a System of Relaxing Meditation*, Melbourne, Australia: Hill of Content 1990.
6. For much of the research detailed in this chapter, I am especially indebted to Larry Dossey and Daniel Benor, who have detailed many of these early studies in their respective books, Dossey's *Be Careful What You Pray For ... You Just Might Get It* and Benor's *Healing Research, Spiritual Healing* and his outstanding, comprehensive website: www.wholistichealingresearch.com.
7. Benor, *Healing Research*, op. cit.: 264.
8. J. Barry, 'General and comparative study of the psychokinetic effect on a fungus culture', *Journal of Parapsychology*, 1968; 32 (94): 237–43.
9. W. H. Tedder and M. L. Monty, 'Exploration of a long-distance PK: A conceptual replication of the influence on a biological system', in W. G. Roll et al. (eds.), *Research in Parapsychology*, Metuchen, NJ: Scarecrow Press, 1981: 90–3. Also see Dossey, *Be Careful What You Pray For*, op. cit.: 169; Benor, *Healing Research*, op. cit.: 268–9.
10. C. B. Nash, 'Test of psychokinetic control of bacterial mutation', *Journal of the American Society for Psychical Research*, 1984; 78: 145–52.
11. Kmetz's study was described in W. Braud et al., 'Experiments with Matthew Manning', *Journal of the Society for Psychical Research*, 1979; 50: 199–223. While the study was promising, in his review of it in *Healing Research*, Benor noted the lack of sufficient detail.
12. Dossey, *Be Careful What You Pray For*, op. cit.: 175–6.
13. Many researchers of alternative medicine maintain the same concerns about studies of Chinese medicine carried out in China. These concerns don't disregard the strong anecdotal evidence about the effectiveness of Traditional Chinese Medicine, only the scientific method of studies of its effectiveness.
14. S. Sun and C. Tao, 'Biological effect of emitted qi with *Tradescantia paludosa* micronuclear technique', *First World Conference for Academic Exchange of Medical Qigong*. Beijing, China, 1988: 61E.
15. *Ibid*.

16. Dossey, *Be Careful What You Pray For*, op. cit.: 176.
17. D. J. Muehsam et al., 'Effects of Qigong on cell-free myosin phosphorylation: Preliminary experiments', *Subtle Energies and Energy Medicine*, 1994; 5 (1): 93–108, also reported in Dossey, *Be Careful What You Pray For*, op. cit.: 177–8.
18. *Ibid.*
19. Benor, *Healing Research*, op. cit.: 253.
20. G. Rein, *Quantum Biology: Healing with Subtle Energy*, Palo Alto, Calif.: Quantum Biology Research Labs, 1992; as reported in Benor, *Healing Research*, op. cit.: 350–2.
21. B. Grad, 'The "laying on of hands": Implications for psychotherapy gentling and the placebo effect', *Journal of the Society for Psychical Research*, 1967; 61 (4): 286–305.
22. C. B. Nash and C. S. Nash, 'The effect of paranormally conditione solution on yeast fermentation', *Journal of Parapsychology*, 1967; 31: 314.
23. Radin, *The Conscious Universe*, op. cit: 130.
24. An entire chapter is devoted to Jacques Benveniste in McTaggart, *The Field*, op. cit.: 59.
25. Description of these results from a telephone conversation with Jacques Benveniste, November 10, 2000.
26. J. M. Rebman et al., 'Remote influence of the autonomic nervous system by focused intention', *Subtle Energies and Energy Medicine*, 1996; 6: 111– 34.W. Braud and M. Schlitz, 'A method for the objective study of transpersonal imagery', *Journal of Scientific Exploration*, 1989; 3 (1): 43–63; W. Braud et al., 'Further studies of the bio-PK effect: Feedback blocking specificity/generalty', in R. White and J. Solfvin (eds.), *Research in Parapsychology*, Metuchen, NJ: Scarecrow Press, 1984: 45–8.
27. C. Watt et al., 'Exploring the limits of direct mental influence: Two studies comparing "blocking" and "co-operating" strategies', *Journal of Scientific Exploration*, 1999; 13 (3): 515–35.
28. J. Diamond, *Your Body Doesn't Lie*, New York: HarperCollins, 1979.
29. J. Diamond, *Life Energy*, New South Wales: Angus & Robertson, 1992: 71.

CHAPTER ELEVEN

Praying for Yesterday

ON THE EVE OF THE MILLENNIUM, Leonard Leibovici, an Israeli profes of internal medicine in Israel and expert on hospital-acquired infections, conducted a study of healing prayer's effect on nearly 4000 adults who had developed sepsis while in the hospital. He set up a strict protocol, using a random number generator to randomize

the participants into two groups, only one of which would be prayed for, and throughout the study maintained impeccable blinding; neither the patients nor the hospital staff knew who was getting treated – or indeed even knew that a study was being carried out. The names of all those in the treatment group were then handed to an individual, who said a short prayer for the well-being and full recovery of the treated group as a whole.

Leibovici was interested in comparing three outcomes between the prayed-for and not-prayed-for groups: the number of deaths in hospital; the overall length of stay in hospital; and the duration of fever. When calculating the results, he was careful to employ several statistical measurements to examine the significance of any differences. As it happened, the group that had been prayed for suffered fewer deaths than the controls (28.1 versus 30.2 per cent), although the difference was not statistically significant. What was scientifically significant, however, were major differences between the prayed-for group and the controls related to the severity of illness and the time it took to heal. Those being prayed for had a far shorter duration of fever and hospital stay and, in general, got better faster than the controls.

The subject of Leibovici's research – the healing effects of prayer – of course was hardly new. But his study offered one novel twist. The patients had been in the hospital between 1990 and 1996. The praying was carried out in 2000 – between 4 and 10 years later.

The study was meant to be a spoof.

The *British Medical Journal* had published it in its Christmas 2001 issue,[1](#) which is generally reserved for light-hearted commentary, next to a reindeer-shaped cluster of rogue cells. But Leibovici was not joking. He was trying to make a serious point in the most graphic way he could. Leibovici had a particular affinity for mathematics and statistics, and used them repeatedly in his reviews and meta-analyses when evaluating particular procedures. He had even come to believe that diseases and the success of treatment could be predicted through mathematical models.[2](#)

But the scientific method, in his view, was being defiled by its careless application to alternative medicine. Two years before, also in the Christmas issue of the *BMJ*, he had published an article claiming that alternative medicine masquerading as scientific medicine was like a cuckoo chick nestling in a reed warbler's nest.[3](#)

The begging noises of the interloper chick are indistinguishable from its warbler counterparts; indeed, as it grows, the cries of the cuckoo are so loud that they match the noise of eight little warblers. The warbler parents ignore any clues that they have an impostor in their midst and continue to nourish the cuckoo chick – to the detriment, even death, of their own offspring. Leibovici was convinced that alternative medicine

could not accommodate the demands of scientific rigour – and that we had no business wasting precious time and resources on the cuckoo in the nest.

But with that article, it seemed that Leibovici was the one wasting his time and breath. Most of his colleagues had missed the point so thoroughly that his only recourse was to *show* them. Two years later, almost to the day, his prayer study appeared in the *BMJ*.

He had intended that the study would illustrate that you simply cannot use the scientific method to explain subjective things like prayer. The problem was that every one had taken the study at face value. Dozens of sceptics derided the study. As one correspondent wrote, if it were possible to violate the arrow of time in this way, it would allow one to go back in time and prevent the Holocaust from happening by murdering Hitler.[4](#)

In support of Leibovici, many scientists interested in psychical research claimed that the study offered proof that prayer was effective at any point in time: Larry Dossey, who has also written extensively on ‘non-local’ consciousness and healing,[5](#) commented that, in a stroke, Leibovici had turned ‘conventional notions of time, space, prayer, consciousness and causality’ on their heads. [6](#)

Many others commented that Leibovici had been undone by the very meticulousness of his study design. Leibovici’s study had used only one supplicant to carry out the prayers and had sent the same prayer at the same time for each patient in the treatment group, so many of those in the alternative medicine camp did not believe the study suffered from some of the same problems in design as the other prayer research. To all the correspondents, Leibovici retorted in the *BMJ* letters section:

The purpose of the article was to ask the following question: Would you believe in a study that looks methodologically correct but tests something that is completely out of people’s frame (or model) of the physical world, for example, retroactive intervention or badly distilled water for asthma?[7](#)

It was wrong, he was saying, because it *had* to be wrong. It was statistics tied up in a knot and gone berserk. So that his motive would be clear, he added:

The article has nothing to do with religion. I believe that prayer is a real comfort and help to a believer. I do not believe it should be tested in controlled trials.

Instead, the true purpose was:

To deny from the beginning that empirical methods can be applied to questions that are completely outside the scientific model of the physical world. Or in a more formal way, if the pre-trial probability is infinitesimally low, the

results of the trial will not really change it, and the trial should not be performed.

Although he had intended to use science to prove the absurdity of alternative medicine, he had actually ended up proving to many people that we can pray today to affect something that occurred yesterday. Leibovici appeared to deeply regret his study and refused to discuss it further.[8](#)

Despite all his efforts throughout his career to apply reason and logic to medicine, this was the study that he would be most remembered for – a study that demonstrated, in effect, that we can go back and change the past.

One of the most basic assumptions about intention is that it operates according to a generally accepted sense of cause and effect: the cause must always precede the effect.

If A causes B, then A must have happened first.

This assumption reflects one of our deepest beliefs, that time is a one-way, forward-moving progression.

This assumption is reinforced every moment of our ordinary lives.

First we order our coffee, then the waitress delivers it to our table.

First we order a book on Amazon, then it arrives in the mail.

Indeed, the most tangible evidence of time's arrow is the physical evidence of our own ageing; first we are born, then we grow old and die.

Similarly, we believe that the consequence of our intentions can only occur in the future. What we do today cannot affect what happened yesterday.

However, a sizeable body of the scientific evidence about intention violates these basic assumptions about causation. Research has demonstrated clear instances of time-reversed effects, where effect precedes cause.

Leibovici's study was unique among prayer research in that it was conducted 'backward in time' – the healing intention was meant to affect events that had already occurred.

But to many frontier scientists, this experiment in 'retro-prayer' simply represented a true-to-life instance of the time-displacement effects regularly seen in the laboratory.

Indeed, some of the largest effects occur when intention is sent out of strict time sequence.

Studies like Leibovici's offer up the most challenging idea of all: that thoughts can affect other things no matter when the thought is made and, in fact, may work better when they are not subject to a conventional time sequence of causation.

Robert Jahn and Brenda Dunne at PEAR discovered this phenomenon when they investigated time displacement in their REG trials. In some 87,000 of these experiments, volunteers were asked to attempt to mentally influence the 'heads' and 'tails' random output of REGs in a specific direction anywhere from three days to two weeks *after* the machines had run.

As a whole, the 'time-displaced' experiments achieved even greater effects than the standard experiments.⁹ Jahn and Dunne had deemed these differences non-significant, only because the number of trials carried out in this manner was tiny compared with the rest of their monumental body of evidence.

Nevertheless, the very idea that intention could work equally well whether 'backward', 'forward' or in sequence, made Jahn realize that all of our conventional notions of time needed to be discarded.¹⁰

The fact that effects were even larger during the time-displaced studies suggested that thoughts have even greater power when their transmission transcends ordinary time and space.

Retro-causation has been explored in great detail by Dutch physicist Dick Bierman and his colleague Joop Houtkooper of the University of Amsterdam,¹¹ and later by Helmut Schmidt, an eccentric physicist at Lockheed Martin who created a elegant variation on time-displaced REG remote influence to determine whether someone's intention could change a machine's output after it had been run.

He rewired his REG to connect it to an audio device so that it would randomly set off a click that would be audiotaped and heard through a set of headphones by either the left or right ear.

He then turned on the machine and tape recorded their output, ensuring that no one, even himself, was listening.

After making copies of this master tape (again, with no one listening), he locked the master tape away, to eliminate the possibility of fraud, and gave medical students the copies a day later.

The volunteers were asked to listen to the tape and send an intention to have more clicks in their left ears.

Schmidt also created control tapes by running the audio device but not asking anyone to attempt to influence the left–right clicks. As expected, the right and left clicks of the controls were distributed more or less evenly.

Once the participants had finished their attempts to influence the tapes, Schmidt had his computer analyse both the student tapes and the master tape that had been hidden away to see if there was any deviation from the typical random pattern. In more than 20,000 trials carried out between 1971 and 1975, Schmidt discovered a significant result: on both the copies and the masters, 55 per cent had more left-hand than right-hand clicks. And both sets of tapes matched perfectly.

Schmidt believed he understood the mechanism for his improbable results. It wasn't that his participants had changed a tape after it had been created; their influence had reached 'back in time' and influenced the machine's output at the moment that it was first recorded.[12](#)

They had changed the output of the machine in the same way they might have if they had been present at the time it was being recorded. They did not *change* the past from what it was; they *influenced the past when it was unfolding as the present* so that it *became* what it was.

Schmidt continually refined the design of his 'retro-PK' studies over 20 years, eventually involving martial arts students, who are trained in mind-control.

In one study, he used a radioactive-decay counter to generate a visual display of random numbers. The students sat in front of this visual display, and attempted mentally to influence the numbers in a particular statistical distribution. Once again, he achieved a highly significant result, with odds against it being a chance occurrence of 1000 to somehow, the intention of the students had reached 'back in time' to affect what occurred in the first place.[13](#)

Time-displaced intention has also been successfully applied to living things. German parapsychologist Elmar Gruber, of the Institut für Grenzgebiete de Psychologie und Psychohygiene in Freiburg, carried out a series of ingenious experiments examining whether the movement of animals and humans can be influenced after the fact.

His first series of tests concerned gerbils running in activity wheels and moving about within a large cage. A special counter kept track of the number of revolutions in the activity wheel. A beam of light in the cage also had a recording device to note whenever the gerbil made contact with it. Similarly, he asked a group of human volunteers to walk around an area across which he had placed a photobeam, which was also attached to a recorder to note every instance that the volunteers ran into it.

Gruber then converted each revolution of the wheel or contact with the photobeam into a clicking sound. Tapes were made of the clicks, which were copied and stored, again to eliminate fraud.

Between one and six days later, volunteers were asked to listen to the tapes and attempt to mentally influence the gerbils to run faster than normal, or the people to run into the beam more often than usual.

Success would be measured by a greater number of clicks than usual.

Gruber carried out each type of trial 20 times, and in each instance, compared the volunteers' tapes with tapes made during sessions when the animals and humans were not subjected to the remote influence.

Four of the six batches of trials achieved significant results, and in three of these, the effect size was larger than 0.44.

An effect size is a statistical figure used in scientific research to demonstrate the size of change or outcome. It is arrived at by a number of factors, usually by comparing two groups, one of which has made the change.

An effect size under 0.3 is considered small, between 0.3 and 0.6 is medium, and anything above 0.6 is considered large. Aspirin, considered one of the most successful heart attack preventives of modern times, has an effect size of just 0.032, more than 10 times smaller than Gruber's overall effect size.

In the case of the activity-wheel gerbil trial, the effect size was a huge 0.7.[14](#) If his results had concerned a drug, Gruber would have discovered one of the greatest lifesavers of all time.

Gruber carried out six more intriguing experiments. In one study he recorded the number of times that people in a Viennese supermarket crossed a photobeam, and then recorded the number of times a photobeam was crossed by cars passing through various tunnels in Vienna during the rush hour.

These again were converted into clicks, and the tapes made of the clicks were stored for one to two months before being played to volunteers, who were asked to influence the speed of the people on foot or in the cars.

This time, he decided to include among his group of influencers some people with psychic ability. He also created similar tapes as controls, which were not exposed to remote intention.

Once again, when compared with sessions that were not subjected to influence, the results were highly significant; all but one of the automobile–tunnel studies had a significant effect size; in two of the studies, the effect sizes (0.52 and 0.74) were enormous.[15](#)

Is it possible to retroactively prevent a disease, after it has infected its host and spread? The Chiron Foundation in the Netherlands designed an intriguing study to test this seemingly impossible proposition.

A large group of rats was randomly divided in two groups, and one group given a parasitic infection of the blood.

The experiment was blinded so that the experimenters themselves did not know which animals were infected and which were controls until after the study was completed. A healer given photographs of the rats after they had been infected with the disease was asked to attempt to prevent the spread of the parasites.

Measurements of the blood cells were taken at several intervals after the animals had been infected. The study was carried out three times, each involving a large number of rats. Two achieved a medium (0.47) effect size.[16](#)

Psychologist William Braud then asked one of the most provocative questions of all: is it possible to ‘edit’ one’s own emotional response to an event? To test this, he designed a batch of studies to test time-displaced influence on nervous activity.

He recorded several tracings of the electrodermal activity (EDA) of volunteers, using standard liedetection equipment – a reasonable gauge of whether a person is calm or agitated. Braud then asked the participants to examine one of their own tracings and to attempt to influence it, by sending an intention either to calm down or activate their own sympathetic nervous system at that earlier point in time.

The other tracings of the participants, which were not exposed to mental influence, were to act as controls. Later, when he compared the tracings with controls, he discovered that those tracings that were exposed to the volunteers’ own retro-influence were calmer than the controls.

Overall, these studies achieved a small, significant effect size (0.37), offering some of the first evidence that human beings might be able to rewrite their own emotional history.[17](#)

Helmut Schmidt successfully employed a similar study design to change his own prerecorded breathing rate, demonstrating that it is possible to retroactively change your own physical state as well.[18](#)

Dean Radin set up an EDA test similar to Braud's, but added remote distance to a test of retroactive influence.

Two months after running the tests, Radin sent copies of the electrodermal readouts to healers in Brazil and asked them to attempt to quiet the readings. After 21 such studies Radin achieved a 0.47 effect size, similar to Braud's.[19](#)

Radin also tested the possibility that, under certain conditions, a future event can influence an earlier nervous-system response.

He made ingenious use of a strange psychological phenomenon called the 'Stroop effect', named after its discoverer, psychologist John Ridley Stroop,[20](#) originator of a landmark test in cognitive psychology.

The Stroop test uses a list of the names of colours (e.g. 'green') printed in different coloured inks. Stroop found that when people are asked to read out the name of a colour as quickly as possible, they take much longer if the name of the colour does not match the colour of the ink used (e.g. if the word 'green' is printed in red ink) than they do if the name and the colour of the ink match (e.g. if the word 'green' is printed in green ink).

Psychologists believe that this phenomenon has to do with the difference in the time it takes the brain to process an image (the colour itself), compared with the time it takes to process a word (the colour name).

Swedish psychologist Holger Klintman devised a variation on the Stroop test. Volunteers were asked first to identify the colour of a rectangle as quickly as they could, then asked whether a colour name matched the colour patch they had just been shown.

A large variation occurred in the time it took his volunteers to identify the colour of the rectangle. Klintman discovered that the identification of the rectangle colour was faster when it matched the colour name shown subsequently.[21](#)

The time it took for people to identify the colour of the rectangle seemed to depend on the second task of determining whether the word matched the rectangle colour. Klintman called his effect 'time-reversed interference'. In other words, the later effect influenced the brain's reaction to the first stimulus.

Radin created a modern version of Klintman's study. His participants sat in front of a computer screen and identified the colours of rectangles that flashed up on the screen as quickly as possible by typing in their first letter.

The image on the screen would then be replaced by the name of a colour, and the volunteer would then have to type either 'y' (yes) to indicate that the name of the colour matched the colour of the rectangle or 'n' (no) to indicate a mismatch. Radin varied the second part of the design, so that, after the participant had identified the colour of the rectangle, he or she would also have to type in the first letter of the actual colour of the letters of the colour's name.

For instance, if the word 'green' flashed up but was coloured blue, he or she would have to type in 'b'.

In four studies of more than 5000 trials, all four showed a retrocausal effect. A significant correlation was observed in two of the studies, with a third marginally significant.[22](#) *Somehow, the time it took to carry out the second task was affecting the time it took to carry out the first one.*

Radin concluded that his studies offered evidence of a time displacement in the nervous system. The implications are enormous. Our thoughts about something can affect our past reaction times.

One scientifically accepted way to examine the overall power of an effect is to pool the results of all the studies together into what is called a 'meta-analysis'. Analysed in this manner, 19 of the retroinfluence studies yielded an extraordinary collective result.[23](#)

William Braud calculated that the overall effect size was 0.32. Although that is considered a small effect on its own, it represents ten times the effect size for most prescription drugs, such as the beta-blocker propranolol, that are recognized as extremely effective.

A different type of analysis of all the best studies of time displacement was carried out in 1996 by Dick Bierman. In statistics, the best way to judge an effect is to work out how much it deviates from the mean, or average.

One method popular with statisticians is to work out the chi-square distribution, which entails plotting the square of each individual score. Any deviation from chance, whether positive or negative, will show up as a large positive deviation in bold relief.

Bierman detected an enormous variance in individual studies, but collectively they produced results whose occurrence by chance alone was an extraordinary 630 billion to one.[24](#)

One interpretation of the laboratory evidence of retro-influence suggests the unthinkable: intention is capable of reaching back down the time line to influence past

events, or emotional or physical responses, at the point when they originally occurred.

The central problem of going 'back to the future' and manipulating our own past are the logical knots the mind gets tied up in when considering them. As British philosopher Max Black argued in 1956, if A causes B, but occurs after B, B often precludes A. Therefore, A cannot cause B.

This conundrum was overlooked in the movie *The Terminator*.

If the Schwarzenegger cyborg goes back in time and kills Sarah Connor so that she cannot give birth to future rebel John Connor, there would be no future revolution between man and machine.

The Terminator no longer has any need to come back in time or, indeed, no longer any purpose for being created.

British philosopher David Wiggins constructed a similar scenario to illustrate the logical problems inherent in the idea of a time machine. Suppose a young man is the grandson of the cruel leader of a fascist movement. He decides to travel back in time to kill his grandfather, to prevent him from taking control. But if he does so, the young man's mother may not be born and he of course would cease to exist.

Nevertheless, physicists no longer consider retro-causation inconsistent with the laws of the universe. More than 100 articles in the scientific literature propose ways in which laws of physics can account for time displacement.[25](#)

Several scientists have proposed that scalar waves, secondary waves in the Zero Point Field, enable people to engineer changes in space-time. These secondary fields, caused by the motion of subatomic particles interacting with the Zero Point Field, are ripples in space-time – waves that can travel faster than the speed of light.

Scalar Field waves possess astonishing power: a single unit of energy produced by a laser in such a state would represent a larger output than all the world's power plants combined.[26](#)

Certain technologies, such as quantum optics, have made use of laser pulses to squeeze the Zero Point Field to such a degree that it creates negative energy.[27](#)

It is well accepted in physics that this negative energy, or exotic matter, is able to bend space-time. Many theoreticians believe that negative energy would allow us to travel through wormholes, travel at warp speed, build time machines and even help human beings to levitate.

When electrons are packed densely together, the density of the spray of virtual particles that are constantly created in the Zero Point Field is increased. These spray densities are organized into electromagnetic waves that flow in two directions, and so may be going 'back and forward' in time.[28](#)

Physicist Evan Harris Walker first proposed that retro-influence can be explained by quantum physics if we just take account of the observer effect.[29](#)

Walker and later Henry Stapp, an elementary particle physicist at the University of California at Berkeley, who acted as an independent monitor of Helmut Schmidt's final martial arts study, believed that a small tweak in quantum theory, making use of 'nonlinear quantum theory', could explain all cases of retro-influence. In a linear system such as current quantum mechanics, the behaviour of a system can be easily described: $2 + 2 = 4$. The system's behaviour is the sum of its parts. In a non-linear system, $2 + 2$ may equal 5 or even 8. The system's behaviour is more than a sum of its parts – by how much more we can't often predict.

In Walker's and then Stapp's view, turning quantum theory into a non-linear system would enable them to include one other element in the equation: the human mind. In Schmidt's martial arts study, the numbers on the visual display remained in their 'potential' state of all possible sets of numbers until they had been observed by the students. At that point, the mental intent of the students and the numbers on the display interacted in a quantum way.

According to Stapp, the physical universe exists as a set of 'tendencies' with 'statistical links' between mental events.

Even though the tape of the numbers has been generated, they divide into a number of channels of all possible outcomes. When a person looks at the numbers, his brain state will also divide into the same number of channels. His intent will select out a particular channel, and through the numbers 'collapse' the channels into a single state.[30](#) Human will – our intention – creates the reality, no matter when.

The other possibility is that all information in the universe is available to us at every moment, and time exists as one giant smeared-out present. Braud has speculated that forebodings of the future might be an act of backward time displacement – a future event somehow reaching back in time to influence a present mind. If you simply reversed presentiment and call it backward influence, so that all future mental activity influences the present, you maintain the same model and results as the retro-causation studies.

All precognition might be evidence of backward- acting influence;[31](#) all future decisions may always influence the past.

There is also the possibility that at the most fundamental layer of our existence there is no such thing as sequential time.

Pure energy as it exists at the quantum level does not have time or space, but exists as a vast continuum of fluctuating charge. We, in a sense, are time and space. When we bring energy to conscious awareness through the act of perception, we create separate objects that exist in space through a measured continuum. By creating time and space, we create our own separateness and indeed our own time.

According to Bierman, what appears to be retro-causation is simply evidence that the present is contingent upon future potential conditions or outcomes, and that non-locality occurs through time as well as space. In a sense, our future actions, choices and possibilities all help to create our present as it unfolds. According to the view, we are constantly being influenced in our present actions and decisions by our future selves.

This explanation was bolstered by a simple thought experiment carried out by Vlatko Vedral and one of his colleagues at the University of Vienna: Caslav Brukner, a Serb who had managed to leave Yugoslavia during the civil war and, like Vedral, spent time at Zeilinger's Viennese lab.

When Brukner joined Vedral in London during a year-long fellowship at Imperial College, he began thinking about quantum computation, and the fact that it is billions of times faster than classical computing. Once a quantum computer is finally perfected it will enable one to scan every last corner of the Internet in half an hour.[32](#)

Could this enormous advantage in speed have some basis in Bell's inequality, the famous test of non-locality? Bell demonstrated that the remote influence maintained between two quantum subatomic particles, even over vast distances, 'violates' our Newtonian view of separation in space.

Could this same test be used to show when temporal constraints – the limits governing time – are also violated? Brukner enlisted Vedral to design a thought experiment with him.

Their experiment rested on a given in science about time: in the evolution of a particle, a measurement taken at a certain point will be utterly independent of a measurement taken later or earlier. In this instance, the 'inequality' of Bell's would refer to the difference between the two measurements when taken at different times.

For their experiment, they no longer needed two particles, and so could utterly eliminate the 'Bob' particle and concentrate on the photon, 'Alice'. The task now was to make theoretical calculations of Alice's polarization at two points of time. If

quantum waves behave like a wriggling skipping rope being shaken at one end, the direction in which the rope is pointed is called polarization. To work out their time sequences mathematically, Brukner and Vedral made use of what is called 'Hilbert', or abstract, space.

First they calculated Alice's polarization, then they measured it moments later. When they had finished their calculations of Alice's current position, they went back and measured her earlier polarization again. They discovered that, between two points of time, Bell's inequality indeed had been violated; they got a different measurement of the first polarization the second time around. The very act of measuring Alice at a later time influenced and indeed *changed* how it was polarized earlier.

The implications of their astonishing discovery were not lost on the scientific community. *New Scientist* included their discoveries in a dramatic cover story: 'Quantum entanglement: How the future can influence the past' and concluded: quantum mechanics seems to be bending the laws of cause and effect ... entanglement in time puts space and time on an equal footing in quantum theory ... Brukner's result suggests that we might be missing something important in our understanding of how the world works.[33](#)

For me, Brukner's thought experiment held a significance far greater than a simple theoretical one. It showed that instantaneous cause and effect not only occurs through space but also back and possibly forward through time. It offered the first mathematical proof that the actions of every moment influenced and *changed* those of our past. It may well be that every action we take, every thought we have in the present, alters our entire history.

Even more significantly, his experiment demonstrated the central role of the observer in creating, and indeed changing, reality. Observing had played an integral part in changing the state of the photon's polarization.

The very act of measuring an entity at one point of time changed its earlier state. This may mean that every observation of ours changes some earlier state of the physical universe. A deliberate thought to change something in our present could also influence our past. The very act of intention, of making a change in the present, may also affect everything that has led to that moment.

This sort of backward influence resembles the non-local correlations found in the quantum world, as if the connections were always there in some underlying arrangement.[34](#) It may be that our future already exists in some nebulous state that we actualize in the present.

This makes sense since subatomic particles exist in a state of potential until observed or thought about. If consciousness operates at the quantum frequency level, it would naturally reside outside space and time, and we would theoretically have access to information – ‘past’ and ‘future’. If humans are able to influence quantum events, they are also able to affect events or moments other than in the present.

Radin discovered more evidence that our psychokinetic influence is operating ‘backwards’ in an ingenious study examining the possible underlying mechanism of intention on the random bits of an REG machine. Radin first ran five REG studies involving thousands of trials, then analysed the experiments through a process called a ‘Markov chain’, which allowed a mathematical analysis of how the REG machine’s output changed over time.

For this process, he made use of three different models of intention: first, as a forward-time casual influence (the mind ‘pushes’ the REG in one direction throughout the influence); second, as a precognitive influence (the mind intuitively hits the REG in its random fluctuations to produce the intended result by ‘looking into the future’ and passively ‘bringing back’ this information to the present); and third, as a true retrocausal influence (the mind first sets the future outcome and applies all the chain of events that will produce it ‘backward’ in time).

Radin’s analysis of the data had one inescapable conclusion: this was not a process running forward in time, in an attempt to hit a particular target, so much as an ‘information’ flow that had travelled back in time.[35](#)

But just how much of the past could we change in the sticks-and-stones world of real life? William Braud had pondered this issue at length. He once observed that those moments in the past most open to change might be ‘seed’ moments when nature has not made up its mind – perhaps the earliest stages of events before they blossomed and grew into something static and unchangeable.[36](#)

These moments were analogous to a sapling that could still be bent and trained before its trunk was too stiff and branches too large; the brain of a child, which is far more open to influence and learning than an adult’s; or even a virus, which is far easier to overcome in its infancy.[37](#)

Random events, decisions with equally likely choices, or illness – all probabilistic moments disposed to early influence where human intention could slightly shift the outcome in a certain direction – might comprise the events in our lives most open to retro-influence. Braud referred to them as ‘open’, or labile, systems – those most open to change.

These systems include many of the workings of living things, which are random processes, much like the quantum systems of random-event generators. Any one of a number of the biological processes in living things requires a cascade of processes, which would be sensitive to the kind of subtle effects on REG machinery observed say, in the PEAR research.[38](#)

In Braud's earlier work, he had discovered that remote influence had its greatest effect when there was a strong need for it.[39](#) The necessity of a particular outcome might be the one quality that moves mountains backward in time.

A clue to the extent of our reach was revealed in Schmidt's discovery of an observer effect in his audio REG experiments that is much like the effect in quantum experiments: it was most important that the person attempting to influence his tapes be the very first listener.

If anyone else heard the tape first and listened with focused attention, it was less susceptible to influence later. A few studies even suggest that observation by any sentient being – human or animal – blocks future attempts at time-displaced influence.

Bierman tested this by rigging up a radioactive source to trigger beeps that were delayed for one second and then observed by a final observer. In about half of the events, another pre-observer was given feedback of this quantum event before the final observer witnessed it. In those instances, the pre-observer's observation resulted in a collapse of the superposition state of the quantum event while, in the other half of cases, the final observer 'produced' the collapse.[40](#)

If this consciousness is the crucial ingredient for 'collapse' to occur, humans – and their ability to 'reduce' reality to limited states – are completely responsible for the idea that time is an arrow in one direction. If our future choice of a particular state is what affects its present 'collapse', the reality may be that our future and present are constantly meeting up with each other.

This accords with what is understood about the observer effect in quantum theory – that the first observation of a quantum entity 'decoheres', or collapses, its pure state of potential into a single state.[41](#) This rather suggests that, if no one had ever seen Hitler, we might have been able to send an intention to prevent the Holocaust.

Although our understanding of the mechanism is still primitive, the experimental evidence of time reversal is fairly robust. This research portrays life as one giant, smeared-out here and now, and much of it – past, present and future – open to our influence at any moment.

But that hints at the most unsettling idea of all. Once constructed, a thought is lit forever.

Note – Chapter 11: Praying for Yesterday

1. L. Leibovici, 'Effects of remote, retroactive intercessory prayer on outcomes in patients with blood stream infection: Randomized controlled trial', *British Medical Journal*, 2001; 323 (7327): 1450–1.
2. S. Andreassen et al., 'Using probabilistic and decision-theoretic methods in treatment and prognosis modeling', *Artificial Intelligence in Medicine*, 1999; 15 (2): 121–34.
3. L. Leibovici, 'Alternative (complementary) medicine: a cuckoo in the nest of empiricist reed warblers', *British Medical Journal*, 1999; 319: 1629–32; Leibovici, 'Effects of remote, retroactive intercessory prayer', op. cit.
4. Letters, BMJ Online, December 22, 2003.
5. L. Dossey, 'How healing happens: exploring the nonlocal gap', *Alternative Therapies in Health and Medicine*, 2002; 8 (2): 12–16, 103–10.
6. B. Oshansky and L. Dossey, 'Retroactive prayer: A preposterous hypothesis?' *British Medical Journal*, 2003; 327: 20–7.
7. Letters, 'Effect of retroactive prayer', *British Medical Journal*, 2002; 324: 1037.
8. Correspondence from Leibovici to author, June 28, 2005.
9. Interview with Jahn and Dunne, July 2005.
10. R. G. Jahn et al., 'Correlations of random binary sequences with pre-stated operator intention: a review of a 12-year program', *Journal of Scientific Exploration*, 1997; 11 (3): 345–67.
11. D. J. Bierman and J. M. Houtkooper, 'Exploratory PK tests with programmable high speed random number generator', *European Journal of Parapsychology*, 1975; 1 (1): 3–14.
12. R. Broughton, *Parapsychology: The Controversial Science*, New York: Ballantine Books, 1991: 175–6.
13. H. Schmidt and H. Stapp, 'Study of PK with prerecorded random events and the effects of preobservation', *Journal of Parapsychology*, 1993; 57: 351.
14. E. R. Gruber, 'Conformance behavior involving animal and human subjects', *European Journal of Parapsychology*, 1979; 3 (1): 36–50.
15. E. R. Gruber, 'PK effects on pre-recorded group behaviour of living systems', *European Journal of Parapsychology*, 1980; 3 (2): 167–75.
16. F. W. J. J. Snel and P. C. van der Sijde, 'The effect of retro-active distance healing on *Babesia rodhania* (rodent malaria) in rats', *European Journal of Parapsychology*, 1990; 8: 123–30.

17. W. Braud, unpublished study, 1993, as reported in W. Braud, 'Wellness implications of retroactive intentional influence: exploring an outrageous hypothesis', *Alternatives Therapies in Health and Medicine*, 2000; 6 (1): 37–48.
18. H. Schmidt, 'Random generators and living systems as targets in retro-PK experiments', *Journal of the American Society for Psychical Research*, 1997; 912 (1): 1–13. D. Radin et al., 'Effects of distant healing intention through time and space: Two exploratory studies', *Proceedings of Presented Papers: The 41st Annual Convention of the Parapsychological Association*, Halifax, Nova Scotia, Canada: Parapsychological Association, 1998: 143–61.
19. J. R. Stroop, 'Studies of interference in serial verbal reactions', *Journal of Experimental Psychology*, 1935; 18: 643, as cited in D. I. Radin and E. C May, 'Evidence for a retrocausal effect in the human nervous system', Boundary Institute Technical Report 2000–1.
20. H. Klintman, 'Is there a paranormal (precognitive) influence in certain types of perceptual sequences? Part I and II', *European Journal of Parapsychology*, 1983; 5: 19–49 and 1984; 5: 125–40, as cited in Radin and May, Boundary Institute Technical Report, op. cit.
21. Radin and May, Boundary Institute Technical Report, op. cit.
22. Braud, 'Wellness implications', op. cit.
23. See <http://www.fourmilab.ch/rpkip/bierman-metaanalysis.html>.
24. Radin and May, Boundary Institute Technical Report, op. cit.
25. G. A. Mourou and D. Umstadter, 'Extreme light', in 'The Edge of Physics' Special edition of *Scientific American*, 2003; 13 (1): 77–83 updated from May 2002 issue.
26. L. H. Ford and T. A. Roman, 'Negative energy, wormholes and warp drive' in 'The Edge of Physics'. Special edition of *Scientific American*, 2003; 13 (1): 85–91 updated from January 2000 issue.
27. J. A. Wheeler and R. P. Reynman, 'Interaction with the absorber as the mechanism of radiation', *Reviews of Modern Physics*, 1945; 17 (2–3): 157–81; J. A. Wheeler and R. P. Reynman, 'Classical electrodynamics in terms of direct interparticle action', *Reviews of Modern Physics*, 1949; 21: 425–33.
28. E. H. Walker, 'The nature of consciousness', *Mathematical BioSciences*, 1970; 7: 131–78.
29. H. P. Stapp, 'Theoretical model of a purported empirical violation of the predictions of quantum theory', *Physical Review A*, 1994; 50 (1): 18–22.
30. Braud, 'Wellness implications', op. cit.
31. L. Grover, 'Quantum computing', *The Sciences*, July/August 1999: 24–30.
32. M. Brooks, 'The weirdest link', *New Scientist*, March 27, 2004; 181(2440): 32–5.
33. D. Bierman, 'Do PSI-phenomena suggest radical dualism?' in S. Hammerof et al. (ed.), *Toward a Science of Consciousness II*, Cambridge, Mass.: MIT Press, 1998: 709–14.

34. D. I. Radin, 'Experiments testing models of mind-matter interaction' *Journal of Scientific Exploration*, 2006; 20 (3), 375–401.
35. Interview with William Braud, October 1999.
36. W. Braud, 'Transcending the limits of time', *The Inner Edge: A Resource for Enlightened Business Practice*, 1999; 2 (6): 16–18.
37. R. D. Nelson, 'The physical basis of intentional healing systems', Technical Report, PEAR 99001, Princeton Engineering Anomalies Research, Princeton New Jersey, January 1999.
38. Braud, interview with author, October 1999.
39. D. Bierman 'Does consciousness collapse the wave packet?' *Mind and Matter*, 2003; 1 (1): 45–58.
40. H Schmidt, 'Additional effect for PK on pre-recorded targets', *Journal of Parapsychology*, 1985; 49: 229–44; 'PK tests with and without preobservation by animals', in L. S. Henkel and J. Palmer (eds.), *Research in Parapsychology*, Metuchen, NJ: Scarecrow Press, 1990: 15–19.

CHAPTER TWELVE

The Intention Experiment

SEEING *ACETABULARIA* FOR THE FIRST TIME takes your breath away. The mesmerizing appearance of this common algae of the Caribbean and the Mediterranean has earned a number of poetic nicknames – 'mermaid's wineglass', or 'sombbrero' in Spanish – and both are fitting. Its slender stem supports a tiny cupped sombrero, like a miniature green umbrella ready to be popped into an underwater tropical cocktail.

For more than 70 years, biology students have marvelled over this tiny plant, not simply for its appearance but for a single bizarre fact of its existence. *Acetabularia* is a freak of nature. From stem to sombrero, the entire plant, measuring up to 5 centimetres, consists of a single cell. Because of this, *Acetabularia*, unlike most living things, can be counted on to behave predictably.

The large nucleus of the cell always sits at the rhizoid, the base of the stalk, and divides only when the plant has reached its full height. This uncomplicated structure has helped to unmask biology's greatest mystery: which portion of the plant engineers its ability to reproduce. In the 1930s, the German scientist Joachim Hammerling elected *Acetabularia* as his perfect 'tool organism' to work out the role of a nucleus in plant genetics.

The simplicity of this single-celled organism with its single giant nucleus not only offered up the secrets of the cell in bold relief, it divulged the whole of the building

plans of plant life. Working with *Acetabularia* allowed one to sit in stunned witness to the complex morphology of life within the totality of a single cell, large enough to be visible to the naked eye.

Acetabularia also represented a model organism for my first intention experiment. Fritz Popp, who was to perform the experiment with me, believed that if we were going to attempt to carry out my proposal, we needed to begin on the ground floor. For this first experiment, I planned to assemble a small group of volunteers in London, and ask them to use their intention to affect an organism in Popp's lab in Germany.

Using *Acetabularia* for our test subject would be analogous to testing a car made of a single moving part. It removes all the variables of a living thing, with its unfathomable number of chemical and energetic processes occurring at every instant.

Humans, for instance, are like a manufacturing plant covering most of the United States. A septillion chemical reactions occur every second in every tablespoon of our cells, tiny explosions that get multiplied by the 50 million million cells of the average human body. In an experiment comparing, say, the growth rates of two sections of the body, it is almost impossible to control for every variable. Growth rates can be altered by food, water, genetics, mood, or even a sudden dip in air temperature.

During our first intention experiment, Popp intended to examine the alteration in the tiny light being emitted from the algae, which was infinitely more subtle than cellular growth rate. Nonetheless, in multicellular living things, even the light that emanates from each cell is subject to a host of influences: the health of the host, the weather and even the activity of the sun.¹ Light can also differ from cell to cell.

With *Acetabularia*, as the light reassuringly derives from its single nucleus, so it is subject to far less fluctuation. With such a primitive organism, Popp explained, it would be possible to demonstrate, with a fair degree of certainty, that any effect, for better or worse, was entirely the result of our remote influence. Only by using such a simple system could we show that our effect was indisputably due to intention and not a dozen other possibilities.

Generally speaking, an increase of photons indicates that a life form is being stressed and a decrease, that its health has improved. If I sent an intention to make the algae healthier, and the photon count went down, it would likely mean that I was having a good effect. If the photon count went up, it was probable that I was, in some way, harming it.

Popp has a number of extremely sensitive photocount detectors at his disposal, which can register an intensity of visible light of about 10–17 watts per square

centimetre, analogous to the light coming from a candle several kilometres away.² This type of ultrasensitive equipment would enable us to record every single hair's breath of difference – even by a single photon – and so determine the extent of our influence.

Popp had reason to be cautious. For 30 years he had faced enormous opposition to his bold assertion that light emanates from living things,³ and had finally won respect from the physics community.

He had set up his international community of likeminded scientists from prestigious centres all over the globe to work on biophoton emissions.⁴ By participating in our experiment, he might risk this hard-won reputation and good will. After all, ultimately I was asking this world-renowned physicist to test whether collective positive thinking could change the physical world.

* * *

The results of a number of experiments had suggested that a 'group' consciousness might possibly exist. In their random-event generator experiments, PEAR's Jahn and Dunne found that the influence of pairs of the opposite sex who knew each other had a powerful complementary effect on the machines – roughly three and a half times that of individuals. Two intensively involved people appeared to create six times the 'order' on a random machine. Some couples even produced a 'signature' result, which did not resemble the effects they generated individually.⁵

There was also evidence that a group all intently focused on the same thought registered as a large effect on a REG machine. Roger Nelson, the chief coordinator of the PEAR lab, had come up with the idea of running REG machines continuously during a particularly engaging event, to examine whether the focused attention of a group had any effect on the random output of the machines.

He and Dean Radin developed what they termed 'FieldREG' devices and ran them during a host of events involving the highly focused attention of an audience: intense or euphoric group workshops; religious group rituals; Wagnerian festivals; theatrical presentations; even the Academy Awards. In most instances, their studies showed that multiple minds holding the same intensely felt thought created some kind of deviation from the norm on the equipment.⁶

Nelson had been fascinated by the possibility of a global collective consciousness. In 1997, he decided to place REGs all over the world, have them run continuously and compare their output with moments of global events with the greatest emotional impact. For his programme, which became known as the Global Consciousness Project, Nelson organized a centralized computer program, so that REGs located in

50 places around the globe could pour their continuous stream of random bits of data into one vast central hub through the Internet.

Periodically, Nelson and his colleagues, including Dean Radin, studied these outpourings and compared them with the biggest breaking news stories, attempting to root out any sort of statistical connection. Standardized methods and analysis revealed any demonstration of order – a moment when the machine output displayed less randomness than usual – and whether the time that it had been generated corresponded with that of a major world event.

By 2006, they had studied 205 top news events, including the death of the Princess of Wales, the millennium celebrations, the death of John F. Kennedy, Jr, and his wife, and the attempted Clinton impeachment. When Nelson analysed four years' worth of data, a pattern emerged. When people reacted with great joy or horror to a major event, the machines seemed to react as well. Furthermore, the degree of 'order' in the machine's output seemed to match the emotional intensity of the event, particularly those that had been tragic: the greater the horror, the greater the order.[7](#)

This trend appeared most notable during the events of 9/11. After the twin towers were destroyed, Nelson, Radin and several colleagues studied the data that had poured in from 37 REGs around the world. Individual statistical analyses were performed by Radin, Nelson, computer scientist Richard Shoup of Boundary Institute and Bryan J. Williams, a psychology undergraduate at the University of New Mexico. According to the results of all four analyses, the effect on the machines during the plane crashes was unprecedented.

Out of any moment in 2001, the greatest variance in the machines away from randomness took place that day. The results also represented the largest daily average correlation in output between each machine than at any other time in the history of the project.[8](#) According to the REGs, the world's mind had reacted with a coherent global horror.

Nelson and three independent analysts took apart the data using a variety of statistical methods. Nelson examined his results through the chi-square distribution method, that statistical technique which plots the square of each of the machine's runs, so that any deviation from chance easily shows up. All of the analysts concluded that an enormous increase in 'order' occurred during time frames relating to key moments in the drama (such as, shortly before the first tower was struck), which were likely to be the most intense periods of horror and disbelief.[9](#) As REGs are designed to control for electrical disturbances, natural electromagnetic fields or increased levels of mobile phone use, the two scientists were able to discard all those possibilities as potential causes.[10](#)

Furthermore, although activity of the REGs was normal in the days leading up to 9/11, the machines became increasingly correlated a few hours *before* the first tower was hit, as though there had been a mass premonition. This similarity in output continued for two days after the first strike. Williams thought of it as kind of psychic signature, a giant unconscious psychokinetic effect created by 6 billion minds set to react in unified horror.[11](#)

The world had felt a collective shudder several hours before the first plane crash, and every REG machine had heard and duly recorded it.

Although not every analyst agreed with these conclusions,[12](#) Nelson, Radin and several of their colleagues eventually were able to publish a summary of their findings in the prestigious physics journal *Foundations of Physics Letters*.[13](#)

Nelson went on to study other events in the wake of 9/11, including the start of the Iraqi war. He compared REG activity with variations in the approval polls of President George W. Bush, to see if he could discover a connection of any kind between the global 'mind' and current American opinions of the president, and whether the REG network reacted most when there were strong feelings of unity and purpose, as the Americans had shared in the wake of 9/11, or when the public mood was polarized, as it had been after the invasion of Iraq and the deposing of Saddam Hussein's regime.

After examining 556 separate polls between 1998 and 2004, Nelson's colleague, Peter Bancel, discovered that peaks in variations followed big public changes of opinion of any variety, either for or against the president. Strong emotion, positive or negative – even to presidential decisions – seemed to produce order.

The results of the FieldREG work and the Global Consciousness Project offer several important clues about the nature of group intention. A group mind appears to have a psychokinetic effect on any random microphysical process, even when not focused on the machinery itself.

The energy from a collective, intensely felt thought appears to be infectious. There also appears to be a 'dose' effect; the effect on an REG of a load of people thinking the same thought is larger than the effect of a single person. Finally, emotional content or degree of focus is important. The thought has to engulf a group of people in a moment of peak attention, so that every member of the group is thinking the same thought at the same time. A catastrophe is certainly an effective way to snap the mind to attention.

The data from the Global Consciousness Project had one serious limitation. However accurately Nelson had taken the temperature of the world mind, his data simply

referred to the effect of mass *attention*.

There had been no *intention* to cause change. What would happen if a number of people were not simply attending to something but also trying to affect it in some way? If the focused attention of a group has a physical impact on sensitive equipment, does the signal get stronger when the group is actually trying to change something?

The only systematic study of group intention concerns the so-called Maharishi Effect of Transcendental Meditation™ (TM), the technique first introduced by Maharishi Mahesh Yogi to the West in the 1960s. Over several decades, the TM organization has carried out more than 500 studies of group meditation, with or without intention, to examine whether meditation has a resonance effect on reducing conflict and suffering.

Maharishi Mahesh Yogi postulated that regularly practising TM enabled you to get in touch with a quantum energy field that connects all things. When a group of meditators was large enough, he claimed, their collective meditations caused 'Super Radiance', a term in physics used to describe the coherence of laser light.

During TM, the theory went, the minds of meditators all become tuned to the same frequency, and this coherent frequency begins to order the disordered frequencies around it. Resolution of individual internal conflict leads to resolution of global conflict.

The TM studies claimed to demonstrate effects from two types of meditation. The first was undirected, the simple consequence of a certain percentage of the population meditating. The second resulted from deliberate intention, and required experience and focus; advanced meditators would target a particular area and direct their meditation to help resolve conflict and lower the rate of violence.

The Maharishi's theory rests entirely on the premise that meditation has a threshold effect. If 1 per cent of the population of a particular area practises TM, he claims, or the square root of 1 per cent of the population practises TM-Sidhi, a more advanced type of meditation, conflict of any variety – rates of murders, crime, drug abuse, even traffic accidents – goes down.

Some 22 studies have tested the positive impact of the Maharishi Effect on crime levels. One study of 24 US cities showed that whenever a city reached a point where 1 per cent of the population was carrying out regular TM, the crime rate dropped to 24 per cent. In a follow-up study of 48 cities, those 24 cities with the requisite threshold percentages of meditators (1 per cent of the population) experienced a 22 per cent decrease in crime, and an 89 per cent reduction in the crime trend. In the other 24

cities without the threshold percentage of meditators, crime increased by 2 per cent and the crime trend by 53 per cent.[14](#)

In 1993, the TM's National Demonstration Project focused on Washington DC during a large upsurge of local violent crime in the first five months of the year. Whenever the local Super Radiance group reached the threshold number of 4000, the rate of violent crime fell and continued to fall, until the end of the experiment. The study was able to demonstrate that the effect had not been due to any other factors, such as police efforts or a special anti-crime campaign. After the group disbanded, the crime rate in the capital rose again.[15](#)

The TM organization has also targeted global conflict. In 1983 a special TM assembly met in Israel to send intentions through meditation to resolve the Palestinian conflict. During their sessions, they made daily comparisons between the number of meditators working on the project and the state of Arab–Israeli relations. On days with a high number of meditators, fatalities in Lebanon fell by 76 per cent. Their reach apparently extended beyond armed conflict; ordinary violence – local crime, traffic accidents and fires – also all decreased. When analysing their results, the TM group claimed to have controlled for confounding influences such as weather.[16](#)

TM adepts have also sought to influence the 'misery index' – the sum of inflation and unemployment rates – in the USA and Canada. And indeed, during one concerted effort between 1979 and 1988, the US index fell by 40 per cent and the Canadian index, by 30 per cent.

Another group of adepts sought to influence the monetary growth and crude-materials price indices as well as the American misery index. In this instance, the misery index fell by 36 per cent, and the crude-materials price index fell by 13 per cent. Although the growth rate of the monetary base was affected, it was only by a small margin.[17](#)

Critics of TM have argued that these effects could easily have been due to other factors – a reduction in the population of young men, say, or better educational programmes in these areas, or even the ebb and flow of the economy – although the TM organization claims to control for such changes.

The problem with these studies, to my mind, is the controversy surrounding the TM organization itself; rumours now abound about data fixing and the infiltration by Maharishi followers into many scientific organizations.

Nevertheless, the TM evidence is so abundant and the studies so thorough that it is difficult to dismiss them completely. Furthermore, the studies are regularly published in peer-reviewed scientific journals, and so must meet some level of scientific rigour

and critical scrutiny. The sheer bulk of the research argues compellingly that a force outside the understanding of orthodox science might be at work.

But even if the results are legitimate, the TM studies, like the REG data, mostly concerns group *attention*. In many instances, the meditators are not people who maintain a focused *intention* to change something else.

For three months in the first quarter of 1998, forest fires raged out of control in the Amazonian state of Roraima, 1500 miles northwest of Brasilia, devastating the rainforest. It had not rained for months – an effect blamed on El Niño – and the ordinarily humid rainforest was bone dry, perfect kindling for the fire that had by that time scorched 15 per cent of the state.

The rains, usually so copious in this part of Brazil, remained elusive. The UN termed the fire a disaster without precedent on the planet. Water-carrying helicopters and some 1500 firefighters, including recruits from neighbouring Venezuela and Argentina, fought the flames to no avail.

In late March, the weather-modification experts were called in: two Caiapo Indian shamans especially flown to the Yanomami reservation, housing the last of what are believed to be Stone Age tribes. They danced around a bit and prayed, and gathered up a few leaves. Two days later, the heavens opened and it began to pour. Up to 90 per cent of the fire was extinguished.[18](#)

The Western equivalent of a rain dance is to hope for good weather, and when carried out as a group intention, it may be just as effective. PEAR's Roger Nelso carried out an ingenious little study, after realizing that the sun shone on graduation day at Princeton for as long as he could remember.

Had the desire of the community for a sunny commencement day had a powerful local effect?

He had gathered weather reports for the past 30 years in Princeton and the surrounding areas for the times around graduation day and statistically compared them; Princeton was drier than usual for that time of year, and drier and sunnier than surrounding communities for just that day. If the figures were to be believed, the collective wish for good weather by the community of Princeton may have created some sort of mental umbrella that only stretched to their borders during that single day.[19](#)

The only other evidence of group mind had been a provocative little double-blind exercise carried out by Dean Radin, who was interested in the claims of Japanese

alternative medicine practitioner Masaru Emoto that the structure of water crystals is affected by positive and negative emotions.[20](#)

Emoto claims to have carried out hundreds of tests showing that even a single word of positive intent or negative intent profoundly changes the water's internal organization.

The water subjected to the positive intent supposedly develops a beautiful, highly complex crystalline structure when frozen, whereas the structure of water exposed to negative emotions became random, disordered, even grotesque. The most positive results supposedly occur with feelings of love or gratitude.

Radin placed two vials of water in a shielded room in his laboratory at the Institute of Noetic Sciences in Petaluma, California. Meanwhile, a group of 200 attendees at one of Emoto's conferences in Japan was shown a photo of the vials and asked to send them a prayer of gratitude.

Radin then froze the water in those vials as well as samples of control water from the same source that had not been exposed to the prayers, and showed the resulting crystals to a panel of independent volunteers. He had carefully blinded the study so that neither he nor his volunteers had any idea which crystals had been grown from the water samples that had been sent intention. A statistically significant number of the volunteer judges concluded that water sent the positive intentions had formed the more aesthetically pleasing crystalline structure.[21](#)

Nelson's Global Consciousness Project effects had been an especially intriguing example of the power of mass thought. In a sense, they showed the same effect captured by Tiller's equipment in his laboratory. Intention appeared to be raising order in the ground state of the Zero Point Field. But was there a magic threshold effect, as the Maharishi maintained? And how many people were required to constitute a critical mass? According to the Maharishi's formula – that the square root of 1 per cent of any population practising advanced meditation will have a positive impact – only 1730 advanced American meditators would be required to have a positive influence on the US, and only 8084 to affect the entire world.

Nelson's work with FieldREGs had suggested that the size of the group was no as important as the intensity of focus; any group, however small, exerted an effect so long as the parties were involved in rapturous attention. But how many people did the group need to exert an effect? How intently focused did we need to be? What were the true limits of our influence – if any? It was time for me to find my own answers.

The original plan for our first intention experiment, as Popp saw it, was to gather a group of experienced meditators in London, and to have them send positive intention

to the *Acetabularia acetabulum* growing in Popp's IIB laboratory in Neuss Germany.

I was deflated after we had discussed the likely target. For our first experiment, I had wanted to help heal burn victims, to save the world from global warming. Single-celled organisms weren't exactly my idea of heroics and high drama.

Then I began to research algae, and quickly changed my mind. Vital algae were being killed off as a result of global warming. Scientists have discovered an inexorable rise in ocean temperatures over the past century.

For the past 30 years, coral reefs, the centrepiece of the sea's ecosystem, have been vanishing off the earth. When oceans warm, the algae hugging coral reefs get sloughed off, and without this protective layer, the coral reefs themselves die. Some 97 per cent of a certain species of coral have disappeared in the Caribbean alone, and the US government has recently declared Elkhorn and Staghorn coral to be endangered species.

According to the United Nations' Intergovernmental Panel on Climate Change, body made up of the world's leading climatologists and other scientists, the predicted level of warming – up to 6°C by the end of this century – will bring on a disaster of biblical proportions: a rise of sea levels by nearly 1 metre; unendurable heat in many parts of the world; a vast increase of vector-borne diseases; raging floods and storms. A change upward of six degrees may not seem like much until one takes on board that lowering it by the same amount would bring on another Ice Age.

The key to warding off all the fires and floods appeared to be algae. Algae and other plants are the firefighters of our overheated oceans. Scientists are presently engaged in studying sediments from the ocean floor to see how the oceans cope with rising levels of gases.

They are especially interested in the reaction of marine plants to global warming, as they are the primary shock absorbers of excess carbon dioxide. Algae provide oxygen and other benefits to plant and animal marine life. Algae offer a little wall of protection to the creatures of the sea from the worse excesses of man.

I reconsidered my resistance to *Acetabularia* as a test subject. Algae might be critical to our survival. The health of most life in the seas depends on these lowly, single-celled creatures, and the seas, like the rainforests, represent the lungs of the earth. As algae goes, so, eventually, do we. Being able to show that mass intention could rescue a sample of algae might demonstrate that our thoughts could combat something as potentially devastating as global warming.

* * *

On 1 March 2006 I travelled to Germany to meet Popp and his colleagues at the IIB laboratory on Museum Island in Hombroich, west of Düsseldorf. The 'island's innovative architecture had first been built to serve the eccentric needs of a millionaire art collector turned Buddhist, Karl Heinrich Müller, who had nowhere to house his vast collection of painting and sculpture. He purchased 650 acres from the American military, and then converted a NATO missile site into an open-air museum.

Müller's ambitions for the island grew to embrace the possibility of an artists' and writers' community. He commissioned a sculptor turned architect named Erwin Heerich and gave him a free hand. Heerich created enormous futuristic brick structures – galleries, a concert hall, working spaces and even residences – and ingeniously placed them to best advantage against the bleak landscape. Nothing had been wasted; even the disused metal bunkers and rocket silos had been converted into studios and working spaces for famous German artists, writers and musicians, including Thomas Kling the lyricist and Joseph Beuys the sculptor.

Past a 'garden' of buildings of different pastels, the eye alighted on a squat building of interlocking squares on a narrow base, like a giant piece of Lego about to take flight – the new official international site of the IIB. Popp politely accepted the building, when it was first offered to him, but found the open, airy loft, its floor-to-ceiling windows staring out on the vast panorama of Museum Island, completely impractical for his purposes. Before long he set up camp in one of the cramped metal bunkers, left from the *raketenstation*, whose small dark rooms are more compatible with the work of counting living light.

There I met Popp's team of eight, which included Yu Yan, a Chinese physicist, Sophie Cohen, a French chemist, and Eduard Van Wijk, a Dutch psychologist. Most of the cramped rooms contained photomultipliers, large modern boxes attached to computers that count photon emissions. One room housed another smaller room, with a bed and a photomultiplier for human subjects.

The pride of place was reserved for a strange homemade contraption of welded metal circles, resembling a David Smith sculpture of scrap metal, which periodically clanged. That, Popp said with pride, was his first photomultiplier, assembled in 1976 by his student, Bernhard Ruth, and still one of the most accurate pieces of equipment in the field. Indeed, he was convinced that it kept improving with age.

When measuring subtle effects, such as the tiny discharges of light from a living thing, it is important to construct a test that will yield a large enough effect to indicate that something has changed.

Our experimental design had to be so robust, said Popp, that a positive result could not be dismissed by *advocatus diaboli*, the scientific process of identifying

weaknesses in a scientific hypothesis and providing a ready explanation for anomalous effects. Or, as Gary Schwartz had put it, if we heard hoof beats, we first had to eliminate horses before leaping to the conclusion that they belonged to zebras.

In our experimental design, we also had to aim for an 'on off, on off' effect, so that we could isolate any changes as being caused by remote influence. Popp suggested that we have our group send intention intermittently at regular intervals: 10 minutes on, then 10 minutes off, so that we would be 'running' intention a few times every hour. If our experiment worked and intention did have an effect, once we plotted our result on a graph it would create an identifiable, zigzag effect.

Popp acquiesced to including dinoflagellates as well as *Acetabularia*. The light emissions of these fluorescent creatures are extraordinarily responsive to change. As he had seen when they had been placed in shaken water, a change of any sort to which a dinoflagellate is exposed readily shows up as a large shift in emissions of light. I made a further appeal for the use of several subjects.

Each would constitute a separate experiment, and then we would have several results to compare. More than one positive finding would be less likely the result of chance. Finally, the scientists agreed. We also added a jade plant, and a human subject whom Eduard felt he could enlist.

As Popp had concluded during his experiment with Dick Blasband, change of any sort is easier to see with something ill that you try to make well, so we needed to stress some of our subjects in some way. The most obvious way to stress a life form is to place it in a hostile medium.

Eduard and Sophie decided to pour some vinegar into the medium of the dinoflagellates. We could stress the jade plant by sticking a needle through one of its fleshy leaves. Eduard ultimately decided to stress our human subject with three cups of coffee, but I agreed not to disclose this fact to my meditators, to see if they could pick up any psychic information about her. We decided to leave the *Acetabularia* alone, to test whether our intentions could also affect a healthy organism. To make it simple, our meditators would send intentions for the biophoton emissions of each organism to decrease and for its health and well-being to improve.

The experiment would run at night, between 3 p.m. and 9 p.m. Eduard and Sophie would turn on the equipment, and I would select three half-hour windows within that time frame, unbeknownst to them, to carry out our group intentions. Although it was impossible to conduct a double-blind trial (all of us in London would of course know when we sent our healing intention), we could create 'singleblind' conditions and control for experimenter effects, by ensuring that neither our human subject nor the

scientists knew when intention was being sent. I would reveal our schedule to them only after the experiment had taken place.

Our study design was constrained by the equipment. A photomultiplier cannot run with the shutter open continually for six hours, so we decided to turn it on from the hour to the half hour, and give it a rest between the half hour and the hour. I would instruct my meditators to send an intention to all four subjects for two 10-minute sessions during the three time windows I'd chosen.

Eduard and Popp planned to look for any qualitative differences in the kind of light being emitted. Any change in the signal or the quantum nature of the photons during the times we were 'running' intention would suggest that change had occurred from an outside influence and that we were having an effect.

I took some photos of our subjects and the scientists. Before leaving, I stole a last look at the *Acetabularia*, growing in small pots in a converted, darkened refrigerator, and the dinoflagellates, which resembled tiny green specks in the water tiny participants about to be stressed, and possibly sacrificed, in the name of science.

A few weeks later, Eduard found a human volunteer in one of his Dutch colleagues, Annemarie Durr,[22](#) a laser biologist and a meditator of long standing. Although rather sceptical of our plan, she was happy to be our first subject. Her agreement to participate was a particularly generous gesture, as it would entail sitting still on a bed in a pitch black room for six hours.

At one of our conferences in mid-March, I asked for volunteers to participate in a first intention experiment from those among our audience who were experienced meditators. I prepared a PowerPoint presentation to brief them on the subjects of our experiment and the experimental protocol, and to reinforce my verbal presentation, and set the day for 28 March at 5:30 p.m., at a university lecture room I had hired for the evening.

That night, there was such a fierce hailstorm when my colleague Nicolette Vuvan and I left our office for the train to central London that we had to take momentary shelter in a doorway.

We were half soaked after battling through a torrent of rain, but I was thrilled with the atmospheric conditions – a dark, stormy night would only aid our activities. Weather this wild often results from geomagnetic or atmospheric disturbance, which I knew enhances psychokinetic effects.

When I checked with America's National Oceanic and Atmospheric Administration's website later that evening, I discovered that they noted 'unsettled' conditions for the

evening, with a fair degree of geomagnetic activity and minor to major storms in space.

Despite the weather, 16 volunteers showed up. I asked them to fill in a collection of forms, which included personal information plus several psychological tests used by Gary Schwartz and Stanley Krippner, including the Arizona Outcome Integrative Scale and the Hartmann Boundary Questionnaire test, to test psychic ability. I wanted as much data as possible in order to gauge whether their state of mind, psychic talent, or health status would have any bearing on our results.

I soon discovered that my volunteers were ideal candidates for an intention experiment. According to the forms they'd filled out for me, they'd meditated for an average of 14 years, and their scores on the psychological tests I'd given them showed that, as a group, they had very thin boundaries, tended toward a highly positive outlook, enjoyed excellent mental, emotional and physical health, and evidenced powerful emotions.

I explained the experiment, offered photographs and details about our four subjects, and then went over the protocol. We would be sending our intentions from 6 to 8:30 p.m. at every hour on the hour to 10 minutes past and from 20 past until the half hour. In between those times we would rest, chat and fill in the forms.

We began at 6 on the dot. As William Tiller had done in his black box experiments, I displayed the intentions in writing on the computer screen as I read them out loud so that all the meditators would be sending exactly the same thought during each meditation.

I led the meditation, directed our focus to each target subject, showing its image on screen, and read aloud the sentence that sent our intention to lower the subject's biophoton emissions and increase its state of health and well-being.

The shared energy immediately felt tangible and increased in power as the evening carried on. Michael, one of our group members, suggested that we call our algae 'Dino' and 'Tabu', to establish some relationship with these little organisms. Although no one had any prior experience in telepathy, some participants began to pick up information about our subjects, notably Annemarie.

Several meditators were convinced that she was an amateur singer, and had a recurrent problem with her throat. Isabel thought she might be suffering from gut problems or something gynaecological. Michael, who was German, kept thinking of a term *in Schutz der Dunkelheit* ('under protection of darkness'), and interpreted it to mean that she was wrapped up in a blanket. Amy said she received a mental image of Annemarie wrapped up in a luxuriously soft blanket on a hard surface and at times

asleep. She was also convinced that she had eaten something disagreeable and that her stomach was upset.

Many meditators felt a connection to the jade plant and 'Tabu', and Peter had a strong sense that *Acetabularia* was responding most to the intentions – but with few exceptions the group had the most difficulty establishing any connection with 'Dino', and this difficulty increased to the point where most felt no connection at all by the final session.

All of us were infused with a strong sense of purpose and momentarily lost a sense of our individual identities.

By the end of the evening, I had cast out my own doubts about the study and the niggling thought that what we were trying to do was faintly ludicrous. Even though we were not healers, we had all felt as if a healing of sorts had occurred. Whatever had happened in there, I thought, heading back out into the stormy night, I grew certain we'd had some kind of effect.

Several days later, I sent Popp our meditation schedule so that his team could compile our results.

I also spoke with Annemarie. Some of our extrasensory impressions had been correct. It was true she sang as a hobby and periodically suffered with a blocked throat. Although she ordinarily did not especially suffer problems in the gut, she had that night because the three cups of coffee Eduard had asked her to drink upset her stomach. Yet even though coffee late in the afternoon usually agitated her and caused insomnia, on the night of our experiment, she drifted off at various points throughout the six hours of the experiment and slept easily that night. She described tingling bodily sensations she had felt periodically through the evening, and the times of their occurrence corresponded with the first and third sessions that we had been 'running' intention. Nevertheless, we had also picked up some 'noise': she was not a vegetarian and never listened to or had sung Vivaldi, as a couple of meditators had felt.

When analysing the data, Eduard studied not only the intensity of light but also its deviation from symmetry: normal emissions from a living thing, when plotted on a graph as a bell curve, are perfectly symmetrical. He also looked at any deviations in the kurtosis, or the customary 'peakedness', of the distribution. High kurtosis means a bell curve that is high around the middle, or mean.

Again, when emissions are plotted on a graph, the normal peak distribution is 0 – the highs and lows cancel each other out. After examining our 12 block periods – the six times we sent intention and the six periods of rest – he found no change in light

intensity. But he did find large changes in the skewness, showing a lack of the customary symmetry (from 1.124 to 0.922), and kurtosis (from 2.403 to 1.581) of the emissions. Something in the light was profoundly altered.

Eduard was excited by the results. They exactly matched those he had observed during his study of healers, when he had tested whether the act of healing has a 'scatter effect' on any other living things in the environment where the healing takes place. In the study, when he had placed some algae with a photon counter in the presence of a healer and his patients and measured the photons of the algae during 36 healings, he had been surprised to discover that the photon count distributions of the algae had 'remarkable' alterations during the healing rituals. Large shifts in the cyclical components of the emissions had occurred.

His tiny study had suggested that healing caused a shift in the light emissions of everything in its path.[23](#) Now he had discovered the same effect when simple intention was sent by ordinary people from 300 miles away.

On 12 April, Fritz Popp sent me data on the algae, the dinoflagellates and the jade plant. Although a first glance at the numbers had convinced him we had had no effect, he changed his mind once he performed his calculations.

Ordinarily, any stressed living thing will begin to accustom itself to the stress, and its light emissions, although initially large, will naturally begin to decrease as the organism gets used to its new circumstances.

Consequently, in order to work out a true demonstration of the effect of change, Popp had to control for this phenomenon. He worked out mathematically a means of starting from zero, so that any deviation from normal behaviour would readily show up. In this way, he would then be able to determine whether any additional change represented an increase or a decrease in the number of biophoton emissions. The number of emissions he then plotted on his graph reflected any excess increase or decrease from the norm.

In all three instances, our subjects registered a significant decrease in biophotons during the meditation sessions, compared with the control periods. The dinoflagellates had been killed by the acid, in the end (one possible reason why they had been so difficult for our meditators to detect).

Nevertheless, Popp said, their response (a lowering of emissions by nearly 140,000) was significantly different from the normal emissions of a dying organism. Among the survivors, the *Acetabularia*, the healthy subject, had evidenced a larger effect than the jade plant, perhaps because it was not overcoming a stress (544 emissions lower

than normal), whereas with the jade plant (which had 65.5 emissions lower than normal), the stress (the pin) remained in the leaf during the experiment.

He plotted the results on a graph, marking out the portions in red that represented the times of our healing intentions and emailed them to me. We had indeed produced a 'zigzag' effect. During meditation, Popp wrote in his report, 'there is a clear preference of dropping down reactions rather than going up', which tracked the times of our intentions. With the *Acetabularia*, we had had an overall decrease over the norm of 573 emissions, and an increase of only 29.

Our little meditation effort had created a major healing effect, a significant decrease in living light. Not only that, but the effect from all that distance was similar to the effect by an experienced healer when healing in the same room. The intention of our group had created the same light as a healer's.

In many ways, it was a crude first effort. We had, after all, tested four subjects, some stressed and some not, and one had died. We had made use of control periods, but not control subjects. Both Eduard and Popp cautioned me not to take too much notice of it: 'We have to be sure that these changes in kurtosis and skewness are real. That means that we have to repeat the experiments a couple of times,' said Eduard. 'Despite the right tendency of the results,' wrote Popp, 'I do not dare to state that it is proof.'

But, despite these caveats, the fact was that we had recorded a significant effect. In the end, achieving a positive result didn't really surprise me. For more than 30 years Popp, Schlitz, Schwartz and all of their fellow scientists have been amassing unimpeachable evidence in other experiments that has stretched credulity. Frontier research into the nature of human consciousness has upended everything that we have hitherto considered scientific certainty about our world.

These discoveries offer convincing evidence that all matter in the universe exists in a web of connection and constant influence, which often overrides many of the laws of the universe that we used to believe held ultimate sovereignty.

The significance of these findings extends far beyond a validation of extrasensory power or parapsychology. They threaten to demolish the entire edifice of present-day science. The discoveries of Tom Rosenbaum, Sai Ghosh and Anton Zeilinger that quantum effects occur in the world of the tangible could signal an end to the divide in modern physics between the laws of the large and the laws of the quantum particle, and the beginning of a single rule book defining all of life.

Our definition of the physical universe as a collection of isolated objects, our definition of ourselves as just another of those objects, even our most basic understanding of

time and space, will have to be recast. At least 40 top scientists in academic centres of research around the world have demonstrated that an information transfer constantly carries on between living things, and that thought forms are simply another aspect of transmitted energy. Hundreds of others have offered plausible theories embracing even the most counter-intuitive effects, such as time-displaced influence, as now consistent with the laws of physics.

We can no longer view ourselves as isolated from our environment and our thoughts the private, self-contained workings of an individual brain. Dozens of scientists have produced thousands of papers in the scientific literature offering sound evidence that thoughts are capable of profoundly affecting all aspects of our lives. As observers and creators, we are constantly remaking our world at every instant. Every thought we have, every judgement we hold, however unconscious, is having an effect. With every moment that it notices, the conscious mind is sending an intention.

These revelations not only force us to rethink what it is to be human, but also how to relate. We may have to reconsider the effect of everything that we think, whether we vocalize it or not. Our relationship with the world carries on, even in our silence.

We must also recognize that these ideas are no longer the ruminations of a few eccentric individuals. The power of thought underpins many well-accepted disciplines in every reach of life, from orthodox and alternative medicine to competitive sport. Modern medicine must fully appreciate the central role of intention in healing. Medical scientists often speak of the 'placebo effect' as an annoying impediment to the proof of the efficacy of a chemical agent. It is time that we understood and made full use of the power of the placebo. Repeatedly, the mind has proved to be a far more powerful healer than the greatest of breakthrough drugs.

We will have to reframe our understanding of our own biology in more miraculous terms. We are only beginning to understand the vast and untapped human potential at our disposal: the human being's extraordinary capacity to influence the world. This potential is every person's birthright, not simply that of the gifted master. Our thoughts may be an inexhaustible and simple resource that can be called upon to focus our lives, heal our illnesses, clean up our cities and improve the planet. We may have the power as communities to improve the quality of our air and water, our crime and accident statistics, the educational levels of our children. One well-directed thought may be a gentle but effective way for every man and woman on the street to take matters of global interest into their own hands.

This knowledge may give us back a sense of individual and collective power, which has been wrested from us, largely by the current world view espoused by modern science, which portrays an indifferent universe populated by things that are separate

and unengaged. Indeed, an understanding of the power of conscious thought may also bring science closer to religion by offering scientific proof of the intuitive understanding, held by most of us, that to be alive is to be far more than an assemblage of chemicals and electrical signalling.

We must open our minds to the wisdom of many native traditions, which hold an intuitive understanding of intention. Virtually all of these cultures describe a unified energy field not unlike the Zero Point Field, holding everything in the universe in its invisible web. These other cultures understand our place in a hierarchy of energy and the value of choosing time and place with care. The modern science of remote influence has finally offered proof of ancient intuitive beliefs about manifestation, healing and the power of thoughts. We would do well to appreciate, as these traditional cultures do, that every thought is sacred, with the power to take physical form.

Both modern science and ancient practices can teach us how to use our extraordinary power of intention. If we could learn how to direct our potential for influence in a positive manner, we could improve every aspect of our world. Medicine, healing, education, even our interaction with our technology, would benefit from a greater comprehension of the mind's inextricable involvement in its world. If we begin to grasp the remarkable power of human consciousness, we will advance our understanding of ourselves as human beings in all our complexity.

But there are still many more questions to ask about the nature of intention. Frontier science is the art of inquiring about the impossible. All of our major achievements in history have resulted from asking an outrageous question. What if stones fall from the sky? What if giant metal objects could overcome gravity? What if there is no end of the earth to sail off? What if time was not absolute, but depends on where you are? All of the discoveries about intention and remote influence have similarly proceeded from asking a seemingly absurd question: what if our thoughts could affect the things around us?

True science, unafraid to explore the dark passages of our ignorance, always begins with an unpopular question, even if there is no prospect of an immediate answer – even if the answer threatens to overturn every last one of our cherished beliefs. The scientists engaged in consciousness research must constantly put forward unpopular questions about the nature of the mind and the extent of its reach.

In our group experiments, we will be asking the most impossible question of all: what if a group thought could heal a remote target? It is a little like asking, what if a thought could heal the world?

It is an outlandish question, but the most important part of scientific investigation is just the simple willingness to ask the question. As Bob Barth of the Office of Prayer Research commented, when asked whether praye research should continue in the wake of the Benson STEP study: 'We can't find the answers if we don't keep asking the questions.' That is how we will begin our own experiments – unafraid to ask the question, whatever the answer.

Note – Chapter 12: The Intention Experiment

1. *Interview with Fritz-Albert Popp, March 1, 2006.*
2. *F.-A. Popp et al., 'Further analysis of delayed luminescence of plants', Journal of Photochemistry and Photobiology B: Biology, 2005, 78: 235–44.*
3. *For a full description of Popp's history, see McTaggart, The Field, op. cit.*
4. *International Institute of Biophysics, see www.lifescientists.de.*
5. *B. J. Dunne, 'Co-operator experiments with an REG device', PEA Technical Note 91005, Princeton Engineering Anomalies Research Princeton, New Jersey, December 1991.*
6. *R. D. Nelson et al., 'FieldREG anomalies in group situations', Journal of Scientific Exploration, 1996; 10 (1): 111–41; R. D. Nelson et al., 'FieldREGII: Consciousness field effects: replications and explorations' Journal of Scientific Exploration, 1998; 12 (3): 425–54.*
7. *D. I. Radin, 'For whom the bell tolls: A question of global consciousness' Noetic Sciences Review, 2003; 63: 8–13 and 44–5; R. D. Nelson et al. 'Correlation of continuous random data with major world events', Foundations of Physics Letters, 2002; 15 (6): 537–50.*
8. *D. I. Radin, 'Exploring relationships between random physical events and mass human attention: Asking for whom the bell tolls', Journal of Scientific Exploration, 2002; 16 (4): 533–47.*
9. *R. D. Nelson, 'Coherent consciousness and reduced randomness: Correlations on September 11, 2001', Journal of Scientific Exploration, 2002; 16 (4): 549–70.*
10. *Ibid.*
11. *Bryan J. Williams, 'Exploratory block analysis of field consciousness effects on global RNGs on September 11, 2001' (<http://noosphere.princeton.edu/williams/GCP911.html>).*
12. *J. D. Scargle, 'Commentary: Was there evidence of global consciousness on September 11, 2001?' Journal of Scientific Exploration, 2002; 16 (4): 571–7.*
13. *Nelson et al., 'Correlation of continuous random data', op. cit.*
14. *M. C. Dillbeck et al., 'The Transcendental Meditation program and crime rate change in a sample of 48 cities', Journal of Crime and Justice, 1981; 4: 25–45.*

15. J. Hagelin et al., 'Effects of group practice of the Transcendental Meditation program on preventing violent crime in Washington, D. C.: Results of the National Demonstration Project, June–July 1993', *Social Indicators Research*, 1999; 47 (2): 153–201.
16. W. Orme-Johnson et al., 'International peace project in the Middle East: the effects of the Maharishi technology of the unified field', *Journal of Conflict Resolution*, 1988; 32: 776–812.
17. K. L. Cavanaugh et al., 'Consciousness and the quality of economic life empirical research on the macroeconomic effects of the collective practice of Maharishi's Transcendental Meditation and TM-Sidhi program.' Paper originally presented at the annual meeting of the Midwest Management Society, Chicago, March 1989, published in R. G. Greenwood (ed.) *Proceedings of the Midwest Management Society*, Chicago: Midwest Management Society, 1989: 183–90; K. L. Cavanaugh et al., 'A multiple input transfer function model of Okun's misery index: An empirical test of the Maharishi Effect.' Paper presented at the Annual Meeting of the American Statistical Association, Washington D. C., August 6–10, 1989, an abridged version of the paper appears in *Proceedings of the American Statistical Association, Business and Economics Statistics Section*, Alexandria, Va.: American Statistical Association, 1989: 565–70; K. L. Cavanaugh and K. D. King, 'Simultaneous transfer function analysis of Okun's misery index: improvements in the economic quality of life through Maharishi's Vedic Science and technology of consciousness.' Paper presented at the Annual Meeting of the American Statistical Association New Orleans, Louisiana, August 22–25, 1988, an abridged version of the paper appears in *Proceedings of the American Statistical Association Business and Economics Statistics Section*, Alexandria, Va.: American Statistical Association, 1988: 491–6; K. L. Cavanaugh, 'Time serie analysis of U.S. and Canadian inflation and unemployment: A test of a field-theoretic hypothesis.' Paper presented at the Annual Meeting of the American Statistical Association, San Francisco, California, August 17–20 1987, published in *Proceedings of the American Statistical Association Business and Economics Statistics Section*, Alexandria, Va.: American Statistical Association, 1987: 799–804.
18. Strong rains fall on fire-ravaged Amazon state, March 31, 1998, Web posted at: 6:46 p.m. EST (2346 GMT), Brasilia, Brazil (CNN) <http://twm.co.nz/>.
19. R. Nelson, 'Wishing for good weather: a natural experiment in group consciousness', *Journal of Scientific Exploration*, 1997; 11 (1): 47–58.
20. M. Emoto, *The Hidden Messages in Water*, New York: Atria, 2005.
21. Interview with Dean Radin, May 3, 2006.
22. Not her real name. I've changed her name at her request. Nevertheless, our meditators were shown her real name and photo.

23. R. Van Wijk and E. P. Van Wijk, 'The search for a biosensor as a witness of a human laying on of hands ritual', *Alternative Therapies in Health and Medicine*, 2003; 9 (2): 48–55.

PART FOUR

The Experiments

Miracles do not happen in contradiction to nature, but only in contradiction to that which is known in nature.

-St Augustine

CHAPTER THIRTEEN

The Intention Exercises

UP UNTIL THIS POINT, *The Intention Experiment* has been concerned with the scientific evidence of the power of intention. What has not been tested is the extent of this power in the cut and thrust of ordinary life. An inordinate number of books have been written about the power of the human being to manifest his or her reality, and, while they have served up many intuitive truths, they offer little in the way of scientific evidence.

Exactly how much power do we possess to shape and mould our daily lives? What can we use this for, individually and collectively? How much power do we possess to heal ourselves, to live lives of greater happiness and purpose?

This is where I would like to enlist your help. Determining the further practical applications of the power of thought is the purpose of the next portion of this book – the part that involves you as a partner in the research.

Although the power of intention is such that any sort of focused will may have some effect, the scientific evidence suggests that you will be a more effective 'intender' if you become more 'coherent', in the scientific sense of the term. To do so to greatest effect, or so the scientific evidence suggests, you will need to choose the right time and place, quiet your mind, learn how to focus, entrain yourself with the object of your intention, visualize and mentally rehearse. Believing that the experiment will work is also essential.

Most of us operate with very little in the way of mental coherence. We walk around immersed in a riot of fragmentary and discordant thought. You will become more coherent simply by learning to shut down that useless internal chatter, which always focuses on the past or the future, never the present. In time, you will become adept at

quietening down your mind and ‘powering up’, much as joggers train their muscles, and each day find that they can perform a little better than the day before.

The following exercises are designed to help you to become more coherent and so more effective in using intention in your life and in our group intention experiments. These have been extrapolated from what has appeared to work best in the scientific laboratory.

Think of intentions in terms of grand and smaller schemes. Take the grand schemes in stages, so that you send out intentions in steps towards achieving the grand scheme. Also start with modest goals – something realizable within a reasonable timeframe. If you are 40 pounds overweight and your goal is to be a size 8 next week, that is not a realistic timeframe. Nevertheless, keep the grand scheme in mind and build towards it as you gain experience. It is also important to overcome your natural scepticism. The idea that your thought can affect physical reality may not fit your current world paradigm, but nor would the concept of gravity if you were living in the Middle Ages.

Choose Your Intention Space

A number of scientific studies suggest that conditioning your space magnifies the effectiveness of your intentions. Choose a place to carry out your intentions that feels comfortable. Clear away extraneous items and make it personal or appealing, with cushions or comfortable furniture, so that whenever you spend time there you will find it an enjoyable refuge, a place where you can sit quietly and meditate. Use candles, soft lights and incense, if you prefer.

Some people find it helpful to create an ‘altar’ of sorts, as a focal point, with objects or photographs that you find inspirational or particularly meaningful. Even if you are not at home, you may find that you will naturally ‘enter’ your intention space by visualizing it whenever you want to send an intention.

Unless you live in the mountains and can open your windows to clean mountain air, you also may want to install an ionizer in your space to increase the number of negative ions in your environment.

The half-life of ions – which is related to the amount of time that ions maintain their effective radiation – depends on the amount of pollutants in the air. The cleaner the air, the longer the half-life of small ions, if there is a source of ionization (e.g. running water) present. The best levels of ions are:

- in the uninhabited country, away from industrialized areas; near running water, whether a shower or a waterfall;

- in natural habitats;
- in clear sunshine – a natural ionizer; after storms;
- in the mountains.

The worst are:

- in enclosed spaces with a gathering of a number of people;
- near television sets and other such electrical appliances, which can give off electric emissions up to 11,000 volts, exposing anything immediately in range to positive charge;
- in cities; near industrial sources; in smog, fog, dust or haze.

As a rule of thumb, the lower the visibility, the lower the ion concentration. Low visibility is due to the presence of a great number of large particles, which air ions readily latch on to. For those among us who are city dwellers, placing plants and some source of water, like an indoor desk fountain, will help to improve ion levels in intention spaces. Keep your space free of electrical gadgets and computers.

Power Up

In order to 'power up' to peak intensity, you must first slow your brain waves down to a meditative, or 'alpha', state of light meditation or dreaming – when the brain emits frequencies (measured on an EEG machine) of 8–13 hertz (cycles per second).

Sit in a comfortable position. Many people like to sit upright in a hard-backed chair, with their hands placed on their knees. You may also sit on the floor cross-legged. Begin breathing slowly and rhythmically in through the nose and out through the mouth (slowly blow all the air out), so that your in-breath is the same length as your out-breath. Allow the belly to relax so that it slightly protrudes, then pull it back slowly as if you were trying to get it to touch your back. This will ensure that you are breathing through your diaphragm.

Repeat this every 15 seconds, but ensure that you are not overexerting or straining. Carry on for 3 minutes and then keep observing it. Work up to 5 or 10 minutes. Begin to focus your attention just on the breath. Practise this repeatedly, as it will form the basis of your meditative practices.

To enter an alpha state, the most important feature, as any Buddhist understands, is to still the mind. Of course, just thinking about nothing is often virtually impossible.

After entering the state by concentrating on the breath or focusing on a single object, most meditation schools recommend some sort of 'anchor', enabling you to keep your chattering mind quiet, so that you are allowed to be more receptive to intuitive information. The usual anchors include focusing on:

the body and its functions, or the breath;

your thoughts, as though they are floating by on a flying carpet, so that they are not 'you';

a mantra, such as used in Transcendental Meditation, is usually a 'word' such as OM ('The Field' in Buddhism), AH (the universal truth of life) or HUM (the physical manifestation of the truth – the universe itself). In the early 1970s, many practitioners of TM were given the mantra AHOM;

numbers, through silent repetitive counting, either backwards or forwards; music – usually something repetitive, such as Bach or chanting;

a single tone, such as that produced by an Australian didgeridoo;

a drum or rattle, the repetitive sounds of which have been used by many traditional cultures to still the mind;

prayer, as with a rosary, since the repetitive sounds still the mind.

Practise until you can comfortably focus on your 'anchor' for 20 minutes or more.

Peak Intensity

Powering up involves developing the ability to attend with peak intensity, moment by moment. One of the surest ways to develop this is to practise the ancient art of mindfulness, espoused as long ago as 1000 BC by Shakyamuni Buddha, founder of modern Buddhism. It is a discipline whereby you maintain clear, moment-to-moment awareness of what is happening internally and externally, rather than in thought.

More than just concentration, mindfulness requires that you police the focus of your concentration and maintain that concentration in the present. With practice, you will be able to silence the constant inner chatter of your mind and concentrate on your sensory experiences, no matter how mundane – whether it is eating a meal, hugging your child, noticing some pain you are experiencing or just picking some lint off your sweater. It is like being a benevolent parent to your mind – selecting what it will focus on and leading it back when it strays.

In time, mindfulness meditation will also heighten your visual perceptions and prevent you from becoming numb to your everyday experience.

One of the difficulties in incorporating mindfulness into ordinary activity is that it is usually taught at retreats, where participants have the luxury of meditating for hours a

day or practising mindfulness by engaging in activities, as it were, in 'slow motion'. Nevertheless, there are ways to adapt many traditional practices for use in your intention meditation.

Once you have achieved your 'alpha state', quietly observe whatever manifests in your mind and body as precisely as you can. Be present and attentive to what is, rather than what your emotions tell you, what you wish were the case, or only what is most pleasant. Do not suppress or banish any negative thoughts, if they are true. One good means of harnessing your mind to the present is to 'come into your body' and feel your body posture.

It is vital that you distinguish mindfulness from mere concentration. The most important distinction is a lack of judgement or reference point about the experience. You attend to every moment in the present without colouring it with preference for the pleasant or distaste for the unpleasant, or even identifying the experience as something happening to you. There is, in short, no 'better' or 'worse'.

Be aware of all the smells, textures, colours and sensual feelings you are experiencing. What does the room smell like? What taste is in your mouth? What does your seat feel like?

Be mindful of what is happening internally and externally. Whenever you catch yourself judging what you see, think to yourself, 'I am thinking', and return to observing with simple attention.

Cultivate the art of simple listening to all sounds in your room: the rumble of a pipe, the honking of a horn, the barking of a dog, a plane flying overhead. Accept all sounds – the noise, chaos or stillness – without judgement.

Notice other sensations in the room: the 'colour' of the day, the light in the room, any movement carrying on in front of you, the sensations of quiet.

Try not to try. Work on eliminating your expectations or striving for (and anxiety over) certain results.

Accept all that happens without judgement. This means putting away all opinions and interpretations of what goes on. Catch yourself clinging to certain views, thoughts, opinions and preferences, and rejecting others. Accept your

own feelings and experiences, even the unpleasant ones.

Try never to rush. If you must rush, be present in the rushing. Feel what it feels like.

Developing Mindfulness in Your Daily Life

Even when you are not using intention, the evidence suggests that you will mould your brain to become better at it if you develop mindfulness in your daily life.

Psychologist Dr Charles Tart, one of the world's experts on altered states of consciousness, has a number of suggestions of ways to do so:[1](#)

Take periodic breaks during the day in which you have quiet time to be mindful of what is happening internally and externally.

Whenever you feel your concentration flitting away in your daily activities, sense your breath – it will help to ground you.

Be mindful of the most mundane of activities, such as brushing your teeth or shaving.

Start with a small exercise, such as fetching your coat and walking, in which you stay focused completely on what you are doing.

Engage in mental noting, in which you label an ongoing activity, for example 'I'm putting on my coat', 'opening the door', 'tying my shoes'.

Use mindfulness in every ordinary situation. When you are preparing dinner or even doing your teeth, be aware of all the smells, textures, colours and sensual feelings you are experiencing.

Learn to really look at your partner and your children, your pets, your friends and work colleagues. Observe them closely during every activity – every part of them without judgement.

During some activity, such as breakfast, ask your children to be mindful (without speaking) of every aspect of it. Concentrate on the taste of your food. Look closely at the texture and the colours of it. How does the cereal crunch? How does their juice feel as it cascades down their throats? Become aware of the smells and sounds around you. While you are watching all this, how are the different parts of your body feeling?

Listen to what your life sounds like – the myriad noises surrounding you every day. When someone speaks to you, listen to the sound of his or her voice as well as the words. Do not think of a reply until he or she has stopped speaking. Practise mindfulness in every activity: walking down the street, driving home, in the garden. If you are practising these exercises and you happen to bump into someone, do not enter into conversation. Just greet the person, shake hands and stay in the present moment.

Use mindfulness when you are extremely busy or under a tight deadline. Observe what it is like to hurry or to be under the gun and what happens when you do. How does it affect your equilibrium? Be an observer of yourself in that situation. Can you stay in your body while you are working hard?

Practise mindfulness while you are standing in line. Experience the feeling of waiting itself, rather than focusing on what you are waiting for. Be aware of your physical movements and your thoughts.

Do not think about or try to work out your problems. Just deal with whatever daily problem solving is immediately in front of you.

Merging with the ‘Other’

Research shows that touch or even focus on the heart or compassionate feelings for the other is a powerful means of causing brain-wave entrainment between people. When two people touch while focusing loving thoughts on their hearts, the ‘coherent’ heart rhythms of one can entrain the brain of the other.[2](#)

Before you set your intention, it may be important to form an empathetic connection with the object of your intention.

Establish connection beforehand by the following techniques:

First send your intention to someone with whom you already have a strong bond – a partner, a child, a sibling, a dear friend.

With someone you do not know, exchange an object or photograph.

Get to know the person. Go for a walk with them or meet them first. Spend half an hour meditating together first.

Ask the person to be open to receiving your intention when you are sending it. If you are sending an intention to something non-human or inanimate, you can also establish some connection. Find out all you can about the object of your intentions, whether a plant, an animal or an inanimate object. Have it near you for a period before sending your intention. It goes without saying that you should

be nice to it – even if ‘it’ is your computer or photocopier.

Be Compassionate

Use the following methods to encourage a sense of universal compassion during your intention session:

Focus your attention to your heart, as though you are sending light to it. Observe the light spreading from your heart to the rest of your body. Send a loving thought to yourself, such as ‘May I be well and free from suffering.’

On the out breath, imagine a white light radiating outward from your heart. As you do, think: ‘I appreciate the kindnesses and love of all living creatures. May all others be well.’ As Buddhists recommend, first think of all those you love, then your good friends. Move on to acquaintances and finally to those people you actively dislike. For each stage, think: ‘May they be well and free from suffering.’

Concentrate on the kindness and compassion of all living things and their contribution to your well-being. Finally, send your message of compassion to all people and living things on earth.

Practise switching roles with some of your loved ones. Imagine what it is like to be your partner or spouse, your parent, your child. Get inside their shoes and imagine what it would be like to see the world through their eyes, with their hopes and fears and dreams. Think how you would respond.

Jerome Stone quotes Sogyal Rinpoche, author of *The Tibetan Book of Living and Dying*,^{[3](#)} who suggests that we open our hearts every day to the suffering around us, with beggars who pass us by, with the poverty, tragedy and grief we see on our television sets:

Don't waste the love and grief it arouses; the moment you feel compassion welling up in you, don't brush it aside, don't shrug it off and try quickly to return to 'normal', don't be afraid of the feeling or embarrassed by it, allow yourself to be distracted from it or let it run aground in apathy. Be vulnerable; use that quick, bright up rush of compassion; focus on it, go deep into your heart and meditate on it, develop it, enhance and deepen it. By doing this you will realize how blind you have been to suffering ...^{[4](#)}

During your intention, if you are sending healing to someone, first try to put yourself in his situation. Imagine what it is like to be him and to be faced with his current crisis. Try to feel and have empathy for your receiver's suffering. Ask yourself how you would feel if you were suffering in this manner and how you would most want to be healed.

Now, direct your loving thoughts to the object of your intention. If he or she is present, hold his or her hand.

Stating Your Intention

In your meditative state, state your clear intention. Although many people use the construction ‘have always been’ – ‘I have always been healthy’ – I prefer the present tense – of sending your intention to its ‘endpoint’ *as a wish that has already been achieved*. For instance, if you are trying to heal back pain, you can say, ‘My lower back and sacrum are free of all pain and now move easily and fluidly.’ Remember to frame your intention as a positive statement; rather than ‘I will not have side effects’, say, ‘I will be free of side effects’.

Be Specific

Specific intentions seem to work best. Be sure to make your intentions highly specific and directed – and the more detailed, the better. If you are trying to heal the fourth finger of your child’s left hand, specify that finger and, if possible, the problem with it.

State your entire intention, and include what it is you would like to change, to whom, when and where. Use the following as a checklist (as news reporters do) to ensure you have covered every specific: who, what, when, where, why and how. It may help if you draw a picture of it, or create a collage from photos or magazine

pictures. Place this somewhere that you can look at often.

The Mental Dry Run

As with elite athletes, the best way to send an intention is to visualize the outcome you desire with all your five senses in real time. Visualization, or guided imagery, involves using images and/or internal messages to obtain a desired goal. It can be used for any desired outcome – to change or improve your living situation, job, relationships, physical condition or health, state of mind (from negative to positive), outlook on life or even a specific aspect of yourself, including your personality. It can also be used to send intentions to someone else. Self-guided imagery is a little like self-hypnosis.

Plan a mental image of the outcome of your intention well ahead of time. When carrying out visualization, many people believe that you must ‘see’ the exact image clearly in your mind’s eye. But for an intention it isn’t necessary to have a sharp internal image or, indeed, any image at all. It is enough to just think about an intention, without a mental picture, and simply to create an impression, a feeling or a thought. Some of us think in images, others through words, still others through sounds, touch or the spatial relationship between objects. Your mental rehearsal will depend on which senses are most developed in your brain.

For our example of healing back pain, imagine yourself free from pain and doing some sort of exercise or movement you enjoy. See yourself walking agilely, free from pain. Remember, feel the feeling of being pain-free and electrically alive. Imagine the internal and external sensations of your limber back. Feel yourself running free. Choose other sensations that support the healing of your back. If you are sending your intention to heal someone else, carry out all the same aspects of the healing, but imagine yourself inside the other person's back. Send your intention to his back.

Practise Visualizing

You can practise visualization first by getting into a meditative state and imagining the following, while recalling or imagining as much as you can about the sight and smells, and your feelings about them:

A favourite recent meal (can you remember some of the smells and tastes you really enjoyed?).

Your bedroom. Walk yourself mentally through it, remembering certain details – the feel of your bedspread, the curtains or carpet. You do not have to see the entire room, just get a detail or impression.

A recent happy moment (with a loved one, or a child). Remember the most vivid sensations and images.

Yourself performing an activity such as running, riding a bike, swimming or working out at the gym. Try to feel what it is like for your body to be moving that way.

Your favourite music (try to 'hear' the music internally).

A recent experience with an intense physical sensation (such as plunging into a pool or the ocean, having a steam bath, feeling snow or rain, or making love).

Try to relive all of the physical sensations.

To visualize your intention, first work it out carefully ahead of time:

Now, create a picture in your mind's eye of the desired result. Imagine it as already existing, with you in that situation.

Try to imagine as much sensory detail as you can about the situation (the look, smell and feel of it).

Think about it in a positive, optimistic, encouraging way; use mental statements, or affirmations, that confirm that it has or is now happening (not that it will happen in the

future). For instance, for someone with a heart problem, 'My heart is healthy and well.'

For healing, try to imagine healing energy (perhaps as a white light or as your personal deity) filling you and observe it healing the portion of your body that is ill – say, turning a diseased organ into a healthy one. If a good-versus-evil 'contest' is most vivid for you, imagine the 'hero' cells battling or eating up the 'bad guys'. Otherwise, visualize diseased cells or tissue changing into healthy cells, healthy cells replacing diseased cells, or imagine your entire body with that specific body part in perfect health. Visualize yourself often as perfectly healthy, carrying out your daily activities. Find an image of the body part on the Internet or in a book as it looks when it is healthy. Imagine your own body part looking like that. Send out the visualization often, both during meditation and throughout the day.

Belief

The copious evidence of the placebo effect demonstrates the extraordinary power of belief. Belief in the power of intention is also vital. Keep firmly fixed in your mind the desired outcome and do not allow yourself to think of failure. Dismiss any it-won't-happen-to-me type of thoughts. If you are attempting to affect someone who does not share your belief that it may be of benefit, speak to them about some of the scientific evidence in *The Intention Experiment* and elsewhere. It is important that both of you share the same beliefs. Herbert Benson believes that his monks were able to achieve their effects because they used words or phrases incorporating their most deeply held beliefs.[5](#)

Move Aside

In studies of meditation, mediumship and healing, those who are successful at intention imagine themselves and the person receiving healing as one with the universe. In your meditative state, enter into a zone where you relax your sense of 'I' and sense a merging with the object of your intention and The Field. Frame your intention, state it clearly and then let go of the outcome. At this point, you may sense that the intention is taken over by some greater force. Close your internal meditation with a request and then move your own ego aside. Remember: this 'power' does not originate with you – you are its conduit. Think of it as a request you are sending to the universe.

Timing

The evidence suggests that mind-over-matter intention (that is, psychokinesis) works best at points of increased geomagnetic activity. You can find out about the geomagnetic levels in your area by consulting several websites. The US Nationa

Oceanic and Atmospheric Administration (NOAA) created a Space Environment Center (SEC), America's official source of space weather activity (www.sec.noaa.gov). The SEC, in turn, set up a special Space Weather Operations (SWO) branch to act as a warning centre for the world concerning disturbances in space. Jointly operated by the NOAA and the US Air Force, SWO provides forecast and warnings of solar and geomagnetic activity.

SWO receives its data in real time from a large number of ground-based observatories and satellite sensors around the world. These data enable the SWO to predict solar and geomagnetic activity, and to make worldwide alerts during heavy storms. For the forecast of the day you plan to carry out your intentions, see <http://sec.noaa.gov/today2.html>.

The SEC has created Space Weather Scales to give lay people an idea of the severity of geomagnetic storms, solar radiation storms and radio blackouts, and their effect on our technological systems (www.sec.noaa.gov/NOAAscales). The numbers attached to them (such as 'G5') indicate the level of severity, with 1 being mild and 5 the most severe.

The Solar and Heliospheric Observatory (SOHO) was set up as a joint project by the European Space Agency and NASA to study the effect of the sun on the earth. For more information, see <http://sohowww.nascom.nasa.gov/>.

For other aspects of space weather, including charts of geomagnetic activity, see <http://sohowww.nascom.nasa.gov/spaceweather/>. This website includes useful charts on geomagnetic activity, solar wind and high-energy proton and X-ray flux.

All geomagnetic activity is measured on a K index, with 0 being the most quiet and 9 the most turbulent. The a index is similar, but uses a larger scale – from 0 to 400.

When you are sending an intention, plan to do so on a day when the K index is 5 or more (or the a index more than 200).

It may also be best to use intention during 1 p.m. local sidereal time (check the web to compute local sidereal time).

Only send intentions on days when you feel happy and well in every way.

Putting It All Together

Your Intention Programme

Enter your intention space. Power up through meditation.

Move into peak focus through mindful awareness of the present.

Get onto the same wavelength by focusing on compassion and making a meaningful connection.

State your intention and make it specific. Mentally rehearse every moment of it with all your senses.

Visualize, in vivid detail, your intention as established fact.

Time it right – check what the sun is doing, and choose days when you feel happy and well.

Move aside – surrender to the power of the universe and let go of the outcome.

Note – Chapter 13: The Intention Exercises

1. 1. See C. T. Tart, 'Initial application of mindfulness extension exercises in a traditional Buddhist meditation retreat setting, 1995', unpublished (www.paradigmsys.com/cttart).
2. R. McCraty et al., 'The electricity of touch: Detection and measurement of cardiac energy exchange between people', in K. H. Pribram (ed.), *Brain and Values: Is a Biological Science of Values Possible?* Mahwah, NJ: Lawrence Erlbaum Associates, 1998: 359–79.
3. S. Rinpoche, *The Tibetan Book of Living and Dying*, San Francisco: HarperSanFrancisco, 1994.
4. S. Rinpoche, as quoted in J. Stone, Instructor's Training Manual, Course Syllabus: Training in Compassionate-Loving Intention, 2003.
5. H. Dienstfrey, *Where the Mind Meets the Body*, London: HarperCollins 1991: 39.

This is part 3 of a multi-part post.

The access to all the posts can be found in this index below...

- [Part 1 of 4 of the Intention Experiment](#)
- [Part 2 of 4 of the Intention Experiment](#)
- [Part 3 of 4 of the Intention Experiment](#) – You are here.
- [Part 4 of 4 of the Intention experiment](#)

Do you want more?

I have many more posts related to this in my MAJestic Index. You can find it here...

[MAJestic](#)

Articles & Links

You'll not find any big banners or popups here talking about cookies and privacy notices. There are no ads on this site (aside from the hosting ads – a necessary evil). Functionally and fundamentally, I just don't make money off of this blog. It is NOT monetized. Finally, I don't track you because I just don't care to.

To go to the MAIN Index;

[Master Index](#)

.

- You can start **reading the articles** by going [HERE](#).
- You can visit the Index Page [HERE](#) to explore by article subject.
- You can also ask the author some questions. You can go [HERE](#) .
- You can find out more about the author [HERE](#).
- If you have concerns or complaints, you can go [HERE](#).
- If you want to make a donation, you can go [HERE](#).

Please kindly help me out in this effort. There is a lot of effort that goes into this disclosure. I could use all the financial support that anyone could provide. Thank you very much.

[wp_paypal_payment]

(Visited 561 times, 1 visits today)